Power, phasors and complex power

Answers and worked solutions · Day 01 · Sat 24 Oct · pack M1

5. Practice

From your course files

Power Circuit Tool WS1 Q1 (workshop).

Hint: $P = 100$ W, $Q = 100$ VAr, $V = 240$ V.

a. The tool shows $v(t) = \sqrt{2} \times 240 \sin(2\pi 50 t)$, so $V_{rms} =$ **240 V** and $V_{pk} = \sqrt{2} \times 240 =$ **339.4 V**. $S = \sqrt{100^2 + 100^2} = 141.4$ VA, so $I_{rms} = S/V = 141.4/240 =$ **0.589 A** and $I_{pk} = \sqrt{2} \times 0.589 =$ **0.833 A**.

b. $Q = +100$ VAr: the current **lags** and the pf is **lagging** (inductive load absorbs Q). $Q = 0$: the current is **in phase** and the pf is **unity**. $Q = -100$ VAr: the current **leads** and the pf is **leading** (capacitive load generates Q).

These match the official answers.

Power Circuit Tool WS1 Q2 (workshop).

a. **Anticlockwise** (the standard direction for phasors, at $\omega = 2\pi 50$ rad/s).

b. The **y-axis** component. The tool's waveforms are written with $\sin$, so the vertical projection of each rotating phasor gives the instantaneous value.

c. With V as the reference at $0°$, look at the angle of the current phasor. Below the voltage phasor (clockwise from it, e.g. $-45°$ for $Q = +100$ VAr) means **lagging**. On top of it means **in phase**. Above it (anticlockwise, $+45°$ for $Q = -100$ VAr) means **leading**.

Power Circuit Tool WS1 Q3 (workshop).

a. $p(t)$ repeats every $T = 10$ ms, so $f = 1/T =$ **100 Hz**, twice the 50 Hz supply frequency.

b. Whenever $v(t) = 0$ or $i(t) = 0$.

(c)(i) **No change.** The average stays at $P = 100$ W for all three values of Q.

ii. $p(t) = P + S\cos(2\omega t + \phi)$, so its minimum is $P - S$. For $Q = 0$: $S = 100$ VA and the minimum is **0 W**. For $Q = \pm 100$ VAr: $S = 141.4$ VA and the minimum is

$100 - 141.4 = -41.4$ **W**. The official sheet reads about −41.2 W off the graph, so the difference is only graph-reading accuracy.

PQ 1.14 (practice questions).

Hint: $\omega = 2\pi n/60$ every time, then $P = \tau\omega$.

a. $\omega = 2\pi(2910)/60 = 304.7$ rad/s, so $\tau = 7500/304.7 =$ **24.61 Nm**.

b. $\omega = 2\pi(50\,000)/60 = 5236$ rad/s, so $P = 4.25 \times 5236 =$ **22.25 kW**.

c. $\omega = P/\tau = 30 \times 10^6/191 \times 10^3 = 157.1$ rad/s, so $n = 60\omega/2\pi =$ **1500 rpm**.

These match the official answers.

PQ 1.21 (practice questions).

Hint: $P_{OUT} = \tau\omega$.

a. $P_{IN} = V_1 I_1 \cos\phi = 110 \times 15 \times 0.8 =$ **1.320 kW**.

b. $\omega = 2\pi(1750)/60 = 183.3$ rad/s, so $P_{OUT} = 6 \times 183.3 =$ **1.100 kW**.

c. $\eta = 1100/1320 =$ **83.30 %**.

d. $P_{LOSS} = 1320 - 1100 =$ **220 W**.

These match the official answers.

PQ 1.22 (practice questions).

Hint: the 240 V is the **phase** voltage, so $I = P_{IN}/(3V_1\cos\phi)$.

a. $\omega = 2\pi(650)/60 = 68.07$ rad/s, so $P_{OUT} = 63 \times 68.07 =$ **4.288 kW**.

b. $P_{IN} = 4288/0.88 =$ **4.873 kW**.

c. $P_{LOSS} = 4873 - 4288 =$ **585 W**.

d. $I_1 = \dfrac{4873}{3 \times 240 \times 0.85} =$ **7.962 A**.

These match the official answers.

PQ 2.6 (practice questions).

Hint: maximum power is at $R_L = R_S$.

a. No current, so no drop: $V_{oc} = E =$ **12.5 V**.

b. $I_{sc} = 12.5/0.02 =$ **625 A**, and $P = I^2 R_S = 625^2 \times 0.02 =$ **7.813 kW**. (Don't short-circuit batteries: this heats them very quickly.)

c. $I = 12.5/(0.02 + 1) = 12.25$ A and $V = IR_L = 12.25$ V. The official sheet gives **12.26 V, 12.26 A**; the exact value is 12.2549, so the difference is rounding.

d. $I = (E - V)/R_S = (12.5 - 8)/0.02 =$ **225 A**, and $P = VI = 8 \times 225 =$ **1.800 kW**.

e. At $R_L = R_S = 0.02\ \Omega$: $V = 6.25$ V, $I = 12.5/0.04 = 312.5$ A, so $P_{max} = E^2/(4R_S) = 12.5^2/0.08 =$ **1.953 kW**.

PQ 2.15 (practice questions, do (f)–(h)).

Hint: in (h), add the currents.

Quick answers for (a)–(e): $24\angle 0°$, $10\angle 0°$, $7.059\angle 0°$, $34\angle 0°$ and $10\angle -90°$ A.

f. $Z = 24 + j24 = 33.94\angle 45°\ \Omega$, so $I = 240/33.94\angle 45° =$ **7.071∠−45° A**.

g. Add the branch currents: $240/24 + 240/(j24) = 10 - j10 =$ **14.14∠−45° A**.

h. $I = 24\angle 0° + 7.071\angle -45° = 24 + 5 - j5 = 29 - j5 =$ **29.43∠−9.782° A**.

These match the official answers.

PQ 2.17 (practice questions).

Hint: in (f)(iii), use V^2/R.

a. $Z = \dfrac{25 \times j125}{25 + j125} =$ **24.51∠11.31° Ω**.

b. $V_1 =$ **240∠0° V**.

c. $I_1 = V_1/Z =$ **9.792∠−11.31° A**.

d. pf $= \cos(11.31°) =$ **0.9806 lag**.

e. $S_1 = V_1 I_1^* = 240 \times 9.792\angle +11.31° =$ **2350∠11.31° VA = 2304 + j460.9 VA**.

f. i. $\mathrm{Re}(S_1) = 2304$ W. (ii) $V_1 I_1 \cos\phi = 240 \times 9.792 \times 0.9806 = 2304$ W. (iii) $V^2/R = 240^2/25 =$ **2.304 kW**. You can't use $I_1^2 R$, because I_1 is the line current, and only part of it flows through the resistor.

g. $Q = \mathrm{Im}(S_1) =$ **+460.9 VAr** (check: $V^2/X = 240^2/125 = 460.8$ VAr).

h. $S = $ **2350 VA**.

i. $2304^2 + 460.9^2 = 2350^2$. The triangle has P = 2304 W along the horizontal, Q = +460.9 VAr upwards, and the angle between S and P is 11.31°.

j. V_1 is along 0°, and I_1 (9.792 A) lags it by 11.31°.

These match the official answers.

PQ 2.18 (practice questions).

Hint: the voltages add at 90°.

a. V_R and V_L are perpendicular, so $V_L = \sqrt{200^2 - 150^2} =$ **132.3 V**.

b. $\phi = -\cos^{-1}(V_R/V_1) = -\cos^{-1}(0.75) =$ **−41.41°** (negative because the current lags). pf = **0.75 lag**.

c. $S = V_1 I_1 = 200 \times 10 =$ **2000 VA**, $P = S\cos\phi =$ **1500 W**, $Q = S\sin(-\phi) =$ **+1323 VAr**.

d. Resistor: **1500 W, 0 VAr**.

e. Inductor: **0 W, +1323 VAr**.

f. $R = P/I^2 = 1500/100 =$ **15 Ω** and $X = Q/I^2 = 1323/100 =$ **13.23 Ω** (reactance is a scalar number).

These match the official answers.

PQ 2.19 (practice questions).

Hint: Q is negative (capacitor).

a. All the real power is in R, which has the full V_1 across it: $R = V^2/P = 140^2/500 =$ **39.2 Ω**.

b. $S = V_1 I_1 = 140 \times 15 =$ **2100 VA**. $Q = -\sqrt{S^2 - P^2} = -\sqrt{2100^2 - 500^2} =$ **−2040 VAr** (negative because of the capacitor).

c. $\phi = +\cos^{-1}(500/2100) =$ **+76.23°**, so pf = **0.2381 leading**.

d. $P/S = 500/2100 = 0.2381$ leading, the same.

e. $X = V^2/Q = 140^2/(-2040) =$ **−9.608 Ω**. (Using the unrounded Q gives −9.610 Ω.)

f. V_1 at 0°. I_R is in phase with V_1. I_C points straight up ($+90^\circ$). $I_1 = I_R + I_C$ is the diagonal, $\phi = 76.23^\circ$ above V_1.

g. P = 500 W horizontal, Q = −2040 VAr downwards, S = 2100 VA as the hypotenuse, with ϕ between S and P.

These match the official answers.

Jun23 Q1a (past exam, 5 marks).

Hint: Playbook A.

(i) On open circuit $I = 0$, so there is no drop across R_S: $E = V_{oc} = \mathbf{13.8\ V}$.

(ii) With the starter motor, $I = 300$ A and $V = 9.5$ V:

$$R_S = \frac{E - V}{I} = \frac{13.8 - 9.5}{300} = \mathbf{0.01433\ \Omega}$$

$$P_{R_S} = I^2 R_S = 300^2 \times 0.01433 = \mathbf{1290\ W}, \qquad P_{motor} = VI = 9.5 \times 300 = \mathbf{2850\ W}$$

Check: $EI = 13.8 \times 300 = 4140$ W, which equals 1290 + 2850.

(iii) $V = E - IR_S = 13.8 - 20 \times 0.01433 = \mathbf{13.51\ V}$.

All of these match the official answers.

Jun23 Q1c (past exam, 9 marks).

Hint: Playbook C. The element values are in the course figure above: $R = 10\ \Omega$, $X_L = 30\ \Omega$ and $X_C = 50\ \Omega$. The given current is $I_1 = 13\angle{+20^\circ}$ A.

Circuit for Jun23 Q1c: series R = 10 Ω, jX_L = j30 Ω, −jX_C = −j50 Ω carrying I_1 = 13∠20° A (values from the course figure)

(i) Ohm's law for each element, using the same current through each:

$$V_R = I_1 R = 13\angle{20^\circ} \times 10 = \mathbf{130\angle{20^\circ}\ V}$$

$$V_L = I_1(jX_L) = 13\angle{20^\circ} \times 30\angle{90^\circ} = \mathbf{390\angle{110^\circ}\ V}$$

$$V_C = I_1(-jX_C) = 13\angle{20^\circ} \times 50\angle{-90^\circ} = \mathbf{650\angle{-70^\circ}\ V}$$

The official sheet's "130 120 V" is a text-extraction artefact of $130\angle{20^\circ}$. V_C (650 V) is larger than the supply voltage. This is normal in a series RLC, because V_L and V_C partly cancel.

(ii) $Z = 10 + j30 - j50 = 10 - j20 = \mathbf{22.36\angle{-63.43^\circ}\ \Omega}$. This is capacitive overall.

$$V_1 = I_1 Z = 13\angle{20^\circ} \times 22.36\angle{-63.43^\circ} = \mathbf{290.7\angle{-43.43^\circ}\ V}$$

(iii) $I_1^* = \mathbf{13\angle{-20^\circ}\ A}$.

$$S = V_1 I_1^* = 290.7\angle -43.43^\circ \times 13\angle -20^\circ = 3779\angle -63.43^\circ = \mathbf{1690 - j3380\ VA}$$

So $P =$ **1690 W** and $Q =$ **−3380 VAr**. The source receives reactive power, because the load is capacitive and the current leads.

Check: $I^2R = 169 \times 10 = 1690$ W and $I^2(X_L - X_C) = 169 \times (-20) = -3380$ VAr. The official answer of −3379 VAr is the same value with rounded intermediate steps.

Jun24 Q1a(ii) (past exam, 2 marks).

$P_{OUT} = 1000 - 300 =$ **700 W** and $\eta = 700/1000 =$ **70 %**. More loss for the same input means η **decreases**. More input for the same loss means η **increases**, because the loss becomes a smaller fraction of the input.

These match the official answers.

Jun24 Q1b (past exam, 3 marks).

Hint: Playbook B. A 3ph nameplate gives line values.

i. $S = \sqrt{3} V_L I_L = \sqrt{3} \times 415 \times 6.8 = 4888$ VA, so pf $= P/S = 4000/4888 =$ **0.8184 lag**.

ii. $\omega = 2\pi(1420)/60 = 148.7$ rad/s, so $P_{OUT} = \tau\omega = 23 \times 148.7 =$ **3.420 kW** and $\eta = 3420/4000 =$ **85.50 %**.

The “4 pole, 50 Hz” information is not needed here. It gives $n_s = 1500$ rpm, which you will use in Ch7.

These match the official answers.

Jun24 Q1c (past exam, 5 marks).

Hint: Playbook D. From the course figure, $R = 100\ \Omega$ and $X_L = 40\ \Omega$.

Circuit for Jun24 Q1c: series R = 100 Ω and jX_L = j40 Ω carrying 2 A (values from the course figure)

i. The same current $I_1 = 2$ A flows through both elements. Resistor: $P = I^2R =$ **400 W**, $Q =$ **0 VAr**. Inductor: $P =$ **0 W**, $Q = I^2X_L =$ **160 VAr**.

ii. V_R is in phase with I_1, and V_L leads it by 90°, so they are perpendicular:

$$V_1 = \sqrt{V_R^2 + V_L^2} = \sqrt{50^2 + 20^2} = \mathbf{53.85\ V}$$

Adding the magnitudes (70 V) is the classic wrong answer.

These match the official answers.

New questions

N1. Single-phase pump motor [5 marks].

a. $P_{IN} = V_1 I_1 \cos\phi = 230 \times 6.2 \times 0.78 =$ **1112 W**. $\eta = 950/1112 =$ **85.41 %** (the loss is 162.3 W).

b. $\omega = 2\pi(2860)/60 = 299.5$ rad/s, so $\tau = 950/299.5 =$ **3.172 Nm**.

c. $S = V_1 I_1 = 230 \times 6.2 =$ **1426 VA**. $Q = S\sin(-\phi)$ with $\phi = -\arccos 0.78 = -38.74°$, so $Q = 1426 \times \sin(38.74°) =$ **+892.4 VAr**. It is positive because the motor is inductive.

N2. Series RC [6 marks].

a. $X_C = 1/(2\pi \cdot 50 \cdot 100 \times 10^{-6}) = 31.83\ \Omega$. $Z = 30 - j31.83 =$ **43.74∠−46.70° Ω**.

b. $I_1 = 240\angle 0°/43.74\angle -46.70° =$ **5.487∠+46.70° A**. This leads, as a capacitive circuit should.

$V_R = I_1 R =$ **164.6∠46.70° V**. $V_C = I_1(-jX_C) = 5.487\angle 46.70° \times 31.83\angle -90° =$ **174.6∠−43.30° V**.

$\sqrt{164.6^2 + 174.6^2} = 240.0$ V, which equals V_1.

c. $S = V_1 I_1^* = 240 \times 5.487\angle -46.70° = 1317\angle -46.70° =$ **903.2 − j958.3 VA**. So $P = 903.2$ W, $Q = -958.3$ VAr and pf $= \cos(46.70°) =$ **0.6859 lead**.

Check: $I^2 R = 5.487^2 \times 30 = 903.2$ W.

N3. One measurement, two models [6 marks].

a. $S = 230 \times 5 =$ **1150 VA**. $Q = +\sqrt{1150^2 - 900^2} =$ **+715.9 VAr** (positive because the load is inductive). pf $= 900/1150 =$ **0.7826 lag**.

b. $R_S = P/I^2 = 900/25 =$ **36.0 Ω**, $X_S = Q/I^2 = 715.9/25 =$ **28.64 Ω**, and $L_S = 28.64/(2\pi \cdot 50) =$ **91.15 mH**.

c. $R_P = V^2/P = 230^2/900 =$ **58.78 Ω**, $X_P = V^2/Q = 230^2/715.9 =$ **73.89 Ω**, and $L_P =$ **235.2 mH**.

Check: converting $36 + j28.64$ to its parallel equivalent through $1/Z$ gives the same 58.78 Ω and 73.89 Ω.

N4. Two loads on one supply (chained) [8 marks].

a. $P_A = 1500/0.80 =$ **1875 W**. $S_A = P_A/\text{pf} = 1875/0.75 =$ **2500 VA**.
$Q_A = S_A \sin(\arccos 0.75) =$ **+1654 VAr**. $I_A = 2500/240 =$ **10.42 A**, with angle $-41.41°$.

b. $Z_B = 40 - j30 = 50\angle -36.87°\ \Omega$, so $I_B = 240/50\angle -36.87° =$ **4.8∠+36.87° A**.

$S_B = VI_B^* = 240 \times 4.8\angle -36.87° =$ **921.6 − j691.2 VA**. Check:
$I^2R = 23.04 \times 40 = 921.6$ W.

c. $P = 1875 + 921.6 =$ **2797 W**. $Q = 1654 - 691.2 =$ **+962.4 VAr**.
$S = \sqrt{2797^2 + 962.4^2} =$ **2958 VA**. pf $= 2797/2958 =$ **0.9456 lag**.
$I = 2958/240 =$ **12.32 A**.

Check by adding the phasors: $10.42\angle -41.41° + 4.8\angle 36.87° = 12.32\angle -18.99°$ A. Note that $|I_A| + |I_B| = 15.2$ A is not the supply current. The capacitive load B partly cancels A's reactive power, which is power-factor correction in disguise (pack M2).

N5. Series RLC with a surprise [7 marks].

a. $X_L = 2\pi(50)(0.04) =$ **12.57 Ω** and $X_C = 1/(2\pi \cdot 50 \cdot 200 \times 10^{-6}) =$ **15.92 Ω**.
$Z = 8 + j(12.57 - 15.92) = 8 - j3.349 =$ **8.673∠−22.72° Ω**. It is **capacitive** overall, because $X_C > X_L$.

b. $I_1 = 230/8.673\angle -22.72° =$ **26.52∠+22.72° A**. $V_R = 212.2\angle 22.72°$ V, $V_L = 333.3\angle 112.72°$ V and $V_C = 422.1\angle -67.28°$ V. $|V_C| = 422$ V is much larger than the 230 V supply. V_L and V_C are $180°$ apart and mostly cancel, so the capacitor must be rated for the larger voltage.

c. $S = 230 \times 26.52\angle -22.72° = 6100\angle -22.72° =$ **5626 − j2355 VA**. So $P = 5626$ W, $Q = -2355$ VAr and pf $=$ **0.9224 lead**.

Check: $I^2R = 26.52^2 \times 8 = 5626$ W. The inductor absorbs $I^2X_L = +8838$ VAr and the capacitor generates $-I^2X_C = -11193$ VAr, which sum to -2355 VAr.

d. Unity pf needs $X_L = X_C$, i.e. $f_0 = \dfrac{1}{2\pi\sqrt{LC}} = \dfrac{1}{2\pi\sqrt{0.04 \times 200 \times 10^{-6}}} =$ **56.27 Hz**.

N6. Three-phase motor nameplate (brief preview of M2) [6 marks].

a. The nameplate power is the output. $\omega = 2\pi(1455)/60 = 152.4$ rad/s, so $\tau = 7500/152.4 =$ **49.22 Nm**.

b. $P_{IN} = 7500/0.88 =$ **8.523 kW**, so $P_{LOSS} =$ **1.023 kW**.

c. $I_L = P_{IN}/(\sqrt{3}V_L\,\text{pf}) = 8523/(\sqrt{3}\times 415\times 0.84) =$ **14.12 A**.
$S = 8523/0.84 = 10.15$ kVA, so $Q = S\sin(\arccos 0.84) =$ **+5.505 kVAr**.

d. Assume star, as the course does when the connection isn't stated: $V_1 = 415/\sqrt{3} =$ **239.6 V**.

6. Quick recall check

Question

1. $\sqrt{2}\times 230 = 325.3$ V.
2. 314.2 rad/s; 954.9 rpm.
3. Leading (Q < 0 means capacitive). $S = 1300$ VA, so pf $= 1200/1300 = 0.9231$ lead.
4. $\sqrt{80^2+60^2} = 100$ V, not 140 V.
5. S is a magnitude, and the two loads' complex powers point in different directions. Only P and Q are conserved, so add those and then use $S = \sqrt{P^2+Q^2}$.

Power-factor correction, 3-phase, equivalent-circuit tests

Answers and worked solutions · Day 02 · Sun 25 Oct · pack M2

5. Practice

From your course files

Jun23 Q1a (past exam, 5 marks).

> *Hint:* $R_S = (13.8 - 9.5)/300$.
>
> i. On open circuit $I = 0$, so there is no drop across R_S: $E = V_{oc} =$ **13.8 V**.
>
> ii. $R_S = (E - V)/I = (13.8 - 9.5)/300 =$ **0.01433 Ω**. Loss: $I^2 R_S = 300^2 \times 0.01433 =$ **1290 W**. To the motor: $VI = 9.5 \times 300 =$ **2850 W**. Check: $EI = 4140$ W $= 1290 + 2850$.
>
> iii. $V = E - IR_S = 13.8 - 20 \times 0.01433 =$ **13.51 V**.
>
> These match the official answers.

Jun23 Q1b (past exam, 6 marks).

> *Hint:* delta power = 3 × star power.
>
> i. Star: $V_1 = 460/\sqrt{3} = 265.6$ V, so $I_1 = 265.6/70 =$ **3.794 A**, $I_L = I_1 =$ **3.794 A**, and $P = 3V_1^2/R = 3 \times 265.6^2/70 =$ **3.023 kW**.
>
> ii. Delta: $V_1 = V_L = 460$ V, so $I_1 = 460/70 =$ **6.571 A**, $I_L = \sqrt{3} I_1 =$ **11.38 A**, and $P = 3 \times 460^2/70 =$ **9.069 kW** (exactly 3 times the star power).
>
> These match the official answers.

Jun24 Q1d (past exam, 8 marks).

> *Hint:* (iii) uses $Q_C = -30$ VAr, 115 V, **60 Hz**.
>
> i. Triangle: $P = 80$ W horizontal, $Q = +60$ VAr up, so $S = 100$ VA. pf $= 80/100 =$ **0.8 lag**.
>
> ii. The capacitor takes no real power, so $P_S =$ **80 W**.
> $Q_S = P_S \tan(\cos^{-1} 0.9) = 80 \times 0.4843 =$ **38.75 VAr**.
> $Q_C = Q_S - Q_L = 38.75 - 60 =$ **−21.25 VAr**.

iii. $C = \dfrac{|Q_C|}{\omega V^2} = \dfrac{30}{2\pi(60) \times 115^2}$ = **6.017 µF**. (Using −21.25 VAr from part (ii) gives 4.26 µF, which is wrong.)

These match the official answers.

Power Circuit Tool WS2 Task 1 (workshop).

Hint: assume the tool's 240 V, 50 Hz supply. The workshop solutions mention a 240 V source; 50 Hz is assumed. If your simulator showed a different frequency, rescale C by $50/f$. (The official sheet gives only qualitative answers.)

$I = (S/V)^* = \big((P + jQ)/240\big)^*.$

Case	P	Q	I_{supply}
1	100	0	$0.4167\angle 0^\circ$ A
2	0	100	$0.4167\angle -90^\circ$ A
3	100	100	$0.5893\angle -45^\circ$ A

1. Same magnitude, but case 2 lags by 90°.
2. Same fluctuating part at 2f, but case 2 has zero average power.
3. $I_3 = I_1 + I_2$ (superposition: P and Q add).

Power Circuit Tool WS2 Task 2 (workshop).

$P = 100$ W and $Q = 200$ VAr, so $S = 223.6$ VA and $I = 0.9317\angle -63.43^\circ$ A (the same for both models, because the tool fixes P and Q).

Case	I_{supply}	R (Ω)	X (Ω)
Series	$0.9317\angle -63.43^\circ$ A	$P/I^2 = 115.2$	$Q/I^2 = 230.4$
Parallel	$0.9317\angle -63.43^\circ$ A	$V^2/P = 576$	$V^2/Q = 288$

1. Series: the **current** is common.
2. R:X = P:Q = **1:2**, because $P = I^2R$ and $Q = I^2X$ share the same I^2.
3. Parallel: the **voltage** is common.
4. R:X = Q:P = **2:1**, because $R = V^2/P$ and $X = V^2/Q$ share the same V^2.

Power Circuit Tool WS2 Task 3 (workshop).

$P = 100$ W, $Q = 140$ VAr. The exact optimum is $C = 140/(314.16 \times 240^2) = 7.74$ µF. With integer steps, 7 µF leaves +13.3 VAr and 8 µF leaves −4.8 VAr, so C_{opt} = **8 µF**.

Case	C (µF)	P_S (W)	Q_S (VAr)	I_S (A)	pf
1	0	100	+140.0	$0.7169\angle -54.46^\circ$	0.581 lag
2	4	100	+67.6	$0.5030\angle -34.07^\circ$	0.828 lag
3	8	100	−4.8	$0.4171\angle +2.73^\circ$	0.999 lead
4	12	100	−77.1	$0.5262\angle +37.65^\circ$	0.792 lead

1. P_S stays at 100 W because a capacitor absorbs no real power.
2. From 0 to C_{opt} the magnitude of I_S falls and its angle rises towards 0. Beyond C_{opt} the magnitude rises again and the angle becomes positive (leading).
3. At C_{opt}, $I_S = 0.4171$ A. The minimum possible current for 100 W at unity pf is $100/240 = 0.4167$ A, almost identical.

PQ 2.20 (practice questions).

Hint: 60 Hz. This is also worked in your Test Reference 6.1.

a. $I_1 = 200/100 =$ **2 A**. $|Z_2| = 22.36\ \Omega$, so $I_2 = 200/22.36 =$ **8.944 A**.
$P = 2^2(100) + 8.944^2(10) = 400 + 800 =$ **1200 W** and $Q = 8.944^2(20) =$ **+1600 VAr**. $S = 2000$ VA, so pf = **0.600 lag**.

b. For pf 0.8 lag at the same P: $Q_{\text{new}} = 1200\tan(\cos^{-1}0.8) = +900$ VAr, so $Q_C = 900 - 1600 = -700$ VAr. $X_C = V^2/Q_C = 200^2/(-700) = -57.14\ \Omega$, so $C = 1/(2\pi \cdot 60 \cdot 57.14) =$ **46.42 µF**.

c. Before: **2000 VA**. After: $\sqrt{1200^2 + 900^2} =$ **1500 VA**.

d. Both triangles share P = 1200 W. Before: Q = +1600 VAr, S = 2000 VA. After: Q = +900 VAr, S = 1500 VA. The capacitor's −700 VAr is the vertical drop between them.

These match the official answers.

PQ 2.6 (practice questions).

Hint: (d) $I = (E - V)/R$.

a. $V_{oc} = E =$ **12.5 V**.

b. $I_{sc} = 12.5/0.02 =$ **625 A**, $P = I^2R =$ **7.813 kW**.

c. $I = 12.5/1.02 = 12.25$ A, $V = 12.25$ V. The official sheet gives **12.26 V, 12.26 A**; the exact value is 12.2549, so the difference is rounding.

d. $I = (12.5 - 8)/0.02 =$ **225 A**, $P = 8 \times 225 =$ **1.800 kW**.

e. At $R_L = R = 0.02\ \Omega$: $P_{max} = E^2/(4R) =$ **1.953 kW** ($V = 6.25$ V, $I = 312.5$ A).

PQ 2.7 (practice questions).

a. $E = V_{oc} =$ **1.61 V** and $R = V_{oc}/I_{sc} = 1.61/3.1 =$ **0.5194 Ω**.

b. $V = E - IR = 1.61 - 0.5 \times 0.5194 =$ **1.350 V** and $P = VI =$ **0.6752 W**.

c. At $R_L = R$: $P_{max} = E^2/(4R) = 1.61^2/(4 \times 0.5194) =$ **1.248 W**.

These match the official answers.

PQ 2.21 (practice questions).

Hint: series uses I^2; parallel uses V^2.

a. $R = P/I^2 = 25/400 =$ **0.0625 Ω**. $|Z| = 5/20 = 0.25\ \Omega$, so $X = \sqrt{0.25^2 - 0.0625^2} =$ **0.2421 Ω**.

b. $R = V^2/P = 110^2/40 =$ **302.5 Ω**. $S = 110 \times 2.4 = 264$ VA, $Q = \sqrt{264^2 - 40^2} = 261.0$ VAr, so $X = V^2/Q =$ **46.36 Ω** (46.37 Ω unrounded).

These match the official answers.

PQ 2.26 (practice questions).

Hint: draw the star and delta first.

a. Star: $V_L = \sqrt{3} \times 110 =$ **190.5 V**, $I_L = I_1 =$ **10 A**.

b. Delta: $V_L = V_1 =$ **110 V**, $I_L = \sqrt{3} \times 10 =$ **17.32 A**.

These match the official answers.

PQ 2.28 (practice questions).

a. $V_1 = 230/\sqrt{3} = 132.8$ V, so $R = 132.8/10 =$ **13.28 Ω**.

b. Delta: $V_1 =$ **230 V**, $I_1 = 230/13.28 =$ **17.32 A**, $I_L = \sqrt{3} \times 17.32 =$ **30 A**.

c. Phase voltage ×$\sqrt{3}$, phase current ×$\sqrt{3}$, supply (line) current ×**3**.

These match the official answers. (The text copy of the answer sheet loses the √ symbols; the PDF itself shows √3, √3 and 3.)

PQ 2.30 (practice questions).

Hint: your Test Reference 7.2.

a. Delta: $V_1 = V_L =$ **415 V**.

b. $Z = 2 + j3 = 3.606\angle 56.31^\circ\ \Omega$, so $I_1 = 415/3.606\angle 56.31^\circ =$ **115.1∠−56.31° A**.

c. $I_L = \sqrt{3} \times 115.1 =$ **199.4 A**.

d. pf $= \cos 56.31^\circ =$ **0.5547 lag**.

e. $S = 3V_1 I_1 = 3 \times 415 \times 115.1 = 143.3$ kVA. The official answer is **143.2 kVA**; the difference is rounding.

f. $P = S \times \text{pf} =$ **79.50 kW** (79.49 kW unrounded).

g. $3I_1^2 R = 3 \times 115.1^2 \times 2 =$ **79.50 kW**, the same.

h. $Q = 3I_1^2 X = 3 \times 115.1^2 \times 3 =$ **+119.2 kVAr**.

PQ 2.31 (practice questions).

In star each phase sees $V_1 = 415/\sqrt{3} = 239.6$ V instead of 415 V.

a. phase voltage ×$1/\sqrt{3}$ (239.6 V). (b) phase current ×$1/\sqrt{3}$ (66.45 A, same angle). (c) line current ×1/3: in star $I_L = I_1 = 66.45$ A, versus 199.4 A before. (d) pf ×1 (the impedance angle is unchanged). (e)–(h) S, P (both ways) and Q all ×1/3, because each goes as V_1^2 per phase.

Check: $P = 3 \times 66.45^2 \times 2 = 26.50$ kW $= 79.50/3$.

These match the official answers (the PDF shows $1/\sqrt{3}$ for (a) and (b); the text copy loses the √).

Jun23 Q4b (past exam, 7 marks), stretch.

Hint: the synchronous motor *generates* its spare vars: $Q_{\text{syn}} = -\sqrt{250^2 - 222.2^2}$.

i. $P_{im} = 200/0.85 =$ **235.3 kW** and $Q_{im} = 235.3 \tan(\cos^{-1} 0.83) =$ **+158.1 kVAr**.

ii. $P_{syn} = 200/0.90 =$ **222.2 kW**. Its 250 kVA rating limits S, so the most reactive power it can supply is $\sqrt{250^2 - 222.2^2} =$ **114.6 kVAr** (generated, so $Q_{syn} = -114.6$ kVAr). Unrounded this is 114.5 kVAr.

iii. $P = 235.3 + 222.2 = 457.5$ kW and $Q = 158.1 - 114.6 = 43.5$ kVAr, so the best plant pf is $457.5/\sqrt{457.5^2 + 43.5^2} =$ **0.9955 lag**.

These match the official answers.

New questions

N1.

a. Delta: $V_1 = V_L = 400$ V, so $I_1 = 400/48 =$ **8.333 A**, $I_L = \sqrt{3} \times 8.333 =$ **14.43 A**, and $P = 3V_1^2/R = 3(400^2)/48 =$ **10.0 kW**.

b. Star: $V_1 = 400/\sqrt{3} =$ **230.9 V**, $I_L = I_1 = 230.9/48 =$ **4.811 A**, and $P = 3(230.9^2)/48 =$ **3.333 kW**. The ratio is $P_\Delta/P_Y =$ **3**.

N2.

a. The drop rises by $12.6 - 11.2 = 1.4$ V for an extra 70 A, so $R_S = 1.4/70 =$ **0.02 Ω**. Then $E = 12.6 + 5(0.02) =$ **12.7 V**.

b. $V = 12.7 - 150(0.02) =$ **9.7 V**, $P_{\text{load}} = 9.7 \times 150 =$ **1455 W**, and $P_{\text{loss}} = 150^2 \times 0.02 =$ **450 W**. Check: $EI = 1905$ W $= 1455 + 450$.

c. $P_{max} = E^2/(4R_S) = 12.7^2/0.08 =$ **2016 W**, at $I = 317.5$ A and $V = 6.35$ V.

N3.

a. $S = 230 \times 6.2 =$ **1426 VA**, $Q = +\sqrt{1426^2 - 1100^2} =$ **+907.5 VAr** (positive because a motor is inductive), and pf $= 1100/1426 =$ **0.7714 lag**.

b. $R_S = P/I^2 = 1100/6.2^2 =$ **28.62 Ω**, $X_S = Q/I^2 = 23.61\ \Omega$, so $L_S = 23.61/314.16 =$ **75.1 mH**.

c. $R_P = V^2/P = 230^2/1100 =$ **48.09 Ω**, $X_P = V^2/Q = 58.29\ \Omega$, so $L_P =$ **185.6 mH**.

d. $C = 907.5/(314.16 \times 230^2) =$ **54.6 µF**, and $I = P/V = 1100/230 =$ **4.78 A** (down from 6.2 A).

N4.

a. Motor input: $P_{im} = 22/0.9 = 24.44$ kW and $Q_{im} = 24.44 \tan(\arccos 0.82) = +17.06$ kVAr. The heater adds 10 kW and 0 kVAr.

$$P = 34.44\ \text{kW}, \quad Q = +17.06\ \text{kVAr}, \quad S = 38.44\ \text{kVA}, \quad \textbf{pf} = \textbf{0.8961 lag}, \quad I_L = \frac{38440}{\sqrt{3} \cdot 415} = \textbf{53.48 A}$$

b. $Q_S = 34.44 \tan(\arccos 0.98) = 6.994$ kVAr, so $Q_C = 6.994 - 17.06 =$ **−10.07 kVAr** (−3.356 kVAr per capacitor).

c. Delta: $C_\Delta = 3356/(314.16 \times 415^2) =$ **62.0 µF**, rated 415 V. Star: $C_Y = 3356/(314.16 \times 239.6^2) =$ **186.1 µF**, rated 240 V. The delta bank needs the higher voltage rating. The new line current is $35.15\ \text{kVA}/(\sqrt{3} \cdot 415) = 48.90$ A.

d. In star each 62.0 µF capacitor sees only 239.6 V, so the bank supplies $Q_C/3 = -3.356$ kVAr. Then $Q_S = 17.06 - 3.356 = 13.71$ kVAr and pf $= 34.44/\sqrt{34.44^2 + 13.71^2} =$ **0.929 lag**.

N5.

a. $S = \sqrt{25^2 + 31^2} = 39.82$ VA, $I = 39.82/60.5 =$ **0.658 A** (Table 5 measured $I_L \approx$ 0.658 A), and pf $= 25/39.82 =$ **0.628 lag**.

b. $\tan(\arccos 0.9) = 0.4843$, so $|Q_S| = 12.11$ VAr.

- Lagging: $Q_S = +12.11$ gives $Q_C = -18.89$ VAr and $C = 18.89/(314.16 \times 60.5^2) =$ **16.4 µF (lag)**.
- Leading: $Q_S = -12.11$ gives $Q_C = -43.11$ VAr and $C =$ **37.5 µF (lead)**.

c. Both give $S = 25/0.9 = 27.78$ VA, so $I_S =$ **0.459 A** in both cases. The minimum current is at unity pf: $25/60.5 =$ **0.413 A**.

d. Reactive power is conserved: $Q_S = Q_L + Q_C = 31.0 - 45.3 = -14.3$ VAr. That is negative, so the supply value must be **−16.6 VAr**. The circuit is over-corrected and the 0.837 pf is **leading**.

6. Quick recall check

Question

1. $415/\sqrt{3} = 239.6$ V.
2. Both rise by a factor of 3. The pf does not change.
3. $S = 100$ VA and $Q = +80$ VAr, so $Q_C = -80$ VAr.
4. $(12.0 - 11.4)/30 = 0.02\ \Omega$.

5. $C_Y = 3C_\Delta$. The star bank needs three times the capacitance, at $1/\sqrt{3}$ of the voltage rating.

Magnetic circuits

Answers and worked solutions · Day 03 · Mon 26 Oct · pack M3

5. Practice

From your course files

PQ 3.3 (practice questions).

$\mathcal{R} = \dfrac{l}{\mu_R \mu_0 A}$ each time, with everything in metres.

a. $A = \pi(0.5 \times 10^{-3})^2 = 7.854 \times 10^{-7}$ m², so $\mathcal{R} = \dfrac{0.1}{1000 \times 4\pi \times 10^{-7} \times 7.854 \times 10^{-7}} =$ **101.3 × 10⁶ At/Wb**.

b. $\mathcal{R} = \dfrac{0.2}{5000 \times 4\pi \times 10^{-7} \times 2 \times 10^{-4}} =$ **159.2 × 10³ At/Wb**.

c. $\mathcal{R} = \dfrac{10^{-3}}{4\pi \times 10^{-7} \times 10 \times 10^{-4}} =$ **795.8 × 10³ At/Wb**. A 1 mm gap has five times the reluctance of the whole 20 cm core in (b).

These match the official answers.

PQ 3.4 (practice questions).

$\text{mmf} = Hl = \dfrac{B}{\mu_R \mu_0} l$ and $I = \text{mmf}/N$.

a. $H = 1/(4\pi \times 10^{-7}) = 795.8$ kA/m, so mmf $= 795.8 \times 10^3 \times 10^{-3} =$ **795.8 At** and $I =$ **7.958 A**.

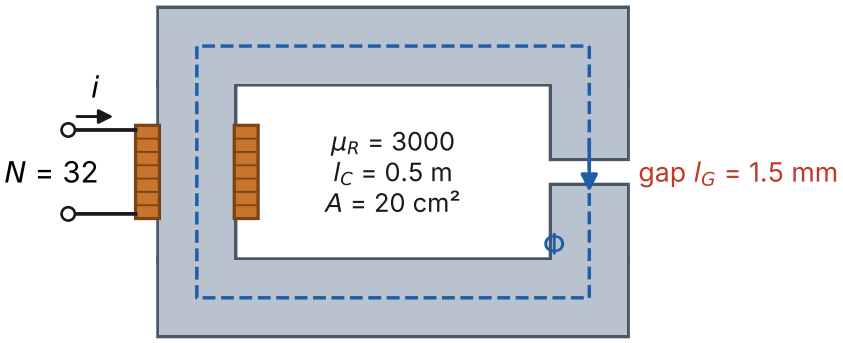
b. $H = 1/(5000 \times 4\pi \times 10^{-7}) = 159.2$ A/m, so mmf $= 159.2 \times 0.2 =$ **31.83 At** and $I =$ **0.3183 A**.

These match the official answers.

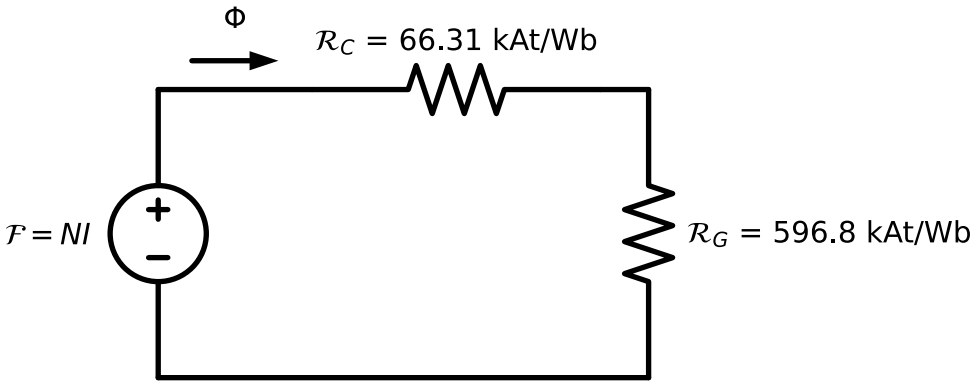
Jun23 Q2a (past exam, 9 marks).

GIVEN: a C-core with $A = 20$ cm² $= 2 \times 10^{-3}$ m², $l_C = 0.5$ m, $\mu_R = 3000$, $N = 32$, and one gap $l_G = 1.5$ mm.

Jun23 Q2a: C-core with one air gap (gap exaggerated)



Jun23 Q2a: C-core with a 32-turn coil and one 1.5 mm gap



Magnetic equivalent circuit for Jun23 Q2a: core and gap reluctances in series

(i) Reluctances and L [4].

$$\mathcal{R}_G = \frac{l_G}{\mu_0 A} = \frac{1.5 \times 10^{-3}}{4\pi \times 10^{-7} \times 2 \times 10^{-3}} = \mathbf{596.8\ kAt/Wb}$$

$$\mathcal{R}_C = \frac{l_C}{\mu_R \mu_0 A} = \frac{0.5}{3000 \times 4\pi \times 10^{-7} \times 2 \times 10^{-3}} = \mathbf{66.31\ kAt/Wb}$$

These are in series: $\mathcal{R}_T = 663.1$ kAt/Wb.

$$L = \frac{N^2}{\mathcal{R}_T} = \frac{32^2}{663.1 \times 10^3} = \mathbf{1.544\ mH}$$

(ii) Maximum energy in the gap only [2]. At $\hat{B} = 1.8$ T, the gap volume is $Al_G = 2 \times 10^{-3} \times 1.5 \times 10^{-3} = 3 \times 10^{-6}$ m³.

$$E = \frac{B^2}{2\mu_0} Al_G = \frac{1.8^2}{2 \times 4\pi \times 10^{-7}} \times 3 \times 10^{-6} = \mathbf{3.867\ J}$$

(For interest: the total $\frac{1}{2}LI^2$ at 1.8 T is 4.30 J. The gap holds 90% of it, matching $\mathcal{R}_G/\mathcal{R}_T = 0.9$.)

(iii) Iron loss at new B and f [3]. Split the 12 W: $P_{H1} = 0.3 \times 12 = 3.6$ W and $P_{E1} = 0.7 \times 12 = 8.4$ W.

$$P_{H2} = 3.6\left(\frac{1.2}{1.5}\right)^2\left(\frac{100}{60}\right) = \mathbf{3.84\ W}$$

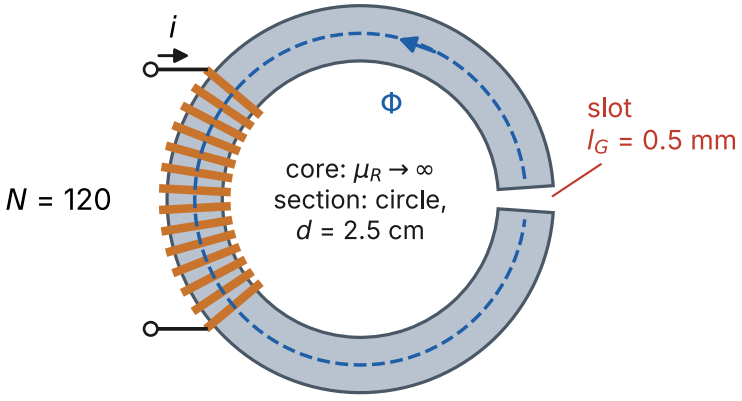
$$P_{E2} = 8.4\left(\frac{1.2}{1.5}\right)^2\left(\frac{100}{60}\right)^2 = \mathbf{14.93\ W}$$

$$P_{FE} = \mathbf{18.77\ W}$$

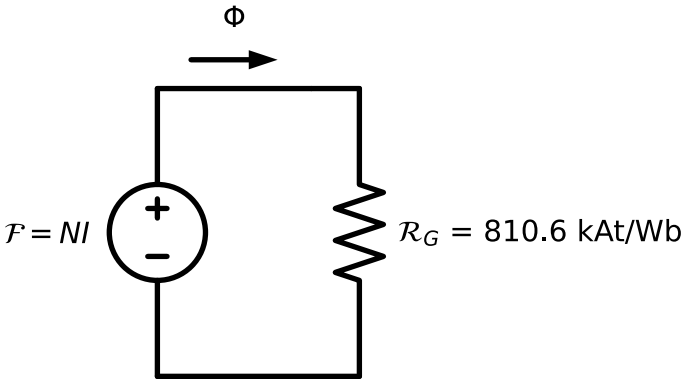
All match the course answers.

Jun24 Q2a (past exam, 7 marks).

GIVEN: a toroid with a circular section of diameter 2.5 cm, so $A = \pi(0.0125)^2 = 4.909 \times 10^{-4}$ m². $N = 120$. The permeability is infinite, so $\mathcal{R}_C = 0$. One slot gives one gap, $l_G = 0.5$ mm.



Jun24 Q2a: toroid of circular section with a 120-turn coil and one 0.5 mm saw-cut slot



$\mathcal{R}_C = 0$ (infinite permeability)

Magnetic equivalent circuit for Jun24 Q2a: with infinite permeability only the gap reluctance remains

(i) [3].

$$\mathcal{R}_G = \frac{0.5 \times 10^{-3}}{4\pi \times 10^{-7} \times 4.909 \times 10^{-4}} = \mathbf{810.6\ kAt/Wb}$$

$$L = \frac{N^2}{\mathcal{R}_G} = \frac{120^2}{810.6 \times 10^3} = \mathbf{17.77\ mH}$$

Note that the mean diameter of the toroid is not given and not needed: with infinite permeability only the gap matters.

(ii) Maximum 50 Hz voltage [2]. Use the back-emf equation. It needs no reluctance.

$$V_{\max} = 4.44\, N f \hat{B}_{\max} A = 4.44 \times 120 \times 50 \times 1.5 \times 4.909 \times 10^{-4} = \mathbf{19.62\ V\ rms}$$

(iii) Force across the slot [2]. There is one gap, so the area is used once.

$$F = \frac{B^2}{2\mu_0} A = \frac{1.5^2}{2 \times 4\pi \times 10^{-7}} \times 4.909 \times 10^{-4} = \mathbf{439.5\ N}$$

All match the course answers.

2023 quiz Q3 (6 marks).

GIVEN: a toroid with mean diameter 80 mm, so $l = \pi(0.08) = 0.2513$ m. $A = 20$ cm² $= 2 \times 10^{-3}$ m², $\mu_R = 1000$, $\hat{B} = 1.5$ T.

(a) Turns for 240 V, 50 Hz [2].

$$N = \frac{240}{4.44 \times 50 \times 1.5 \times 2 \times 10^{-3}} = 360.4 \ \Rightarrow\ \mathbf{N = 360\ (or\ 361)}$$

Rounding up to 361 keeps $\hat{B} \le 1.5$ T. The answer sheet accepts both.

(b) L and X for N = 200 [3].

$$\mathcal{R} = \frac{0.2513}{1000 \times 4\pi \times 10^{-7} \times 2 \times 10^{-3}} = 100.0\ \text{kAt/Wb}$$

$$L = \frac{200^2}{100 \times 10^3} = \mathbf{0.400\ H}, \qquad X = 2\pi(50)(0.4) = \mathbf{125.7\ \Omega}$$

(c) Current at 100 V [1]. $I = V/X = 100/125.66 = \mathbf{0.796\ A}$. The course gives 0.7955 A because it used X rounded to 125.6 Ω. Unrounded, the answer is 0.7958 A.

2023 quiz Q4 (4 marks).

GIVEN: two poles with one 4 mm gap between them. The pole area is 20 cm², $N = 75$, $B = 1.1$ T, and the core is infinitely permeable.

(a) [2]. All the mmf is across the gap: $NI = H_G l_G = (B/\mu_0) l_G$.

$$I = \frac{B l_G}{\mu_0 N} = \frac{1.1 \times 0.004}{4\pi \times 10^{-7} \times 75} = \mathbf{46.69\ A}$$

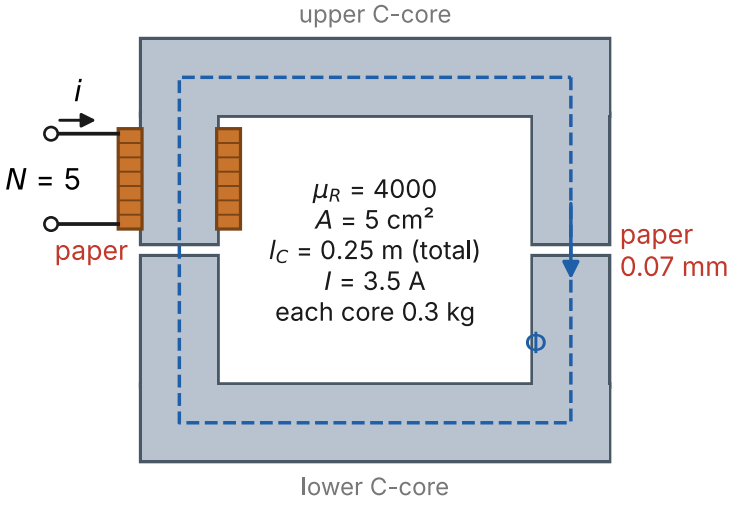
(b) [2]. There is one gap, so one pole-face pair:

$$F = \frac{1.1^2}{2\mu_0} \times 20 \times 10^{-4} = \mathbf{962.9\ N}$$

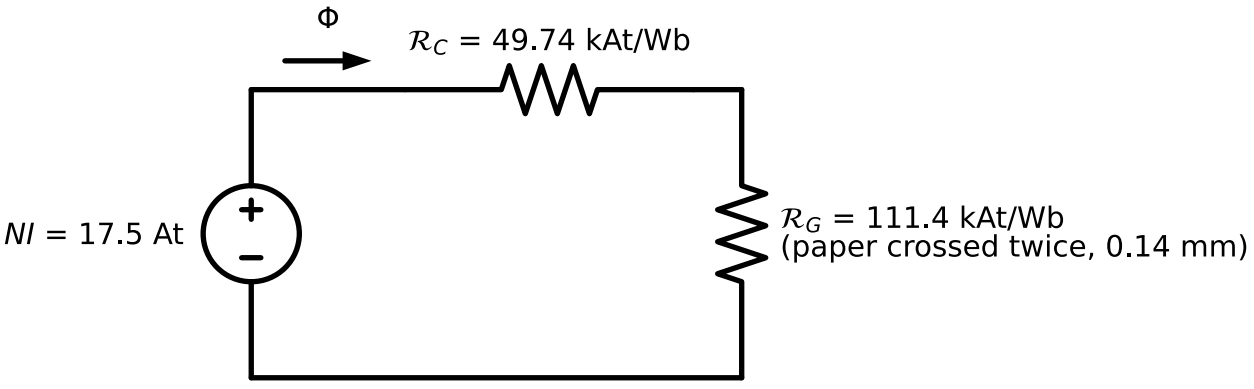
2024 test Q4 (7 marks).

GIVEN: two identical C-cores. $A = 5$ cm² $= 5 \times 10^{-4}$ m², $\mu_R = 4000$, the total iron path is $l_C = 0.25$ m, $N = 5$, $I = 3.5$ A (so $NI = 17.5$ At), and each core has $m = 0.3$ kg.

2024 test Q4: two C-cores with paper between them



2024 test Q4: two C-cores with a 5-turn coil, separated by 0.07 mm paper at both joints



Magnetic equivalent circuit for 2024 test Q4(b): iron and paper-gap reluctances in series

(a) No gap [2]. $NI = H_C l_C = B l_C/(\mu_R \mu_0)$, so

$$B = \frac{\mu_R \mu_0 NI}{l_C} = \frac{4000 \times 4\pi \times 10^{-7} \times 17.5}{0.25} = \mathbf{0.3519\ T}$$

(b) 0.07 mm paper [3]. The flux crosses the paper twice per loop, so $l_G = 0.14$ mm.

$$NI = B\left(\frac{l_C}{\mu_R \mu_0} + \frac{l_G}{\mu_0}\right) = B(49.74 + 111.4) \times 10^3$$

$$B = \frac{17.5}{161.1 \times 10^3} = \mathbf{0.1086\ T}$$

A sheet of paper has more than twice the reluctance of the whole iron path.

(c) Minimum B to hold the lower core [2]. The weight is $mg = 0.3 \times 9.81 = 2.943$ N. Both pole faces pull, so the force area is $2A = 10^{-3}$ m².

$$B = \sqrt{\frac{2\mu_0 mg}{2A}} = \sqrt{\frac{2 \times 4\pi \times 10^{-7} \times 2.943}{10^{-3}}} = \mathbf{0.0860\ T}$$

This is below 0.1086 T, so the magnet still holds with the paper in place.

2024 test Q5 (5 marks).

GIVEN: a 240 V/48 V, 50 Hz transformer with $A = 40$ cm² $= 4 \times 10^{-3}$ m² and $\hat{B}_{\max} = 1.6$ T. The path length and μ_R are not needed here.

(a) [2].

$$N_1 = \frac{240}{4.44 \times 50 \times 1.6 \times 4 \times 10^{-3}} = 168.9 \ \Rightarrow\ \mathbf{169\ turns}$$

This gives $\hat{B} = 1.599$ T, which is just under the limit.

(b) [2]. $A_w = \pi(0.25 \times 10^{-3})^2 = 1.963 \times 10^{-7}$ m².

$$R = \frac{l}{\sigma A_w} = \frac{40}{59 \times 10^6 \times 1.963 \times 10^{-7}} = \mathbf{3.453\ \Omega}$$

(c) [1]. $N_2 = 169 \times 48/240 = 33.8 \ \Rightarrow\ \mathbf{34\ turns}$.

2023 quiz Q7 (4 marks).

- **Process.** The back-emf equation links the primary voltage, frequency, turns and peak core flux. For a given core area and supply, choose the number of primary turns so that the peak flux density sits just below the saturation level of the core material.
- **Too few turns.** The peak flux density must be higher. The core saturates and the magnetising current becomes very large and distorted. The winding overheats.
- **Too many turns.** The core runs at a low flux density, so it is under-used. The longer wire has more resistance, which raises copper loss and lowers efficiency, and can also cause overheating.

(Exam instruction: words only, so no equations or sketches.)

PQ 3.10 (practice questions).

Hint: $\mathcal{R}_C = 0$; $l_G = \mu_0 N^2 A/L$; $I_{\max} = N\Phi_{\text{sat}}/L$.

a. Only the gap has reluctance, so $L = N^2/\mathcal{R}_G = \mu_0 N^2 A/l_G$:

$$l_G = \frac{\mu_0 N^2 A}{L} = \frac{4\pi \times 10^{-7} \times 50^2 \times 5 \times 10^{-4}}{10^{-3}} = \mathbf{1.571\ mm}$$

b. $\mathcal{R}_G = 10^{-3}/(4\pi \times 10^{-7} \times 5 \times 10^{-4}) = 1.592 \times 10^6$ At/Wb, so $N = \sqrt{L\mathcal{R}_G} = \sqrt{1592} = 39.89$, which rounds to **40 turns**.

c. $\Phi_{sat} = B_{sat}A = 0.3 \times 5 \times 10^{-4} = 1.5 \times 10^{-4}$ Wb. From $N\Phi = LI$: $I_{\max} = 50 \times 1.5 \times 10^{-4}/10^{-3} =$ **7.5 A**.

These match the official answers.

PQ 3.13 (practice questions).

Hint for (c): the loss factor at 400 Hz is $(8 + 64)/2 = 36$ at fixed B, so $\hat{B} = 1/\sqrt{36}$.

a. Both loss types go as $\hat{B}^2$: $3 \times 1.5^2 =$ **6.75 W**.

b. The frequency rises ×8. (i) $P_H \propto f$: $3 \times 8 =$ **24 W**. (ii) $P_E \propto f^2$: $3 \times 64 =$ **192 W**. (iii) 1.5 W of each: $1.5 \times 8 + 1.5 \times 64 = 12 + 96 =$ **108 W**.

c. Need $108\,\hat{B}^2 = 3$, so $\hat{B} = \sqrt{3/108} = 1/6 =$ **0.1667 T**.

These match the official answers.

PQ 3.14 (practice questions).

a. In the **airgap** (it has most of the reluctance, so most of the mmf and energy).

b. **No**, because air is not conductive (no eddy currents) and has no magnetic domains (no hysteresis).

c. **Eddy-current** loss and **hysteresis** loss.

d. **Eddy-current** loss, because it is proportional to frequency squared, so it dominates at high frequencies.

e. To **reduce eddy-current losses**: the insulated laminations break up the eddy-current paths.

f. It **increases rapidly and becomes distorted** (peaky), because μ_R collapses and much more current is needed for each extra bit of flux.

These match the official answers.

New questions

N1. Gapped rectangular core [5 marks].

Magnetic equivalent circuit for New question 1

$A = 1.2 \times 10^{-3}$ m².

a. $\mathcal{R}_C = \dfrac{0.6}{2500\mu_0 \times 1.2 \times 10^{-3}} = 159.2\ \text{kAt/Wb}, \qquad \mathcal{R}_G = \dfrac{0.8 \times 10^{-3}}{\mu_0 \times 1.2 \times 10^{-3}} = 530.5\ \text{kAt/Wb}$

$\mathcal{R}_T = 689.7$ kAt/Wb, so $L = 250^2/689.7 \times 10^3 = \mathbf{90.6\ mH}$. Without the gap it would be 393 mH.

b. $\Phi = 1.2 \times 1.2 \times 10^{-3} = 1.44$ mWb. Then $NI = \Phi\mathcal{R}_T = 993.1$ At, so $I = \mathbf{3.97\ A}$.

The gap takes $\mathcal{R}_G/\mathcal{R}_T = \mathbf{76.9\%}$ of the mmf (764 At of 993 At).

N2. Shell (E) core with parallel paths [7 marks].

Magnetic equivalent circuit for New question 2: centre-limb and gap reluctances in series with the two outer-path reluctances in parallel

a. The circuit is: source NI in series with $\mathcal{R}_{\text{centre}}$, $\mathcal{R}_G$, and the two outer reluctances in parallel.

$$\mathcal{R}_{\text{centre}} = \frac{0.10}{2000\mu_0(8 \times 10^{-4})} = 49.74\ \text{k}, \quad \mathcal{R}_G = \frac{10^{-3}}{\mu_0(8 \times 10^{-4})} = 994.7\ \text{k}, \quad \mathcal{R}_{\text{outer}} = \frac{0.30}{2000\mu_0(4 \times 10^{-4})} = 298.4\ \text{k}$$

All are in At/Wb. The two equal outer paths in parallel give $\mathcal{R}_{\parallel} = 298.4/2 = 149.2$ kAt/Wb.

$$\mathcal{R}_T = 49.74 + 994.7 + 149.2 = \mathbf{1194\ kAt/Wb}$$

b. $L = 200^2/1.194 \times 10^6 = \mathbf{33.5\ mH}$.

c. $\Phi = 1.2 \times 8 \times 10^{-4} = 0.96$ mWb. Then $NI = \Phi\mathcal{R}_T = 1146$ At, so $I = \mathbf{5.73\ A}$.

The flux splits equally: 0.48 mWb in each outer limb. So $B_{\text{outer}} = 0.48 \times 10^{-3} / 4 \times 10^{-4} = \mathbf{1.2\ T}$. Shell cores make the outer limbs half the centre area for exactly this reason: the flux density stays uniform.

N3. Back-emf design and loss scaling [8 marks].

a. $N = 230/(4.44 \times 50 \times 1.5 \times 1.5 \times 10^{-3}) = 460.5$, so round up to **461 turns**. Then $\hat{B} = 230/(4.44 \times 461 \times 50 \times 1.5 \times 10^{-3}) = \mathbf{1.498\ T}$.

b. $\hat{B} \propto 1/f$ at fixed V, so $\hat{B} = 1.498 \times 50/60 = \mathbf{1.249\ T}$.

The base split is $P_{H1} = 8$ W and $P_{E1} = 12$ W.

- $P_H = 8 \times (5/6)^2 \times (6/5) = \mathbf{6.67\ W}$ (this is the $\propto 1/f$ case).
- $P_E = 12 \times (5/6)^2 \times (6/5)^2 = \mathbf{12.0\ W}$ (unchanged).
- Total **18.67 W**.

c. At 40 Hz and 230 V, $\hat{B} = 1.498 \times 50/40 = 1.87$ T. That is past the knee, so the core saturates and the magnetising current becomes large and distorted, overheating the winding.

$V_{\max} = 4.44 \times 461 \times 40 \times 1.5 \times 1.5 \times 10^{-3} = \mathbf{184.2\ V}$. This is the constant-V/f rule: $230 \times 40/50 = 184$ V.

At $\hat{B} = 1.5$ T, 40 Hz: $P_H = 8 \times (1.5/1.498)^2 \times 0.8 = 6.42$ W and $P_E = 12 \times (1.5/1.498)^2 \times 0.64 = 7.70$ W, for a total of **14.1 W**. Treating $\hat{B}$ as unchanged gives 14.08 W, which is also acceptable.

N4. Electromagnetic lifter, two gaps [6 marks].

Magnetic equivalent circuit for New question 4: two gap reluctances in series, iron reluctance zero

a. There are two pole faces, so the force area is $2A = 1.2 \times 10^{-3}$ m².

$$B = \sqrt{\frac{2\mu_0 F}{2A}} = \sqrt{\frac{2\mu_0 (150)}{1.2 \times 10^{-3}}} = 0.5605\ \text{T}$$

The total gap is $l_G = 3$ mm, so

$$I = \frac{B l_G}{\mu_0 N} = \frac{0.5605 \times 0.003}{4\pi \times 10^{-7} \times 400} = \mathbf{3.35\ A}$$

b. $\mathcal{R}_G = 0.003/(\mu_0 \times 6 \times 10^{-4}) = 3.979$ MAt/Wb, so $L = 400^2/3.979 \times 10^6 = \mathbf{40.2\ mH}$.

$E = \frac{B^2}{2\mu_0} A l_G = \mathbf{0.225\ J}$. Check: $\frac{1}{2} L I^2 = 0.225$ J.

c. The gap falls by a factor of 3 at the same NI, so B rises ×3 to 1.68 T and F rises ×9 to about 1350 N. But 1.68 T is at or beyond the knee of the B–H curve. The iron saturates, so the real force will be lower than 1350 N and the "infinite permeability" assumption fails.

N5. Gapped inductor design, chained [8 marks].

a. $N\hat{B}A = L\hat{I}$, so $N = L\hat{I}/(\hat{B}A) = (5 \times 10^{-3} \times 10)/(1.2 \times 6 \times 10^{-4}) = 69.4$. Round up to **70 turns**.

b. $L = N^2 \mu_0 A / l_G$, so

$$l_G = \frac{\mu_0 N^2 A}{L} = \frac{4\pi \times 10^{-7} \times 70^2 \times 6 \times 10^{-4}}{5 \times 10^{-3}} = \mathbf{0.739\ mm}$$

c. $\hat{B} = \mu_0 N \hat{I} / l_G = \mathbf{1.190\ T}$, which is at or below 1.2 T.

$\frac{1}{2} L \hat{I}^2 = \frac{1}{2}(5 \times 10^{-3})(100) = 0.250$ J, and $\frac{\hat{B}^2}{2\mu_0} A l_G = 0.250$ J. They are equal because all the energy is in the gap when $\mu_R \to \infty$.

d. $E = 4.44 \times 70 \times 50 \times 1.190 \times 6 \times 10^{-4} = \mathbf{11.10\ V\ rms}$.

Check: $\omega L I_{\text{rms}} = 2\pi(50)(5 \times 10^{-3})(10/\sqrt{2}) = 11.11$ V. The tiny difference comes from rounding $\sqrt{2}\pi$ to 4.44.

6. Quick recall check

Question

1. At/Wb (A/Wb, or H^{-1}). mmf NI is the analogue of emf (voltage).
2. L roughly halves. B roughly halves (since $\Phi = NI/\mathcal{R}$). The force falls to about a quarter ($F \propto B^2$).
3. ×0.25. Both P_H and P_E go as $\hat{B}^2 \propto V^2$ (Jun24 Q2c(iii)).
4. $\hat{B} \times 5/6$. $P_H \propto \hat{B}^2 f$ gives $\times 5/6$. $P_E \propto \hat{B}^2 f^2$ gives ×1 (unchanged).
5. 120 V. $\hat{B} \propto V/f$, so the full 240 V at 25 Hz would double $\hat{B}$ and saturate the core.

Transformer equivalent circuit

Answers and worked solutions · Day 04 · Tue 27 Oct · pack M4

5. Practice

From your course files

Jun24 Q2b (past exam, 5 marks).

> This is worked in full as **Example 1** in section 3. Redo it without looking, then check.
>
> *Hint:* $Z_2 = n^2 Z_1$ and Z_3 adds the 15 Ω.
>
> i. $Z_1 =$ **0.3 Ω**, $Z_2 = 36 \times 0.3 =$ **10.8 Ω**, $Z_3 = 15 + 10.8 =$ **25.8 Ω**.
>
> ii. $I_1 = 12/25.8 =$ **0.4651 A**, $I_2 = 6I_1 =$ **2.791 A**, $V_1 = I_1 Z_2 =$ **5.023 V**, $V_2 = V_1/6 =$ **0.8372 V**.
>
> These match the official answers.

PQ 3.20 (practice questions).

> $n = 5$, $n^2 = 25$.
>
> a. $R'_L = n^2 R_L = 25 \times 10 =$ **250 Ω**.
>
> b. $I_1 = 240/(12 + 250) = 240/262 =$ **0.9160 A**.
>
> c. $V_1 = I_1 R'_L = 0.9160 \times 250 =$ **229.0 V**.
>
> d. $I_2 = nI_1 =$ **4.580 A** and $V_2 = V_1/n =$ **45.80 V**.
>
> e. $P_{OUT} = I_2^2 R_L = 4.580^2 \times 10 =$ **209.8 W**.
>
> f. $P_{12\Omega} = I_1^2 \times 12 =$ **10.07 W**.
>
> g. $\eta = 209.8/(240 \times 0.9160) = 209.8/219.8 =$ **95.42 %**.
>
> These match the official answers.

PQ 3.21 (practice questions).

> *Hint:* $5^2 \times 5 = 125\,\Omega$, then a divider with 12 Ω.
>
> $R'_L = 25 \times 5 = 125\ \Omega$. Divider: $V_1 = 240 \times \dfrac{125}{125 + 12} = 219.0$ V. Then $V_2 = V_1/5 =$ **43.8 V**.

> This matches the official answer.

Jun23 Q2b (past exam, 5 marks).

> *Hint:* SC $Q = 256.1$ VAr, OC $Q = 308.7$ VAr.
>
> **SC test (series R + jX).** $R = P/I^2 = 55/9.7^2 =$ **0.5845 Ω**. $|Z| = 27/9.7 = 2.784\ \Omega$, so $X = \sqrt{2.784^2 - 0.5845^2} =$ **2.721 Ω**. (Equivalently $Q = \sqrt{(27 \times 9.7)^2 - 55^2} = 256.1$ VAr and $X = Q/I^2$.)
>
> **OC test (parallel R_C and jX_M).** $R_C = V^2/P = 240^2/45 =$ **1280 Ω**.
> $S = 240 \times 1.3 = 312$ VA, so $Q = \sqrt{312^2 - 45^2} = 308.7$ VAr and
> $X_M = V^2/Q = 240^2/308.7 =$ **186.6 Ω**.
>
> All referred to the primary (the tests were measured there). These match the official answers.

Jun23 Q2c (past exam, 6 marks).

> i. $V_2' = V_1 - I_2'(R_1 + R_2' + jX_1 + jX_2')$.
>
> ii. $V_2' = V_1 \dfrac{R_L'}{R_L' + R_1 + R_2' + jX_1 + jX_2'}$. The shunt branch ($R_C$, jX_M) sits directly across V_1, so it doesn't enter the divider.
>
> iii. **Lagging.** The series reactance makes I_2' lag V_1, and V_2' is in phase with I_2' because the load is resistive.
>
> *Note:* the official sheet writes "V_1'" and "R_L". The primary voltage needs no prime, and the load should be the referred R_L' (the question defines it that way). This is a notation slip, not a different answer.

PQ 3.27 (practice questions).

> a. $n = 240/48 = 5$, $n^2 = 25$. $R = R_1 + n^2 R_2 = 0.6 + 25 \times 0.02 =$ **1.1 Ω**.
> $X = X_1 + n^2 X_2 = 2 + 25 \times 0.06 =$ **3.5 Ω**. The shunt branch is $R_C =$ **3200 Ω** in parallel with $X_M =$ **550 Ω**, placed across V_1.
>
> b. No load: only the shunt branch draws current. $I_R = 240/3200 = 0.075$ A and $I_M = 240/550 = 0.4364$ A are perpendicular, so $I_0 = \sqrt{0.075^2 + 0.4364^2} =$ **0.4427 A**. $P_0 = V^2/R_C = 240^2/3200 =$ **18.0 W**.
>
> c. Short circuit at full voltage: $|Z| = |1.1 + j3.5| = 3.669\ \Omega$, so $I = 240/3.669 =$ **65.42 A** and $P = I^2 R = 65.42^2 \times 1.1 =$ **4707 W** (a destructive overload, nearly 16

× rated current).

d. Rated input current $= 1000/240 =$ **4.167 A**. The supply voltage that drives it into the short circuit is $4.167 \times 3.669 =$ **15.29 V** (6.4 % of rated voltage).

These match the official answers.

PQ 3.28 (practice questions).

OC (from (b)): 240 V, 0.4427 A, 18.0 W. $R_C = V^2/P = 240^2/18.0 = 3200\ \Omega$. $S = 240 \times 0.4427 = 106.25$ VA, $Q = \sqrt{106.25^2 - 18^2} = 104.7$ VAr, so $X_M = V^2/Q = 550.1\ \Omega$.

SC (from (c)): 240 V, 65.42 A, 4707 W. $R = P/I^2 = 4707/65.42^2 = 1.100\ \Omega$. $|Z| = 240/65.42 = 3.669\ \Omega$, so $X = \sqrt{3.669^2 - 1.100^2} = 3.500\ \Omega$.

These recover the 3.27 values. The official answers are $R_C = 3300\ \Omega$, $X_M = 549.6\ \Omega$, $R = 1.100\ \Omega$, $X = 3.501\ \Omega$. The small differences in X_M and X come from rounding the 3.27 results. The 3300 Ω can't come from 18.0 W at 240 V ($240^2/18 = 3200$ exactly), so it is almost certainly a typo for 3200 Ω.

PQ 3.29 (practice questions, do (a)–(c)).

Hint: $V_1 = V_2' + I_2'(R + jX)$.

a. SC: $R = 55/9.5^2 =$ **0.6094 Ω**, $|Z| = 27/9.5 = 2.842\ \Omega$, $X = \sqrt{2.842^2 - 0.6094^2} =$ **2.776 Ω**. OC: $R_C = 240^2/45 =$ **1280 Ω**; $S = 288$ VA, $Q = \sqrt{288^2 - 45^2} = 284.5$ VAr, $X_M = 240^2/284.5 =$ **202.5 Ω**.

b. Rated output voltage referred to the primary: $V_2' =$ **240∠0° V**. $I_2' = 2400/240 = 10$ A at $\phi = -\cos^{-1} 0.87 = -29.54°$, so $I_2' =$ **10∠−29.54° A**.

c. $V_1 = 240 + 10\angle{-29.54°} \times (0.6094 + j2.776) = 259.9\angle 4.67°$ V, so $|V_1| =$ **259.9 V**.

d. is efficiency (pack M5). The official answers are $P_O = 2088$ W, $P_{CU} = 60.94$ W, $P_{FE} = 52.77$ W and $\eta = 94.84\%$.

These match the official answers.

2024 test Q6 (4 marks).

Hint: both tests are line values on a star connection, so convert to per-phase values first.

Per phase (SC): $V_1 = 710/\sqrt{3} = 409.9$ V, $I_1 = 390$ A, $P_1 = 140/3 = 46.67$ kW.

$R = P_1/I_1^2 = 46\,667/390^2 =$ **0.3068 Ω**. $|Z| = 409.9/390 = 1.051\ \Omega$, so $X = \sqrt{1.051^2 - 0.3068^2} =$ **1.005 Ω**.

I disagree with the answer sheet's $R = 0.3086\,\Omega$. $46\,667/390^2 = 0.3068$, so the sheet has transposed two digits (your Test Reference flags the same thing). Its $X = 1.005\ \Omega$ and its intermediate values ($V_1 = 409.9$ V, $P_1 = 46.67$ kW, $Q_1 = 152.9$ kVAr) agree with this working.

2024 test Q7 (4 marks).

Hint: generating Q means leading, so the angle is positive.

Referred to the primary, rated output voltage is the primary rating: $V_2' = 14\,000/\sqrt{3} =$ **8.083∠0° kV** (per phase, star assumed).

$|I_2'| = S/(3V_2') = 150 \times 10^6/(3 \times 8083) = 6186$ A. The load generates Q, so $Q = -35$ MVAr and $\sin(-\phi) = -35/150$, giving $\phi = +13.49°$. So $I_2' =$ **6.186∠+13.49° kA**.

These match the official answers.

2024 test Q8 (4 marks).

- **Ideal transformer.** The output voltage stays constant, because it is set only by the input voltage and the turns ratio and is not affected by the load current. The output current halves, by Ohm's law, because the load resistance has doubled.
- **Real transformer.** It has series winding resistance and leakage reactance. When the load current falls, the voltage drop across these series terms falls, so the output voltage rises slightly. The output current therefore falls to slightly more than half of its original value.

(Exam instruction: words only, so no equations or sketches.) These are the course's model points.

2023 quiz Q6 (9 marks).

a. $V_2' = 3300/\sqrt{3} =$ **1.905∠0° kV** per phase. $S = P/\text{pf} = 27/0.78 = 34.62$ kVA, so $|I_2'| = 34\,620/(3 \times 1905) = 6.056$ A at $-\cos^{-1} 0.78 = -38.74°$: $I_2' =$ **6.056∠−38.74° A**.

b. Circuit: R_C (18 kΩ) and jX_M (j2.3 kΩ) across V_1, then $4 + j35\ \Omega$ in series to V_2'. The shunt branch doesn't affect the KVL:

$$V_1 = V_2' + I_2'(R + jX) = 2100 + 7.7\angle -25° \times (4 + j35) = 2.254\angle 5.889°\ \text{kV per phase}$$

Line voltage: $\sqrt{3} \times 2.254 =$ **3.904 kV**.

I disagree with the sheet's 3.883 kV. $\sqrt{3} \times 2.254 = 3.904$, and the sheet's own phase value ($2.254 \angle 5.889^\circ$ kV) agrees with mine.

PQ 3.30 (practice questions, overflow: do (a), (b)).

a. $V_2' = 415/\sqrt{3} =$ **239.6∠0° V**. $I_2' = 3000/(3 \times 239.6) =$ **4.174∠0° A** (unity pf).

b. $V_1 = 239.6 + 4.174 \times (0.95 + j8.5) = 246.1 \angle 8.29^\circ$ V per phase, so the line voltage is $\sqrt{3} \times 246.1 =$ **426.3 V**. The official answer is 426.4 V ($V_1 = 246.2$ V); the difference is rounding.

(c)–(d) are pack M5. The official answers are $P_{CU} = 49.65$ W, $P_{FE} = 55.10$ W and $\eta = 96.62\%$.

PQ 3.31 (practice questions, overflow: do (a), (b)).

Hint: both tests are line values on a star connection, so per phase use $95/\sqrt{3}$ V and $12/3$ kW, then $415/\sqrt{3}$ V and $800/3$ W. Note that the OC test was **not** at rated voltage, so X_M is the low-flux value (see Example 3).

a. SC per phase: 54.85 V, 500 A, 4000 W. $R = 4000/500^2 =$ **0.01600 Ω**, $|Z| = 54.85/500 = 0.1097\ \Omega$, $X =$ **0.1085 Ω**. OC per phase: 239.6 V, 15 A, 266.7 W. $R_C = 239.6^2/266.7 =$ **215.3 Ω**; $S = 3594$ VA, $Q = 3584$ VAr, $X_M = 239.6^2/3584 =$ **16.02 Ω**.

b. $V_2' = 15\,000/\sqrt{3} =$ **8.660∠0° kV**. $|I_2'| = (50/0.8) \times 10^6/(3 \times 8660) = 2406$ A, so $I_2' =$ **2.406∠−36.87° kA**.
$V_1 = 8660 + 2406 \angle -36.87^\circ \times (0.016 + j0.1085) = 8.849$ kV per phase ($8.850 \angle 1.20^\circ$ kV unrounded), so the line voltage is **15.33 kV**.

c. is pack M5. The official answer is $\eta = 97.34\%$ ($P_{FE} = 1.091$ MW, $P_{CU} = 0.2779$ MW).

These match the official answers.

New questions

New Question 1 (easy) [6 marks].

a. $n = 8$. $Z_L' = 64(0.4 + j0.3) =$ $\mathbf{25.6 + j19.2\ \Omega}$ $(= 32 \angle 36.87^\circ\ \Omega)$.

b. $Z_{total} = 8 + 25.6 + j19.2 = 33.6 + j19.2 = 38.70 \angle 29.74^\circ\ \Omega$.

$I_1 = 240/38.70 \angle 29.74^\circ =$ $\mathbf{6.202 \angle -29.74^\circ\ A}$.

$V_1 = I_1 Z_L' = 6.202 \angle -29.74^\circ \times 32 \angle 36.87^\circ =$ $\mathbf{198.5 \angle 7.13^\circ\ V}$.

c. $I_2 = nI_1 = \mathbf{49.61\angle -29.74^\circ}$ **A**. $V_2 = V_1/n = \mathbf{24.81\angle 7.13^\circ}$ **V**.

$P = |I_2|^2 \times 0.4 = 49.61^2 \times 0.4 = \mathbf{984.6}$ **W**.

Check from the secondary side: refer the source resistance across, $8/64 = 0.125\,\Omega$, and use the source $240/8 = 30$ V. Then $I_2 = 30/(0.525 + j0.3) = 49.61\angle -29.74^\circ$ A. ✓

New Question 2 (medium) [5 marks].

$n = 415/110 = 3.773$, so $n^2 = 14.23$.

SC (already on the primary side): $S = 22 \times 12.05 = 265.1$ VA.
$Q = \sqrt{265.1^2 - 120^2} = 236.4$ VAr.

$R = 120/12.05^2 = \mathbf{0.8264\,\Omega}$. $X = 236.4/12.05^2 = \mathbf{1.628\,\Omega}$.

(12.05 A is the rated primary current, $5000/415$.)

OC (on the LV side): $S = 110 \times 1.8 = 198$ VA. $Q = \sqrt{198^2 - 55^2} = 190.2$ VAr.

$R_{C,LV} = 110^2/55 = 220\,\Omega$. $X_{M,LV} = 110^2/190.2 = 63.61\,\Omega$.

Refer to the primary: $R_C = 14.23 \times 220 = \mathbf{3131\,\Omega}$. $X_M = 14.23 \times 63.61 = \mathbf{905.5\,\Omega}$.

Answer circuit for New Question 2: 3131 Ω ‖ j905.5 Ω across the 415 V input, then 0.8264 Ω + j1.628 Ω

`CIRCUIT:` $3131\,\Omega \parallel j905.5\,\Omega$ across the 415 V input terminals, then $0.8264\,\Omega$ and $j1.628\,\Omega$ in series to V_2'.

(Check: 415 V across these shunt values draws 0.477 A on the HV side. That equals $1.8/n$. ✓)

New Question 3 (medium) [5 marks].

a. **OC, per phase:** $V_1 = 11000/\sqrt{3} = 6351$ V, $I_1 = 1.2$ A, $P_1 = 1900/3 = 633.3$ W.

$S_1 = 7621$ VA, $Q_1 = \sqrt{7621^2 - 633.3^2} = 7595$ VAr.

$R_C = 6351^2/633.3 = \mathbf{63.68\,k\Omega}$. $X_M = 6351^2/7595 = \mathbf{5.311\,k\Omega}$.

SC, per phase: $V_1 = 480/\sqrt{3} = 277.1$ V, $I_1 = 26.24$ A, $P_1 = 5200/3 = 1733$ W.

$S_1 = 7272$ VA, $Q_1 = 7062$ VAr.

$R = 1733/26.24^2 = \mathbf{2.517\,\Omega}$. $X = 7062/26.24^2 = \mathbf{10.26\,\Omega}$.

b. $I_{rated} = 500\text{k}/(\sqrt{3} \times 11\text{k}) = 26.24$ A. ✓ The SC voltage is $480/11000 = 4.4\%$ of rated, consistent with "about 10% or less".

New Question 4 (hard, chained) [8 marks].

a. $R'_L = n^2 R_L = 14.23 \times 2.42 = \mathbf{34.45\ \Omega}$.

b. $V'_2 = V_1 - I'_2(R + jX)$, with $R = R_1 + R'_2$ and $X = X_1 + X'_2$.

c. $$V'_2 = 415 \times \frac{34.45}{34.45 + 0.8264 + j1.628} = 415 \times \frac{34.45}{35.27 + j1.628}$$

$V'_2 = \mathbf{404.9 \angle -2.64^\circ\ V}$ (lagging V_1).

Actual: $V_2 = 404.9/3.773 = \mathbf{107.3\ V}$ (down from 110 V on no load).

d. $I'_2 = V'_2/R'_L = 11.75 \angle -2.64^\circ$ A.

$I_{sh} = 415/3131 + 415/(j905.5) = 0.1325 - j0.4583 = 0.4771 \angle -73.87^\circ$ A.

$I_1 = I'_2 + I_{sh} = \mathbf{11.92 \angle -4.82^\circ\ A}$.

e. $Z'_L = 34.45 \angle -36.87^\circ\ \Omega$. The divider gives $V'_2 = 418.3 \angle -3.02^\circ$ V. **The magnitude rises above 415 V, but V'_2 still lags.** The student is wrong. The phase of $V'_2/V_1 = Z'_L/(Z'_L + R + jX)$ is a lag whenever the load angle is less inductive than $\tan^{-1}(X/R) = 63^\circ$, which covers every resistive or capacitive load.

New Question 5 (hard, saturation) [5 marks].

a. **120 V:** $S = 6.0$ VA, $Q = \sqrt{6^2 - 2.2^2} = 5.582$ VAr. $R_C = 120^2/2.2 = \mathbf{6545\ \Omega}$. $X_M = 120^2/5.582 = \mathbf{2580\ \Omega}$.

240 V: $S = 100.8$ VA, $Q = 100.4$ VAr. $R_C = 240^2/8.6 = \mathbf{6698\ \Omega}$. $X_M = 240^2/100.4 = \mathbf{573.5\ \Omega}$.

b. Predicted: $I_M = 240/2580 = 0.093$ A and $I_C = 240/6545 = 0.037$ A, so $|I| = 0.100$ A. $P = 240^2/6545 = 8.8$ W.

Measured: 0.42 A and 8.6 W. **The loss prediction is good (within 2%), but the current is about 4 times too low.**

c. At 240 V the peak flux density is twice as high (back-emf equation), and the core is in saturation. The permeability falls, so L_M falls from 8.2 H to 1.8 H and the magnetising current rises sharply. Core loss still scales as V^2, so R_C stays roughly constant. Use the **rated-voltage (240 V) values**. This is the same effect as your Prac 2.

6. Quick recall check

Question

1. The applied voltage is small (about 10% of rated). The shunt currents are proportional to V, so they are negligible next to the short-circuit current, and the circuit reduces to $R + jX$ in series.
2. $n^2 Z_L = 100 \times 2 = 200\,\Omega$.
3. $X_M = V^2/Q$ (parallel). $X = Q/I^2$ (series). Both use $Q = \sqrt{(VI)^2 - P^2}$.
4. It lags. The series X makes I_2' lag V_1, and $V_2' = I_2' R_L'$ is in phase with I_2'.
5. $V_1 = 6600/\sqrt{3} = 3811$ V, $I_1 = 3$ A, $P_1 = 2$ kW.

Regulation, efficiency, 3-phase transformers

Answers and worked solutions · Day 05 · Wed 28 Oct · pack M5

5. Practice

From your course files

Jun24 Q2c (past exam, 8 marks).

> This is worked in full as **Example 2** in Section 3 (with its circuit). Do it without looking first.
>
> i. Series branch $R = 0.08 + 0.06 = 0.14\ \Omega$ and $jX = j0.28\ \Omega$; shunt branch $250\ \Omega \parallel j14\ \Omega$ at the input.
>
> ii. Star: $V_1 = 16\,000/\sqrt{3} = 9237.6$ V. $Z = 250 \parallel j14 = 13.98\angle 86.79^\circ\ \Omega$, so $|I_1| = 9237.6/13.98 =$ **660.9 A** and $P = 3V_1^2/R_C =$ **1.024 MW**.
>
> iii. **×0.5, ×0.25, ×0.25, ×1.** All agree with the course answers.

2023 quiz Q5 (quiz, 5 marks).

> Hint: P_{IN} is given, so subtract the losses. Assume star, so $V_1 = 11\,700/\sqrt{3} = 6755$ V per phase.
>
> - Copper loss: $P_{CU} = 3I_2'^2R = 3(6.8)^2(8.5) =$ **1.179 kW**.
> - Iron loss: $P_{FE} = 3V_1^2/R_C = 3(6755)^2/100\,000 =$ **1.369 kW**.
> - Efficiency: $\eta = \dfrac{P_{IN} - P_{CU} - P_{FE}}{P_{IN}} = \dfrac{115 - 1.179 - 1.369}{115} =$ **97.79 %**.
>
> Course answer: 1.179 kW, 1.369 kW, 97.79 % (the answer sheet notes the factor of three). X and X_M are not needed.

2023 quiz Q6 (quiz, 9 marks).

> **(a)** $V_2' = 3300/\sqrt{3}\angle 0^\circ =$ **1.905∠0° kV**. $S = P/\text{pf} = 27\,000/0.78 = 34.62$ kVA, so
>
> $$|I_2'| = \frac{S}{3V_2'} = \frac{34\,615}{3(1905.3)} = 6.056\ \text{A}, \qquad I_2' = \mathbf{6.056\angle -38.74^\circ\ A}$$
>
> ($\cos^{-1} 0.78 = 38.74^\circ$, negative because lagging.)
>
> **(b)** Circuit: $18\,000\ \Omega \parallel j2300\ \Omega$ across V_1, then $4\ \Omega$ and $j35\ \Omega$ in series carrying I_2' to V_2' (the same layout as Section 1, with these values).
>
> $$V_1 = V_2' + I_2'(R + jX) = 2100 + 7.7\angle -25^\circ(4 + j35) = 2241.8 + j231.2 = 2.254\angle 5.889^\circ\ \text{kV}$$

Line voltage $= \sqrt{3} \times 2253.7 =$ **3.904 kV**.

> **NOTE ON THE OFFICIAL ANSWER.** The course gives $V_1 = 2.254\angle 5.889^\circ$ kV (which this matches exactly) but a line voltage of "3.883 kV". That does not follow from its own phase value: $\sqrt{3} \times 2253.7 = 3.904$ kV. I think 3.904 kV is correct and 3.883 is a slip. (The text version of the paper reads "21000 V"; the paper itself says 2100∠0° V.)

2023 quiz Q7 (quiz, 4 marks).

Model answer (following the course marking points):

- The number of primary turns sets the peak flux in the core for a given voltage and frequency, through the back-emf equation. The turns are chosen so that the operating peak flux density is below the saturation value of the core material.
- **Too few turns:** a higher peak flux density is needed to support the voltage. The core saturates, the magnetising current becomes very large (and distorted), and the winding may overheat.
- **Too many turns:** the core runs at a low flux density, so the core is under-used. The extra turns mean more winding length and resistance, so the copper loss is higher, the efficiency is lower, and the winding may overheat.

PQ 3.29 (practice questions).

Single phase, so there is no factor of 3. Use V_1 for the iron loss.

(a) SC: $R = 55/9.5^2 = 0.6094\ \Omega$, $|Z| = 27/9.5 = 2.842\ \Omega$,
$X = \sqrt{2.842^2 - 0.6094^2} = 2.776\ \Omega$. OC: $R_C = 240^2/45 = 1280\ \Omega$;
$S = 240 \times 1.2 = 288$ VA, $Q = \sqrt{288^2 - 45^2} = 284.5$ VAr,
$X_M = 240^2/284.5 = 202.5\ \Omega$.

(b) $V_2' =$ **240∠0° V**. $|I_2'| = 2400/240 = 10$ A and $\cos^{-1} 0.87 = 29.54^\circ$ lagging, so $I_2' =$ **10∠−29.54° A**.

(c) $V_1 = 240 + 10\angle -29.54^\circ (0.6094 + j2.776) = 259.0 + j21.1 =$ **259.9∠4.67° V**.

(d) $P_O = 2400 \times 0.87 =$ **2088 W**. $P_{CU} = 10^2 \times 0.6094 =$ **60.94 W**.
$P_{FE} = 259.85^2/1280 =$ **52.75 W**. $\eta = 2088/(2088 + 60.94 + 52.75) =$ **94.84 %**.

Course answers: 0.6094, 2.776, 1280, 202.5 Ω; 240∠0° V, 10∠−29.54° A; 259.9 V; 2088 W, 60.94 W, 52.77 W, 94.84 %. The iron loss differs only in the last digit because the official answer rounds V_1 first.

PQ 3.30 (practice questions).

Unity pf, and 3000 VA is the three-phase total. Assume star.

(a) $V_2' = 415/\sqrt{3} =$ **239.6∠0° V**. $I_2' = 3000/(3 \times 239.6) =$ **4.174∠0° A**.

(b) $V_1 = 239.6 + 4.174(0.95 + j8.5) = 243.57 + j35.48 = 246.1\angle 8.29^\circ$ V per phase. Line: $\sqrt{3} \times 246.1 =$ **426.3 V**.

(c) $P_{CU} = 3(4.174)^2(0.95) =$ **49.64 W**. $P_{FE} = 3(246.1)^2/3300 =$ **55.08 W**.

(d) $\eta = 3000/(3000 + 49.64 + 55.08) =$ **96.62 %**.

Course answers: 239.6∠0° V, 4.174∠0° A; 426.4 V [V_1 = 246.2 V]; 49.65 W, 55.10 W; 96.62 %. The last-digit differences come from rounding V_1.

PQ 3.31 (practice questions).

Convert the test values to per phase first. The OC test is at 415 V, not rated voltage.

(a) SC per phase: $V = 95/\sqrt{3} = 54.85$ V, $I = 500$ A, $P = 4000$ W. $R = 4000/500^2 =$ **0.01600 Ω**; $|Z| = 54.85/500 = 0.1097\ \Omega$, so $X =$ **0.1085 Ω**. OC per phase: $V = 415/\sqrt{3} = 239.6$ V, $P = 266.7$ W, $S = 239.6 \times 15 = 3594$ VA, $Q = 3584$ VAr. $R_C = 239.6^2/266.7 =$ **215.3 Ω**, $X_M = 239.6^2/3584 =$ **16.02 Ω**.

(b) $V_2' = 15\,000/\sqrt{3} =$ **8.660∠0° kV**. $S = 50/0.8 = 62.5$ MVA, $|I_2'| = 62.5\text{M}/(3 \times 8660) = 2406$ A, so $I_2' =$ **2.406∠−36.87° kA**. $V_1 = 8660 + 2406\angle -36.87^\circ(0.016 + j0.1085) = 8847.7 + j185.8 = 8.850\angle 1.20^\circ$ kV per phase, so the line value is **15.33 kV**.

(c) $P_{CU} = 3(2406)^2(0.016) = 0.2778$ MW. $P_{FE} = 3(8849.6)^2/215.3 = 1.091$ MW (the core runs at rated voltage here, far above the 415 V test). $\eta = 50/(50 + 0.2778 + 1.091) =$ **97.34 %**.

Course answers: 0.01600, 0.1085, 215.3, 16.02 Ω; 8.660∠0° kV, 2.406∠−36.87° kA, 15.33 kV [V_{PH} = 8.849 kV]; 97.34 %. All agree.

Extension: at rated load $P_{CU,FL} = 3(3849)^2(0.016) = 711$ kW. With the 1.091 MW iron loss, max η is at $x = 1.24$.

Jun24 Q4c(ii) (past exam, 3 marks).

- **Copper (winding) loss:** resistive heating of the primary and secondary windings by the load current. It is modelled by the series winding resistance (the primary resistance plus the referred secondary resistance). Because it depends on the square of the current, it rises with the square of the output.
- **Iron (core) loss:** hysteresis and eddy-current losses in the magnetic core, caused by the alternating flux. It is modelled by the core-loss resistance in the shunt branch

across the supply. It depends on the flux density and frequency, which are set by the supply voltage and frequency, so it is essentially constant whatever the output.
- Hence at light load the iron loss dominates, at heavy load the copper loss dominates, and the efficiency is greatest where the two are equal.

New questions

N1. Loss scaling and maximum efficiency [5 marks].

a. $P_{OUT} = 10\text{k} \times 0.8 = 8000$ W. $\eta = 8000/(8000 + 180 + 120) =$ **96.39 %**.

b. $x = 0.5$, so $P_{CU} = 0.5^2 \times 180 = 45$ W, while P_{FE} stays at 120 W. $P_{OUT} = 4000$ W. $\eta = 4000/(4000 + 45 + 120) =$ **96.04 %**.

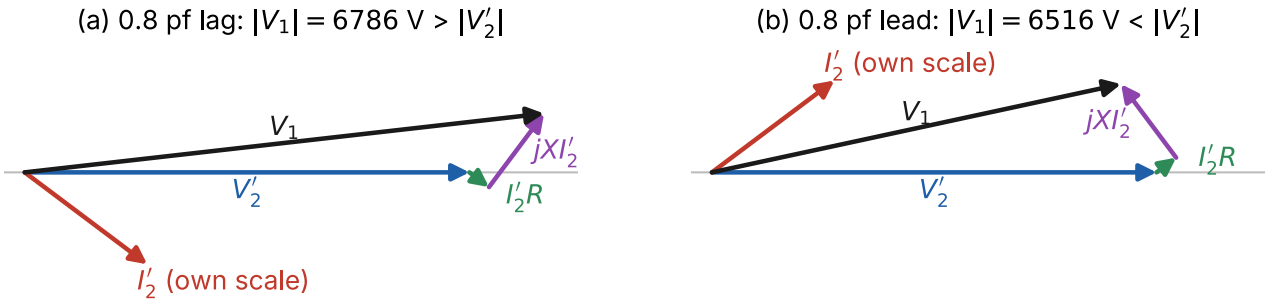
c. $x = \sqrt{120/180} = 0.8165$, so the load is **8.165 kVA**. At unity pf, $P_{OUT} = 8165$ W and the total loss is $2 \times 120 = 240$ W. $\eta_{max} = 8165/(8165 + 240) =$ **97.14 %**.

N2. Regulation with lagging and leading loads [6 marks].

$V_2' = 6600\angle 0°$ V and $|I_2'| = 50\,000/6600 = 7.576$ A.

a. $I_2' = 7.576\angle -36.87°$ A.
$V_1 = 6600 + 7.576\angle -36.87°(8 + j30) = 6784.8 + j145.5 =$ **6786 ∠1.23° V**.
Regulation $= (6786.4 - 6600)/6600 =$ **2.82 %**. Check:
$7.576(8 \times 0.8 + 30 \times 0.6) = 184.8$ V, which is 2.80 %.

b. $I_2' = 7.576\angle +36.87°$ A. $V_1 = 6600 + 7.576\angle 36.87°(8 + j30) =$ **6516 ∠1.92° V**.
Regulation $= (6515.8 - 6600)/6600 =$ **−1.28 %**. It is negative because the leading current rotates the jXI_2' drop so that it partly opposes V_2'. The output voltage is then higher than the no-load value (Prac 2 handout, Fig. 6).



New question 2, $V_2' = 6600\angle 0°$ V. Series drops drawn 6 times larger than scale.

Phasor diagrams for New question 2: (a) lagging load, V_1 larger than V_2′; (b) leading load, V_1 smaller than V_2′

c. Zero regulation needs $R\cos\phi = X\sin\phi$, so $\tan\phi = R/X = 0.2667$. That gives $\phi = 14.9^\circ$, a pf of **0.966 leading**. The exact phasor solution gives 0.961 leading.

N3. Delta-connected primary on open circuit [6 marks].

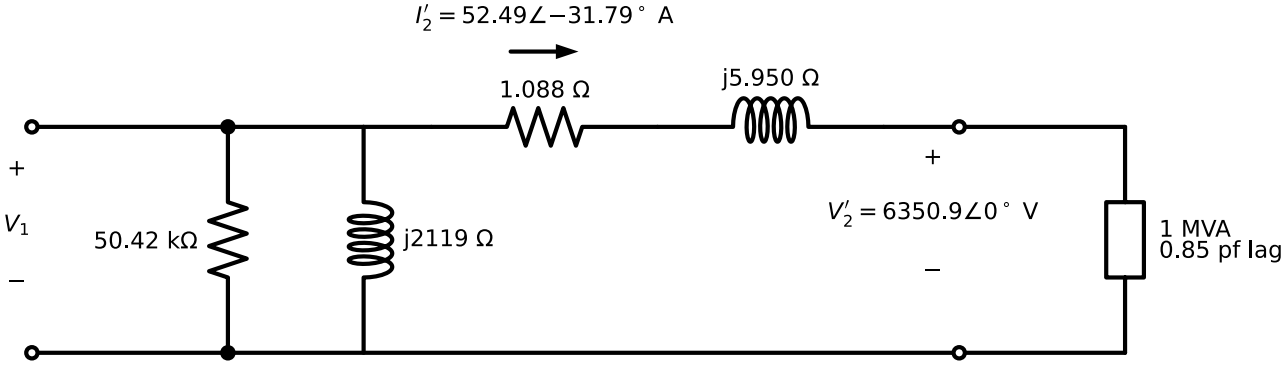
a. For delta, $V_1 = V_L = 11\,000$ V. $Z = 200\text{k} \parallel j30\text{k} = 29.67\angle 81.47^\circ$ kΩ, so $I_1 = 11\,000/Z =$ **0.371 ∠−81.5° A** per phase. Line current $I_L = \sqrt{3} \times 0.3708 =$ **0.642 A**. $P = 3V_1^2/R_C = 3(11\,000)^2/200\,000 =$ **1.815 kW**. $Q = 3V_1^2/X_M = 12.1$ kVAr. pf $= \cos 81.47^\circ =$ **0.148 lagging**.

b. Rated line current $= 500\text{k}/(\sqrt{3} \times 11\text{k}) = 26.24$ A. So $0.642/26.24 =$ **2.4 %**. This is typical of a large transformer (the Prac 2 handout quotes 2 to 4 %).

c. The model gives ×0.5, ×0.25, ×0.25, ×1. In practice the core runs further below saturation at half voltage, so X_M is larger and the current drops by **more** than half. The pf would also change. The model assumes constant R_C and X_M.

N4. Full chained analysis from test data [10 marks].

a. OC per phase: $V_1 = 11\,000/\sqrt{3} = 6350.9$ V and $P = 800$ W, so $R_C = 6350.9^2/800 =$ **50.42 kΩ**. $S = 6350.9 \times 3 = 19\,053$ VA and $Q = \sqrt{S^2 - P^2} = 19\,036$ VAr, so $X_M = V_1^2/Q =$ **2119 Ω**. SC per phase: $V_1 = 317.5$ V and $P = 3000$ W. $R = 3000/52.5^2 =$ **1.088 Ω**. $|Z| = 317.5/52.5 = 6.048\ \Omega$, so $X = \sqrt{6.048^2 - 1.088^2} =$ **5.950 Ω**.



Per-phase circuit for New question 4 with the part (a) parameters and the 1 MVA, 0.85 pf lagging load

b. $V_2' = 6350.9\angle 0^\circ$ V. Rated current $= 10^6/(\sqrt{3} \times 11\,000) = 52.49$ A, so $I_2' =$ **52.49 ∠−31.79° A**. $V_1 = 6350.9 + 52.49\angle -31.79^\circ(1.088 + j5.950) = 6568\angle 2.05^\circ$ V. Line voltage $= \sqrt{3} \times 6568.1 =$ **11.38 kV**. Regulation $= (6568.1 - 6350.9)/6350.9 =$ **3.42 %**.

c. $P_{OUT} = 0.85$ MW. $P_{CU} = 3(52.49)^2(1.088) = 8.995$ kW. $P_{FE} = 3(6568.1)^2/50\,417 = 2.567$ kW, using V_1. $\eta = 850/(850 + 8.995 + 2.567) =$ **98.66 %**.

d. The SC test was done at rated current, so $P_{CU,FL} = 9.0$ kW. $P_{FE} = 2.4$ kW at rated voltage. $x = \sqrt{2.4/9.0} = 0.516$, so the load is **≈ 517 kVA** (27.1 A). $\eta_{max} = 0.516 \times 850/(0.516 \times 850 + 2 \times 2.4) =$ **98.92 %**. Here the peak is well below rated load, which is the normal design for a power transformer.

N5. Short answer: no-load current and the Prac 2 evidence [4 marks].

a. The core has finite permeability, so a magnetising current is needed to supply the mmf that sets up the core flux. In the circuit this is the current through the magnetising reactance, and it lags the voltage by nearly 90°. The changing flux also produces hysteresis and eddy-current losses in the core. These need a real-power component of current, which is modelled by the core-loss resistance. Together these make up the no-load current, which is mostly reactive.

b. The peak core flux density is proportional to the applied voltage. Near rated voltage the core approaches saturation, and its permeability falls sharply. A much larger magnetising current is then needed for each extra increment of flux, so the magnetising reactance falls (from about 3980 Ω to about 366 Ω in Prac 2). The current rises far faster than the voltage and becomes distorted. The core-loss resistance changed much less, because iron loss depends on flux density and frequency rather than on permeability.

6. Quick recall check

Question

1. Copper loss ×0.25 (it goes as I^2). Iron loss ×1 (it stays constant at fixed V and f).
2. Iron loss uses V_1 ($P_{FE} = 3V_1^2/R_C$). Copper loss uses I_2' ($P_{CU} = 3I_2'^2 R$).
3. $P_{CU} = P_{FE}$, which occurs at $x = \sqrt{P_{FE}/P_{CU,FL}}$.
4. ×0.5, ×0.25, ×0.25, ×1.
5. $V_2' = 33\text{k}/\sqrt{3} = 19.05\angle 0°$ kV. Supply line voltage $= \sqrt{3}|V_1|$.

Machine basics, DC machines, rotating fields

Answers and worked solutions · Day 07 · Fri 30 Oct · pack M6

5. Practice

From your course files

Jun24 Q3a (past exam, 6 marks).

The official answer is only “synchronous, DC, induction; sketch curves from Sections 4.1 and 7.9; halve V and f / halve V / halve V and f”. (On the paper the three plots are stacked top to bottom; above they are shown left to right in the same order.)

(i) [3 marks]

1. **Synchronous machine:** a vertical line at n_S. At half speed it is a vertical line at $n_S/2$.
2. **DC machine:** a line sloping down from n_{NL} at τ = 0. At half speed the new line starts at $n_{NL}/2$ and is **parallel** to the original. The speed drop at any torque is unchanged, so the curve slides left.
3. **Induction machine:** a steep line sloping down from n_S, shown in its linear region near n_S. At half speed it starts at $n_S/2$ and is **parallel** to the original. The slip speed at a given torque is unchanged.

The figure in Section 1.3 shows all three, with the halved versions dashed.

(ii) [3 marks]

- **Synchronous:** halve the supply frequency f, which halves $n_S = 60f/p$. **Also halve V**, to keep V/f and hence the peak flux constant. This avoids core saturation and large currents.
- **DC: halve the terminal voltage V.** The no-load speed V/k halves.
- **Induction: halve both f and V** (constant V/f), for the same reason as the synchronous machine.

In practice the AC machines need a variable-voltage, variable-frequency inverter. The DC machine needs a variable DC supply, such as a PWM chopper like the cordless-drill example in Lecture 4.1.6.

PQ 4.3 (practice questions).

a. The rating is the output: **500 W**.

b. $\omega_R = 10\,000 \times 2\pi/60 = 1047.2$ rad/s, so $\tau_R = 500/1047.2 =$ **0.4775 Nm**.

c. The speed drops 2000 rpm at rated torque, so only 1000 rpm at half torque: $n = 12\,000 - 1000 =$ **11 000 rpm**.

d. $P = \tau\omega = (0.4775/2)(11\,000 \times 2\pi/60) =$ **275 W**. The torque is half, but the speed is higher than rated speed, so the power is more than half of rated power.

Course answers: 500 W, 0.4775 Nm, 11 000 rpm, 275 W. All agree.

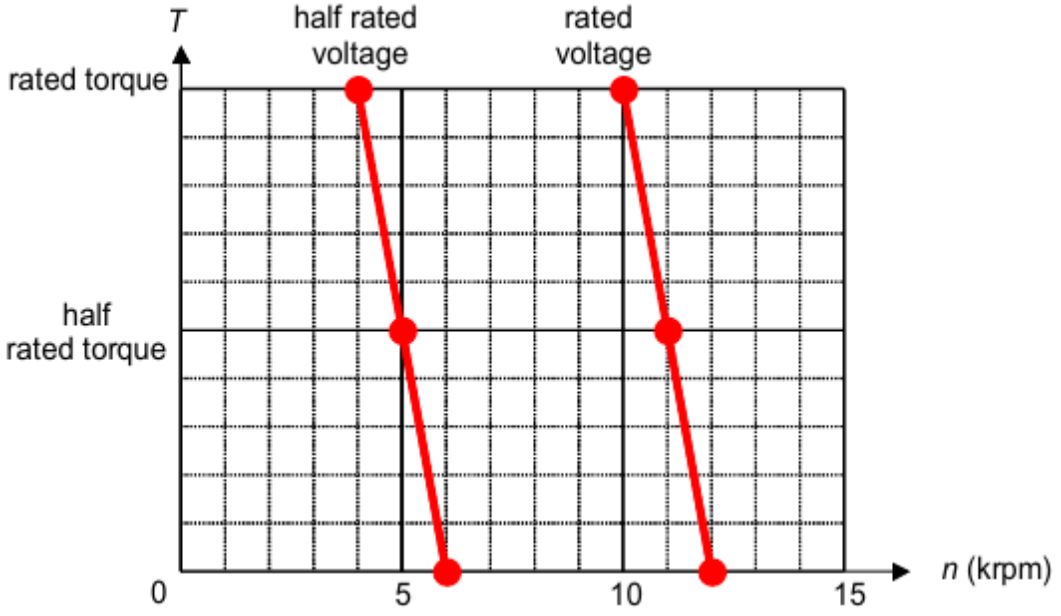
PQ 4.4 (practice questions).

a. The no-load speed is proportional to the supply voltage: $12\,000/2 =$ **6000 rpm**.

b. The 2000 rpm drop at full-load torque does not depend on the voltage: $6000 - 2000 =$ **4000 rpm**.

c. Half torque gives half the drop: $6000 - 1000 =$ **5000 rpm**.

d. Two parallel straight lines: rated voltage from (12 krpm, 0) through (11 krpm, half) to (10 krpm, rated); half voltage from (6 krpm, 0) through (5 krpm, half) to (4 krpm, rated).



PQ 4.4 official answer sketch (from the practice questions): the rated- and half-rated-voltage lines with the three operating points on each

PQ 4.7 (practice questions).

a. $n_S \propto f$, so $f = 60 \times 900/1200 =$ **45 Hz**. Keep V/f constant: $V = 110 \times 45/60 =$ **82.5 V**.

b. The slip speed at a given torque stays the same: 100 rpm at rated torque, 50 rpm at half torque.

c. $900 - 100 =$ **800 rpm**. (ii) $900 - 50 =$ **850 rpm**.

Course answers: 82.5 V, 45 Hz; 800 rpm, 850 rpm.

PQ 4.11 (practice questions).

> $p = 60f/n_S = 60 \times 50/630 = 4.762$ pole-pairs. The closest integer is 5 pole-pairs, so **10 poles**, giving $n_S = 60 \times 50/5 =$ **600 rpm**. (8 poles gives 750 rpm and 12 poles gives 500 rpm, both further away.) Here you round to the nearest, because the question asks for the closest speed, not a speed above a rated speed.

PQ 4.13 (practice questions).

> a. $n_S = 120 \times 50/8 =$ **750 rpm**.
> b. A synchronous machine runs at constant speed: **750 rpm**.
> c. $\tau_R = 11\,000/(750 \times 2\pi/60) = 11\,000/78.54 =$ **140.1 Nm**.
> d. Still **750 rpm**.
> e. The speed is constant, so the power is proportional to the torque: **5.5 kW**.

PQ 5.8 (practice questions).

> $k = 0.24$ V/(rad/s), $R = 0.5\ \Omega$, $V = 12$ V.
>
> a. No load: τ = 0, so I = 0 and E = V. $\omega_{NL} = 12/0.24 = 50$ rad/s, i.e. **477.5 rpm**.
>
> b. $\eta = n/n_{NL} = 400/477.5 =$ **83.77 %**.
>
> c. At stall ω = 0: (i) $E =$ **0 V**; (ii) $I = V/R =$ **24 A**; (iii) $\tau = kI =$ **5.760 Nm**; (iv) $P_{IN} = VI =$ **288 W**; (v) $P_{OUT} =$ **0 W** (n = 0); (vi) **0 %**.
>
> d. $I = 0.2/0.24 = 0.8333$ A, $E = 12 - 0.8333 \times 0.5 = 11.58$ V, $\omega = 11.58/0.24 = 48.26$ rad/s, so **460.9 rpm**.
>
> For reference, PQ 5.7: $E = 0.24 \times 41.89 = 10.05$ V, $I = (12 - 10.05)/0.5 = 3.894$ A (official 3.9 A, rounded), $\tau = 0.936$ Nm, $P_{OUT} \approx 39.2$ W, $P_{IN} = 46.8$ W, η = 83.78 % / 83.75 % (the official answers differ only by that rounding of I).

PQ 5.10 (practice questions).

> a. $k = \tau/I = 15/35 = 0.4286$ Nm/A (= V/(rad/s)). At no load $E = V$, so $\omega_{NL} = 24/0.4286 = 56$ rad/s, i.e. **534.8 rpm**.
>
> b. At stall E = 0, so $R = V/I = 24/35 =$ **0.6857 Ω**.
>
> c. At 400 rpm: $E = 0.4286 \times (2\pi \times 400/60) = 17.95$ V, $I = (24 - 17.95)/0.6857 = 8.820$ A, $P = EI =$ **158.3 W**.

> **NOTE:** the PQ answer (158.4 W) and the WS5 handwritten answer (158.1 W) both differ only through intermediate rounding. Any of these will get the marks.

PQ 5.19 (practice questions).

a. $k = E/\omega = 10.5/(150 \times 2\pi/60) = 10.5/15.71 =$ **0.6684 V/(rad/s)**.

b. $E = 14.6$ V: $\omega = 14.6/0.6684 = 21.84$ rad/s = **208.6 rpm**. With $E = V$ the current is **0 A**.

c. Generator: $E = V + IR = 14.6 + 10 \times 0.4 = 18.6$ V, so $\omega = 18.6/0.6684 = 27.83$ rad/s = **265.7 rpm**.

d. For a DC generator $\eta = V/E = 14.6/18.6 =$ **78.49 %**.

e. At low wind speed $E < 14.6$ V, so without the diode current would flow from the battery into the machine. That discharges the battery and makes the machine run as a motor, turning the wind turbine into a fan. The diode blocks this reverse current.

PQ 6.4 (practice questions).

Read the order in which the phases reach their positive peaks.

a. V_A, then V_C, then V_B: **acb** (or cba or bac).
b. V_A, then V_B, then V_C: **abc** (or bca or cab).

PQ 6.5 (practice questions).

Swap any two of the three supply leads. This reverses the phase sequence, so the rotating field (and the rotor) turns the other way. Relabelling the phases cyclically does not change the sequence.

PQ 6.7 (practice questions).

Poles come in N–S pairs, so the number must be an even integer: (a) **yes** (b) **no** (odd) (c) **no** (not an integer) (d) **yes**.

PQ 6.9 (practice questions).

a. The grid is 50 Hz and the fewest possible poles is 2 ($p = 1$): $n_S = 60 \times 50/1 =$ **3000 rpm, two poles**.

b. $p = 60f/n_S = 60 \times 50/100 = 30$ pole-pairs, so **60 poles**.

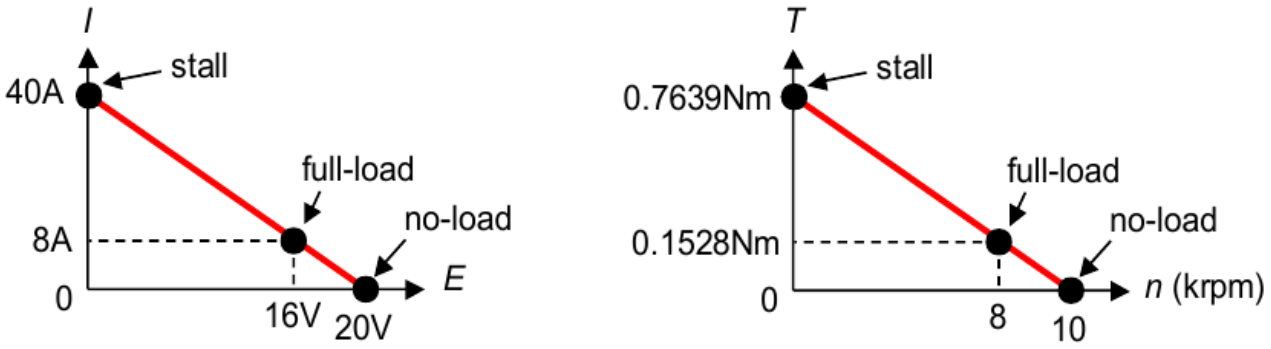
PQ 5.11 (practice questions, extra).

$k = 20/(2\pi \times 10\,000/60) = 0.01910$ V/(rad/s).

a. Zero speed: E = 0, so $I = V/R = 20/0.5 =$ **40 A** and $\tau = kI =$ **0.7639 Nm**.

b. At 8000 rpm $E = 20 \times 8000/10\,000 = 16$ V, so $I = (20 - 16)/0.5 =$ **8 A** and $\tau = kI =$ **0.1528 Nm**.

c. Both characteristics are straight lines: I falls from 40 A at E = 0 (stall) to 0 at E = 20 V (no load), passing 8 A at 16 V (full load); τ falls from 0.7639 Nm at n = 0 to 0 at 10 krpm, passing 0.1528 Nm at 8 krpm.

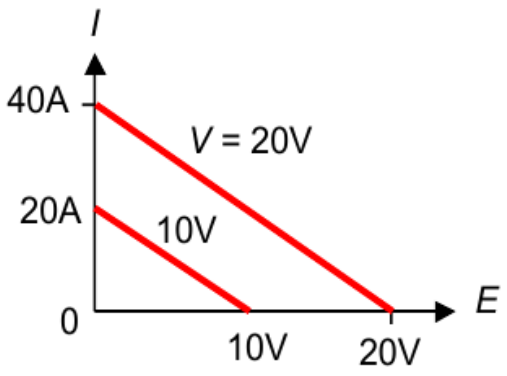


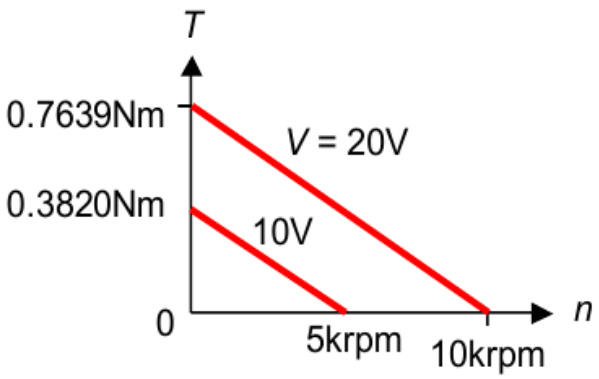
PQ 5.11 official answer (from the practice questions): the I–E and torque–speed lines with stall, full-load and no-load points

d. Half rated torque is half the full-load current and half the speed drop: **4 A, 9000 rpm**.

PQ 5.12 (practice questions, extra).

The rated-voltage lines come from PQ 5.11 (stall 40 A and 0.7639 Nm; no load 20 V and 10 krpm). At half voltage (10 V) the no-load induced voltage and speed halve (10 V, 5 krpm), and the stall current V/R halves to **20 A**, so the stall torque halves to **0.3820 Nm**. The half-voltage lines are parallel to the rated ones.





PQ 5.12 official answer (from the practice questions): rated (20 V) and half (10 V) voltage lines on the I–E and torque–speed axes

New questions

N1. [4 marks]

a. $p = 60 \times 60/1140 = 3.158$, which rounds down to p = 3. So $n_S = 3600/3 = \mathbf{1200\ rpm}$, with **6 poles**.

b. $\eta_{max} = n_R/n_S = 1140/1200 = \mathbf{95.0\ \%}$.

c. The slip speed at rated torque is 60 rpm. $n = 1200 - 0.6 \times 60 = \mathbf{1164\ rpm}$.

N2. [6 marks]

a. $\omega_{NL} = 2\pi(3000)/60 = 314.16$ rad/s. At no-load, E = V, so $k = 48/314.16 = \mathbf{0.1528\ V/(rad/s)}$.

b. $\tau_R = kI_R = 0.1528 \times 12 = \mathbf{1.833\ Nm}$. $E_R = 48 - 12(0.25) = 45$ V, so $\omega_R = 45/0.1528 = 294.5$ rad/s, which is **2812.5 rpm**. $\eta = E/V = 45/48 = \mathbf{93.75\ \%}$. Check: $P_{out} = EI = 540$ W and $P_{in} = 576$ W.

c. $I_S = V/R = 48/0.25 = \mathbf{192\ A}$. $\tau_S = kI_S = \mathbf{29.34\ Nm}$, which is **16 times** τ_R (= 192/12).

d. The maximum is at $n_{NL}/2 = \mathbf{1500\ rpm}$, where $I = I_S/2 = 96$ A and E = 24 V. So $P_{max} = 24 \times 96 = \mathbf{2304\ W}$. That is 8 times the rated current, at only 50 % efficiency, so the motor would overheat.

N3. [6 marks] (chains from N2)

a. $n_{NL} \propto V$: **1500 rpm**. Rated torque means I = 12 A still, so $E = 24 - 12(0.25) = 21$ V, giving $\omega = 21/0.1528 = 137.4$ rad/s, which is **1312.5 rpm**. (Check: the drop is

187.5 rpm, the same as at 48 V.) $\eta = E/V = 21/24 = \mathbf{87.5\,\%}$. The same I^2R loss is now a bigger fraction of a smaller power.

b. $E = k\omega = 0.1528 \times 2\pi(1000)/60 = 16.0$ V. So $V = E + IR = 16 + 3 = \mathbf{19.0\ V}$.

c. I = 6 A, so $E = 24 - 1.5 = 22.5$ V, which gives **1406 rpm**. (Equivalently, $1500 - 0.5 \times 187.5 = 1406.25$ rpm.)

N4. [6 marks]

a. I = 0 when E = V: $\omega = 24/0.5 = 48$ rad/s, which is **458.4 rpm**.

b. Generating, so $E = V + IR = 24 + 5(0.8) = 28$ V. Then $\omega = 56$ rad/s, which is **534.8 rpm**. $\tau = kI = \mathbf{2.5\ Nm}$ (input torque). $P_M = EI = \mathbf{140\ W}$. $P_E = VI = \mathbf{120\ W}$. $\eta = V/E = \mathbf{85.71\,\%}$.

c. $E = 0.5 \times 2\pi(400)/60 = 20.94$ V < 24 V. The current flows **from the battery into the machine**: $I = (24 - 20.94)/0.8 = \mathbf{3.820\ A}$. The torque is $kI = \mathbf{1.910\ Nm}$ of *motoring* torque. The battery is being discharged (91.7 W) to spin the turbine as a fan. A series diode prevents this (Lecture 5.3.2, PQ 5.19e).

N5. [5 marks]

a. The slip speed at rated torque is $1500 - 1440 = 60$ rpm, and it stays constant. So the required $n_S = 900 + 60 = 960$ rpm. $f = 960 \times p/60 = 960 \times 2/60 = \mathbf{32.0\ Hz}$, and $V = 415 \times 32/50 = \mathbf{265.6\ V}$.

b. $n = 960 - 30 = \mathbf{930\ rpm}$.

c. A synchronous machine has no slip, so $n_S = 900$ rpm. $f = 900 \times 3/60 = \mathbf{45\ Hz}$, and $V = 415 \times 45/50 = \mathbf{373.5\ V}$.

N6. [4 marks] (words only)

a. **Still clockwise.** The terminal sequence A, B, C now sees b, c, a, which is a cyclic shift of abc and so the same phase sequence. Only swapping *two* leads (giving acb) reverses the field and the rotor.

b. $p = 60f/n_S = 60 \times 50/100 = 30$ pole-pairs, so **60 poles**. At 60 Hz, p = 36, so **72 poles**.

c. **2.** One electrical cycle moves the field by one pole-pair, which is half a revolution in a 4-pole machine. The field turns at $n_S = 1500$ rpm while the current cycles at 3000 cycles per minute.

6. Quick recall check

Question

1. $n_S = 60 \times 60/4 = 900$ rpm.
2. $n_{NL} = 1500$ rpm. $n_R = 1200$ rpm (the drop of 300 rpm is unchanged).
3. E = 0 and $I_S = V/R$, which is very large. $\tau_S = kI_S$.
4. Peak flux is proportional to V/f (from $E = 4.44Nf\hat{\Phi}$). If f falls with V fixed, the flux rises, the core saturates and the current becomes excessive.
5. p = 2 (60 × 50/1470 = 2.04), so $n_S = 1500$ rpm and $\eta = 1470/1500 = 98\,\%$.

Induction machine fundamentals

Answers and worked solutions · Day 10 · Mon 2 Nov · pack M7

5. Practice

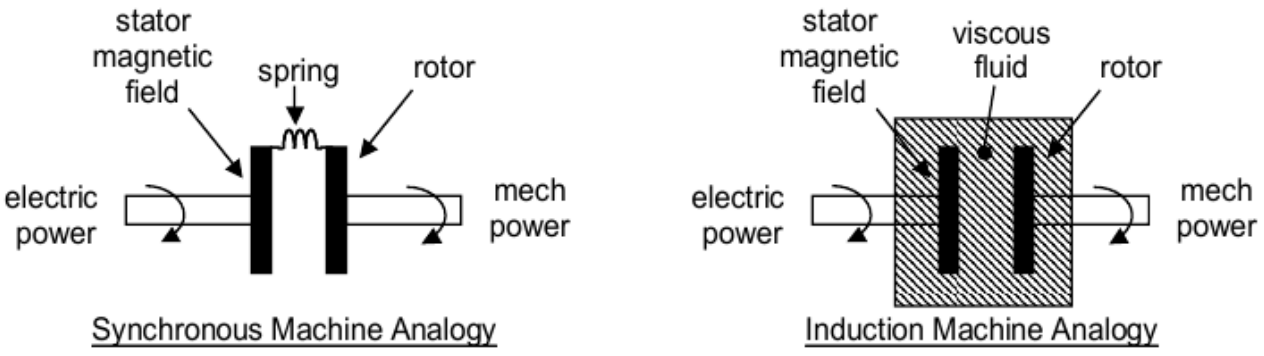
From your course files

PQ 7.1 (practice questions).

a. A **short-circuited three-phase winding**.
b. Any three of: low cost, self-starting, high efficiency, reliable, available in a wide range of sizes.
c. **Squirrel-cage** rotor. The bars are normally **aluminium**.
d. The **slip speed** $n_S - n$.
e. Generator when the rotor speed is **greater** than n_S. Motor when it is **less**.
f. A **transformer** (with a short-circuited secondary).
g. A **resistor**: $R_2'(1-s)/s$.
h. **Starting torque** is the torque at zero speed ($s = 1$). **Pull-out torque** is the maximum torque the motor can produce at any speed.

PQ 7.2 (practice questions).

- **Synchronous machine:** join the disks with a **spring**. Both turn at exactly n_S. The twist of the spring (the rotor angle) sets the torque.
- **Induction machine:** put a **viscous fluid** between the disks. The rotor disk lags the field disk. The torque is proportional to the speed difference (slip speed). Power flows from the faster disk to the slower one.



PQ 7.2 official answer (from the practice questions): a spring between the disks for the synchronous machine, a viscous fluid around the disks for the induction machine

PQ 7.3 (practice questions).

$n_S = 120f/\text{poles}$:

a. $120(50)/4 =$ **1500 rpm**. (b) $120(60)/8 =$ **900 rpm**. (c) $120(50)/2 =$ **3000 rpm**. (d) $120(800)/4 =$ **24 000 rpm**. (e) $120(50)/24 =$ **250 rpm**.

PQ 7.4 (practice questions).

Try $f = 50$ and 60 Hz in poles $= 120f/n_S$ and keep the answer that is an even integer:

a. 1500 rpm: 50 Hz gives 4 (60 Hz gives 4.8). **4 poles, 50 Hz.**
b. 1800 rpm: 60 Hz gives **4 poles, 60 Hz**.
c. 750 rpm: 50 Hz gives **8 poles, 50 Hz** (60 Hz gives 9.6).
d. 1200 rpm: 60 Hz gives **6 poles, 60 Hz** (50 Hz gives 5, which is odd).
e. 360 rpm: 60 Hz gives **20 poles, 60 Hz** (50 Hz gives 16.67).
f. 150 rpm: **40 poles at 50 Hz** or **48 poles at 60 Hz**. Both are valid.

PQ 7.5 (practice questions).

Method: $p = \lfloor 60f/n_R \rfloor$, $n_S = 60f/p$, slip speed $= n_S - n_R$ (and, for PQ 7.8, $s_R = (n_S - n_R)/n_S$, $f_2 = s_R f$). The voltage is irrelevant.

Motor	$60f/n_R$	p	Poles	n_S (rpm)	Slip speed (rpm)	s_R (PQ 7.8)	f_2 (PQ 7.8)
(a) 415 V, 50 Hz, 1440 rpm	2.083	2	**4**	**1500**	**60**	**0.04000**	**2.000 Hz**
(b) 3.3 kV, 60 Hz, 3520 rpm	1.023	1	**2**	**3600**	**80**	**0.02222**	**1.333 Hz**
(c) 135 V, 50 Hz, 680 rpm	4.412	4	**8**	**750**	**70**	**0.09333**	**4.667 Hz**
(d) 230 V, 50 Hz, 950 rpm	3.158	3	**6**	**1000**	**50**	**0.05000**	**2.500 Hz**

All values match the PQ answers.

PQ 7.8 (practice questions).

These are the last two columns of the PQ 7.5 table: (a) 0.04, 2 Hz (b) 0.02222, 1.333 Hz (c) 0.09333, 4.667 Hz (d) 0.05, 2.5 Hz. For example (a): $s_R = 60/1500 = 0.04$ and $f_2 = 0.04 \times 50 = 2$ Hz.

PQ 7.6 (practice questions).

Use the linear torque–speed line through $(n_S, 0)$ and (n_R, τ_R) (Lecture 7.2.2), so $\tau = \tau_R\,(n_S - n)/(n_S - n_R)$.

a. $60(50)/2910 = 1.031$, so $p = 1$, $n_S =$ **3000 rpm**. Rated slip speed $=$ **90 rpm**.

b. At no load τ = 0, so slip speed = 0 and $n =$ **3000 rpm** (ideally).

c. Rated torque occurs at rated speed, **2910 rpm**.

$$\omega_R = 2910 \times \frac{2\pi}{60} = 304.7 \text{ rad/s}, \qquad \tau_R = \frac{7500}{304.7} = \mathbf{24.61 \text{ Nm}}$$

d. $2910/3000 =$ **97 %**. Yes: the speed drops only 3 % from no load to full load, so "almost fixed speed" is fair.

e. Half torque means half the slip speed, 45 rpm. $n = 3000 - 45 =$ **2955 rpm**.

f. $P = \tau\omega = 12.31 \times 2955 \times 2\pi/60 =$ **3.808 kW**, which is **50.77 %** of rated. It is more than half because the speed (2955 rpm) is higher than rated speed. In fact $P/P_R = 0.5 \times 2955/2910$.

Disagreement with the PQ answer: the PQ gives 3.801 kW and 50.68 %. Those numbers are $12.31 \times 2950\,\text{rpm}$, i.e. the answer used 2950 rpm instead of its own 2955 rpm from part (e). With 2955 rpm the correct values are 3.808 kW and 50.77 %. The explanation (higher speed) is the same.

g. Slip speed $= 90 \times 20/24.61 = 73.14$ rpm, so $n = 3000 - 73.14 =$ **2927 rpm**.

h. 3070 rpm is above n_S, so it is a **generator**. Slip speed $= -70$ rpm.

$$\tau = 24.61 \times \frac{-70}{90} = \mathbf{-19.14 \text{ Nm}}$$

The negative sign means the shaft drives the machine.

PQ 7.7 (practice questions).

$60(50)/1470 = 2.04$, so $p = 2$, $n_S = 1500$ rpm and the rated slip speed is 30 rpm.

a. Half torque means half the slip speed (15 rpm): $n = 1500 - 15 =$ **1485 rpm**.

b. A generator at rated torque has a slip speed of −30 rpm: $n = 1500 + 30 =$ **1530 rpm**.

PQ 7.9 (practice questions).

$60(50)/960 = 3.125$, so $p = 3$, $n_S = 1000$ rpm. $R_2' = 1\,\Omega$.

Case	n (rpm)	$s = (1000 - n)/1000$	$f_2 = s \times 50$	R_2'/s
(a) locked rotor	0	**1**	**50 Hz**	**1 Ω**
(b) no load	1000	**0**	**0 Hz**	**∞ (open circuit)**
(c) rated	960	**0.040**	**2 Hz**	**25 Ω**
(d) generator	1050	**−0.050**	**−2.5 Hz**	**−20 Ω**

A negative R_2'/s means the "resistor" delivers power: mechanical power is absorbed and appears as electrical power. All values match the PQ answers.

PQ 7.10 (practice questions).

a. **Greater** than synchronous speed.
b. The difference between synchronous speed and rotor speed, $n_S - n$.
c. In the induction machine the secondary is short-circuited, the effective rotor resistance R_2'/s depends on slip, and the magnetising current is much larger because of the airgap.
d. $n = 0.9n_S$, so $s =$ **0.1** and $f_2 = 0.1 \times 50 =$ **5 Hz**.
e. The **airgap power** P_{AG}.
f. **Stator copper loss** and **(stator) iron loss**.
g. The **rotor copper loss**, sP_{AG}. (The question says "less than"; it means P_{AG} is *greater* than P_M.)
h. The rotor frequency sf is very low. Iron loss rises with frequency (eddy-current loss with f^2), so the rotor iron loss is negligible.
i. R_1 from a DC resistance test on the stator. $R_1 + R_2'$ from the locked-rotor test, so $R_2' = R_{LR} - R_1$ (detail in pack M8).
j. i. Low slip: $\tau \propto s$. (ii) High slip: $\tau \propto 1/s$.
k. Squirrel cage: R_2' is fixed. Wound rotor: R_2' can be increased with external three-phase resistors through slip-rings.
l. Changing R_2' changes the **speed (slip)** at which maximum torque occurs, but not its **value**.
m. **Starting** torque.
n. Ideal $\eta = 1 - s = n/n_S =$ **85 %**.
o. At no load $I_2' \approx 0$, so the input current is almost all magnetising current through X_M. That current lags V_1 by nearly 90°, so the pf is low.
p. About **2**.
q. $R_2'/s = R_2' + R_2'(1 - s)/s$. R_2' is the real rotor-winding resistance: the power in it (sP_{AG}) is heat. $R_2'(1 - s)/s$ is a fictitious resistance: the power in it ($(1 - s)P_{AG}$) is

the electrical power converted to mechanical power.

PQ 7.11 (practice questions).

This is the Type D method in Section 2.

a. $60(50)/1420 = 2.11$, so $p = 2$ and $n_S =$ **1500 rpm**.

b. $\omega_R = 2\pi \times 1420/60 = 148.7$ rad/s, so $\tau_R = 2200/148.7 =$ **14.80 Nm**.

c. i. $P_{IN} = \sqrt{3} V_L I_L \cos\phi = \sqrt{3}(415)(4.9)(0.78) =$ **2.747 kW**. (ii) $P_{IN} = P_{OUT}/\eta = 2200/0.80 =$ **2.750 kW**. They agree.

d. Ideal $\eta = n_R/n_S = 1420/1500 =$ **94.67 %**. The ideal efficiency only counts the rotor copper loss. Stator copper loss, iron loss and friction and windage make the real efficiency (80 %) lower.

Course answers: 1500 rpm, 14.80 Nm, 2.747 kW and 2.750 kW, 94.67 %.

PQ 4.7 (practice questions, only if you skipped M6).

a. $f = 60 \times 900/1200 =$ **45 Hz** and, keeping V/f constant, $V = 110 \times 45/60 =$ **82.5 V**.

b. The slip speed at a given torque is unchanged (100 rpm rated, 50 rpm at half torque): (i) **800 rpm** (ii) **850 rpm**. Full working in pack M6.

New questions

N1. [3 marks]

$n_S = 120(60)/6 = 1200$ rpm. $s = (1200 - 1164)/1200 =$ **0.03**. $f_2 = 0.03 \times 60 =$ **1.8 Hz**.

$R_2'/s = 0.45/0.03 =$ **15 Ω**, which splits into $R_2' =$ **0.45 Ω** (rotor copper loss) and $R_2'(1-s)/s =$ **14.55 Ω** (mechanical power).

N1: R2′/s = 15 ohm split into R2′ = 0.45 ohm and R2′(1-s)/s = 14.55 ohm

N2. [3 marks]

$n_S = 1000$ rpm, $s = 0.03$.

$P_{AG} = P_{\text{rotor Cu}}/s = 400/0.03 =$ **13.33 kW**. $P_M = (1-s)P_{AG} =$ **12.93 kW**.

$\tau = P_{AG}/\omega_S = 13333/104.72 =$ **127.3 Nm**. Check: $P_M/\omega = 12933/101.58 = 127.3$ Nm.

N3. [5 marks]

a. Delta, so $V_1 = V_L = 400\angle 0^\circ$ V. $s = 45/1500 = 0.03$, $R_2'/s = 33.33\,\Omega$.

$$Z = (1.2 + 33.33) + j4 = 34.76\angle 6.607^\circ\ \Omega, \qquad I_1 = I_2' = \frac{400}{34.76\angle 6.607^\circ} = \mathbf{11.51\angle -6.607^\circ\ A}$$

Line current $I_L = \sqrt{3} \times 11.51 =$ **19.93 A**.

b. $P_{AG} = 3(11.51)^2(33.33) =$ **13.24 kW**. $\tau = 13239/157.08 =$ **84.28 Nm**.
$P_{IN} = 3(400)(11.51)\cos 6.607^\circ = 13.72$ kW. $P_M = 0.97 \times 13.24 = 12.84$ kW.
$\eta = 12.84/13.72 =$ **93.63 %**.

Trap: if you use $400/\sqrt{3}$ (star thinking), you get $\tau = 28.09$ Nm, exactly one third.

N4. [10 marks]

a. $60(50)/1440 = 2.08$, so $p = 2$, $n_S = 1500$ rpm, $\omega_S = 157.1$ rad/s. $s = 60/1500 =$ **0.04**. $R_2'/s =$ **15 Ω**. Circuit: $V_1 = 415/\sqrt{3} = 239.6\angle 0^\circ$ V across $500\,\Omega \parallel j40\,\Omega$, then the series branch $0.8\,\Omega + j3\,\Omega + 15\,\Omega$.

N4(a) circuit with values: 239.6 V, 500 ohm parallel j40 ohm, then 0.8 ohm, j1.5 ohm, j1.5 ohm and R2′/s = 15 ohm

b. $Z = 15.8 + j3 = 16.08\angle 10.75^\circ\ \Omega, \qquad I_2' = \dfrac{239.6}{16.08\angle 10.75^\circ} = \mathbf{14.90\angle -10.75^\circ\ A} = 14.64 - j2.779$ A

$$I_M = \frac{239.6}{500} - j\frac{239.6}{40} = 0.4792 - j5.990\ \text{A}$$

$I_1 = 15.12 - j8.769 = \mathbf{17.48\angle -30.12^\circ\ A}, \qquad \text{pf} = \cos 30.12^\circ =$ **0.8650 lagging**

c. $P_{IN} = 3(239.6)(17.48)(0.8650) =$ **10.87 kW**. $P_{AG} = 3(14.90)^2(15) =$ **9.988 kW**.
$\tau = 9988/157.08 =$ **63.59 Nm**. The torque formula gives the same:
$3(239.6)^2(15)/\{157.08[(15.8)^2 + 3^2]\} = 63.59$ Nm.

d. $P_M = (1-s)P_{AG} = 0.96 \times 9988 =$ **9.589 kW**. $\eta = 9.589/10.87 =$ **88.25 %**. Ideal $\eta = 1 - s = 96\,\%$. Losses: stator Cu $3(14.90)^2(0.8) = 532.7$ W, iron $3(239.6)^2/500 = 344.5$ W, rotor Cu $sP_{AG} = 399.5$ W. Then $9589 + 399.5 + 532.7 + 344.5 = 10\,865$ W $= P_{IN}$. The ideal figure counts only the rotor copper loss. The stator copper and iron losses (877 W) take it from 96 % down to 88.25 %.

N5. [5 marks]

a. $\tau \propto V_1^2$ at fixed slip: $\tau = 0.9^2 \times 63.59 =$ **51.51 Nm**.

b. At low slip, $\tau \approx 3V_1^2 s/(\omega_S R_2')$. For constant τ, $s \propto 1/V_1^2$: $s \approx 0.04/0.81 = 0.04938$, so $n \approx 1500(1 - 0.04938) =$ **1426 rpm**.

c. Put $x = R_2'/s$ and $V_1 = 0.9 \times 239.6 = 215.6$ V into the torque formula:

$$3V_1^2 x = \tau\omega_S \left[(R_1 + x)^2 + X^2\right] \;\Rightarrow\; 9988x^2 - 123\,521x + 96\,287 = 0$$

The roots are $x = 11.53\,\Omega$ (the stable, low-slip root) and $x = 0.836\,\Omega$ (beyond pull-out). So $s = 0.6/11.53 =$ **0.05204** and $n =$ **1422 rpm**. Since τ is unchanged, $P_{AG} = \tau\omega_S = 9988$ W is also unchanged. Rotor Cu $= sP_{AG} = 0.05204 \times 9988 =$ **519.7 W** (up from 399.5 W, +30 %), and I_2' rises from 14.90 A to 16.99 A. Comment: a voltage sag on a constant-torque load makes the motor run slower, draw more rotor current and run hotter. The linear estimate is close (1426 vs 1422 rpm) but slightly optimistic, because R_1 and X are not negligible.

6. Quick recall check

Question

1. $s = 30/1500 = 0.02$, so $f_2 = 0.02 \times 50 =$ **1 Hz**.
2. R_2': rotor copper loss (heat), sP_{AG}. $R_2'(1 - s)/s$: power converted to mechanical, $(1 - s)P_{AG}$.
3. Rotor Cu = **500 W**, P_M = **9.5 kW**.
4. $\tau = P_{AG}/\omega_S$ (synchronous). $\tau = P_M/\omega$ (rotor). Both give the same τ.
5. Delta, so $V_1 = V_L =$ **690 V**.

Machine tests, maximum torque, generator mode

Answers and worked solutions · Day 11 · Tue 3 Nov · pack M8

5. Practice

From your course files

Jun24 Q3b (past exam, 4 marks).

$R_1 = 2\ \Omega$. This is given per phase, so no halving is needed.

Blocked rotor, per phase: $V_1 = 80/\sqrt{3} = 46.19$ V, $I_1 = 5$ A, $P_1 = 390/3 = 130$ W.

$$R = R_1 + R_2' = \frac{130}{5^2} = 5.2\ \Omega \quad \Rightarrow \quad R_2' = 5.2 - 2 = \mathbf{3.2\ \Omega}$$

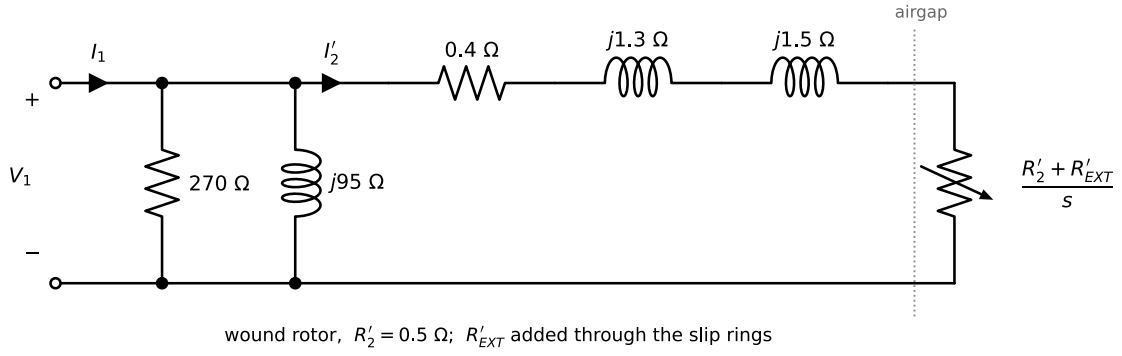
$$S_1 = 46.19 \times 5 = 230.9\ \text{VA}, \quad Q_1 = \sqrt{230.9^2 - 130^2} = 190.9\ \text{VAr}, \quad X = \frac{190.9}{5^2} = \mathbf{7.635\ \Omega}$$

So $X_1 = X_2' = 3.818\ \Omega$.

> **CHECK AGAINST THE COURSE ANSWERS:** R1 = 2 Ω, R2′ = 3.2 Ω, X = 7.635 Ω. All three agree.

The no-load test is not needed for the series parameters, but it gives $R_C = 239.6^2/90 = 637.9\ \Omega$ and $X_M = 239.6^2/833.8 = 68.85\ \Omega$ (using $P_1 = 90$ W, $S_1 = 838.6$ VA, $Q_1 = 833.8$ VAr).

Jun23 Q3b (past exam, 5 marks).



wound rotor, $R_2' = 0.5\ \Omega$; R_{EXT}' added through the slip rings

Jun23 Q3b circuit: 270 ohm parallel j95 ohm, then 0.4 ohm, j1.3 ohm, j1.5 ohm and (R2′ + REXT′)/s

(i) [2] $X = 1.3 + 1.5 = 2.8\ \Omega$. X_M and R_C are not needed, because they are at the terminals in the simplified circuit.

For maximum torque at $s = 1$:

$$\frac{R_2' + R_{EXT}'}{1} = |R_1 + jX| = \sqrt{0.4^2 + 2.8^2} = 2.828\ \Omega$$

$$R_{EXT}' = 2.828 - 0.5 = \mathbf{2.328\ \Omega}$$

> **CHECK AGAINST THE COURSE ANSWER:** R2EXT′ = 2.328 Ω. This agrees.

(ii) [3] Sketch. Draw three curves on τ vs n, all from $\tau = 0$ at $n = n_S$, all with **the same peak height**:

- $R_{EXT} = 0$: peak at $s = 0.5/2.828 = 0.177$ (close to n_S). Lowest starting torque.
- $R_{EXT} = 0.5$ pu $= 1.164\ \Omega$: peak at $s = (0.5 + 1.164)/2.828 = 0.588$. Higher starting torque.
- $R_{EXT} = 1$ pu $= 2.328\ \Omega$: peak at $s = 1$ ($n = 0$). Starting torque equals τ_{max}.

Near n_S the higher-resistance curves are less steep, because there $\tau \approx 3V_1^2 s/(\omega_S R_2')$. See the figure in Section 1.5 for the shape.

PQ 7.11 (practice questions).

a. $60(50)/1420 = 2.11$, so $p = 2$ and $n_S =$ **1500 rpm**.
b. $\omega_R = 2\pi(1420)/60 = 148.7$ rad/s, so $\tau_R = 2200/148.7 =$ **14.80 Nm**.
c. i. $P_{in} = \sqrt{3} V_L I_L \text{ pf} = \sqrt{3}(415)(4.9)(0.78) =$ **2.747 kW**. (ii) $P_{out}/\eta = 2200/0.8 =$ **2.750 kW**.
d. Ideal efficiency $= 1 - s = n_R/n_S = 1420/1500 =$ **94.67 %**. It only includes the rotor copper loss; stator copper, iron, and friction and windage losses make the actual efficiency (80 %) lower.

Course answers: 1500 rpm, 14.80 Nm, 2.747 kW, 2.750 kW, 94.67 %. (Full working is also in pack M7.)

PQ 7.12 (practice questions).

(a) Blocked rotor: $V_1 = 65/\sqrt{3} = 37.53$ V, $I_1 = 4.8$ A, $P_1 = 116.7$ W.
$R = 116.7/4.8^2 = 5.064\ \Omega$, so $R_2' = 5.064 - 3 = 2.064\ \Omega$. $S_1 = 180.1$ VA, $Q_1 = 137.2$ VAr, $X = 137.2/4.8^2 = 5.957\ \Omega$, and $X_1 = X_2' = 2.978\ \Omega$.

No load: $V_1 = 239.6$ V, $P_1 = 68.33$ W, $S_1 = 718.8$ VA, $Q_1 = 715.5$ VAr.
$R_C = 239.6^2/68.33 = 840.1\ \Omega$ and $X_M = 239.6^2/715.5 = 80.23\ \Omega$.

(b) Draw the course simplified circuit (Section 1): $840.1\ \Omega \parallel j80.23\ \Omega$ across $V_1 = 239.6$ V, then $R_1 = 3\ \Omega$, $jX_1 = jX_2' = j2.978\ \Omega$ and R_2'/s in series.

(c) $p = 2$, $n_S = 1500$ rpm, $s = 45/1500 = 0.03$, $R_2'/s = 68.79\ \Omega$, $\omega_S = 157.08$ rad/s.

$I_2' = 239.6/(3 + 68.79 + j5.957) = 3.326\angle -4.74^\circ$ A.

$I_1 = I_2' + 239.6/840.1 + 239.6/(j80.23) = 4.858\angle -42.2^\circ$ A, so pf $= 0.741$ lag.

$P_{IN} = 3(239.6)(4.858)(0.741) = 2.588$ kW.

$\tau = 3(3.326)^2(68.79)/157.08 = 14.53$ Nm, and
$P_{OUT} = 14.53 \times (2\pi \times 1455/60) = 2.215$ kW, so $\eta = 85.6\ \%$.

(d) $P_{Cu} = 3(3.326)^2(3 + 2.064) = 168.1$ W, $P_{Fe} = 3(239.6)^2/840.1 = 205.0$ W, and $\eta = 2215/(2215 + 373.1) = 85.6\ \%$. This is the same as (c), which confirms the power balance.

Course answers: $R_2' = 2.064\ \Omega$, $X = 5.956\ \Omega$, $R_C = 840.2\ \Omega$, $X_M = 80.23\ \Omega$; (c) [$s = 0.030$, $I_2' = 3.323\angle -4.738^\circ$ A] 4.855 A, 0.7408 lag, 2.585 kW, 14.52 Nm, 2.212 kW, 85.58 %; (d) Cu 167.8 W, Fe 205.0 W, total 372.8 W, 85.58 %. These differ from the working above only in the third or fourth figure (intermediate rounding). *Hint:* copper loss uses I_2' through $R_1 + R_2'$.

PQ 7.13 (practice questions).

Everything per phase. $R_1 =$ **1 Ω** (given per phase).

Blocked rotor: $V_1 = 90/\sqrt{3} = 51.96$ V, $I_1 = 8$ A, $P_1 = 90$ W.
$R_1 + R_2' = 90/8^2 = 1.406\ \Omega$, so $R_2' =$ **0.406 Ω**. $S_1 = 415.7$ VA,
$Q_1 = \sqrt{415.7^2 - 90^2} = 405.8$ VAr, $X = 405.8/8^2 =$ **6.341 Ω** ($X_1 = X_2' = 3.171\ \Omega$).

No load: $V_1 = 239.6$ V, $I_1 = 5.4$ A, $P_1 = 100$ W. $R_C = 239.6^2/100 =$ **574.1 Ω**.
$S_1 = 1293.8$ VA, $Q_1 = 1290.0$ VAr, $X_M = 239.6^2/1290.0 =$ **44.50 Ω**.

Course answers: 0.406, 6.341, 574.1, 44.50 Ω. All agree.

PQ 7.15 (practice questions).

$V_1 = 460/\sqrt{3} = 265.6$ V. 60 Hz and 1760 rpm give $p = 2$, $n_S = 1800$ rpm, $\omega_S = 188.5$ rad/s. $X = 2.3\ \Omega$ and $|R_1 + jX| = |0.25 + j2.3| = 2.314\ \Omega$.

(a) $s_{\tau max} = 0.2/2.314 = 0.08645$, so $n = 1800(1 - 0.08645) = 1644$ rpm.

(b) (i) At n_S, $s = 0$ and $\tau = 0$. (ii) $s = 40/1800 = 0.02222$, $R_2'/s = 9.0\ \Omega$:
$\tau = 3(265.6)^2(9.0)/\{188.5[(9.25)^2 + 2.3^2]\} = 111.2$ Nm. (iii) $R_2'/s = 2.314\ \Omega$:
$\tau = 218.9$ Nm. (iv) $s = 1$: $\tau = 3(265.6)^2(0.2)/\{188.5[0.45^2 + 2.3^2]\} = 40.88$ Nm.

(c) Sketch: τ = 40.88 Nm at 0 rpm, rising to the 218.9 Nm peak at 1644 rpm, then falling steeply through 111.2 Nm at 1760 rpm to 0 at 1800 rpm.

(d) The total rotor resistance is $2.314\ \Omega$, so the external part is $2.114\ \Omega$. Starting torque $= \tau_{max} = 218.9$ Nm.

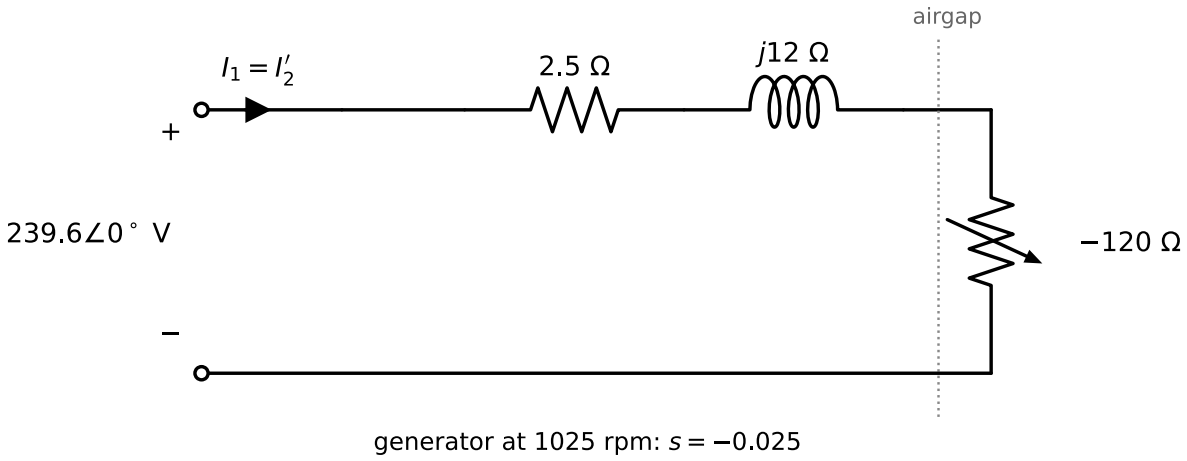
Course answers: (a) 1644 rpm (b) 0, 111.2, 218.9, 40.88 Nm (d) 2.314 Ω, 218.9 Nm. *Hint:* 1760 rpm at 60 Hz means 4 poles; in (d) the question wants the *total* rotor resistance.

Jun24 Q3c (past exam, 10 marks).

(i) [3] 960 rpm at 50 Hz: $p = 3000/960 = 3.125$, so $p = 3$ (6 poles).
$n_S = 60 \times 50/3 = \mathbf{1000}$ rpm and $\omega_S = 104.72$ rad/s.

$$s = \frac{1000 - 1025}{1000} = \mathbf{-0.025}, \qquad \frac{R_2'}{s} = \frac{3}{-0.025} = \mathbf{-120\ \Omega}$$

(ii) [5] Circuit: V_1, then $R_1 = 2.5\ \Omega$, then $jX = j12\ \Omega$, then $R_2'/s = -120\ \Omega$, all in series (no shunt branch).



Jun24 Q3c circuit: 239.6 V, 2.5 ohm, j12 ohm and R2′/s = -120 ohm in series, no shunt branch

$V_1 = 415/\sqrt{3} = 239.6\angle 0^\circ$ V.

$$Z = 2.5 - 120 + j12 = -117.5 + j12 = 118.1\angle 174.2^\circ\ \Omega$$

$$I_1 = \frac{239.6\angle 0^\circ}{118.1\angle 174.2^\circ} = \mathbf{2.029\angle -174.2^\circ}\ \text{A}$$

$$P_{IN} = 3V_1 I_1 \cos\phi = 3(239.6)(2.029)\cos(-174.2^\circ) = \mathbf{-1451}\ \text{W}$$

The negative sign means 1.451 kW flows *into* the supply. The machine is generating. ($Q_{IN} = +148$ VAr, so it still absorbs reactive power.)

(iii) [2] With no shunt branch, $I_2' = I_1$:

$$P_{AG} = 3I_2'^2\frac{R_2'}{s} = 3(2.029)^2(-120) = \mathbf{-1.481}\ \text{kW}, \qquad \tau = \frac{P_{AG}}{\omega_S} = \frac{-1481}{104.72} = \mathbf{-14.15}\ \text{Nm}$$

Power flow check: shaft input $= |P_{AG}|(1-s) = 1481 \times 1.025 = 1519$ W. Rotor copper loss $= 3(2.029)^2(3) = 37$ W, leaving $|P_{AG}| = 1481$ W. Stator copper loss $= 3(2.029)^2(2.5) = 31$ W. Electrical output $= 1481 - 31 = 1451$ W. This matches $|P_{IN}|$.

> **CHECK AGAINST THE COURSE ANSWERS:** 1000 rpm, −0.025, −120 Ω; $I_1 = 2.029\angle -174.2^\circ$ A; $P_{in} = -1451$ W; $P_{ag} = -1.482$ kW; $\tau = -14.15$ Nm. All agree. (The course's −1.482 kW and this −1.481 kW differ only in rounding: the unrounded value is −1481.5 W.)

PQ 7.14 (practice questions, if time: parts (d)–(f)).

$V_1 = 415/\sqrt{3} = 239.6$ V, $p = 1$, $n_S = 3000$ rpm, $\omega_S = 314.16$ rad/s, $X = 0.6\ \Omega$. Rated slip $s = 210/3000 = 0.07$, $R_2'/s = 2.286\ \Omega$. Parts (a)–(c) are needed for (d).

(a) $s = 1$: $I = 239.6/|0.36 + j0.6| =$ **342.4 A**; $P = 3(342.4)^2(0.36) =$ **126.6 kW**; $\tau = 3(342.4)^2(0.16)/314.16 =$ **179.2 Nm**. (If you get 93.7 A you have used rated speed, not standstill.)

(b) $I_2' = 239.6/(0.2 + 2.286 + j0.6) = 93.70\angle -13.57^\circ$ A; $I_1 = I_2' + 239.6/(j12) =$ **100.3 A**, pf = **0.9083 lag**, $P_{IN} = 3(239.6)(100.3)(0.9083) =$ **65.47 kW**.

(c) $\tau_R = 3(93.70)^2(2.286)/314.16 =$ **191.6 Nm**; $P_{OUT} = \tau_R\,\omega_S(1-s) =$ **55.99 kW**; $\eta = 55.99/65.47 =$ **85.52 %**.

(d) $\tau_{start}/\tau_R = 179.2/191.6 =$ **0.935** (official 0.9353, from rounded values). $I_{start}/I_R = 342.4/100.3 =$ **3.414** (starting current without X_M, as in (a)).

(e) $s_{\tau max} = 0.16/|0.2 + j0.6| = 0.2530$, so $n = 3000(1 - 0.2530) =$ **2241 rpm**. $\tau_{max} = \dfrac{3V_1^2}{2\omega_S(R_1 + |R_1 + jX|)} =$ **329.3 Nm**, which is $329.3/191.6 =$ **1.719** times rated torque.

(f) At no load $I_2' \approx 0$, so $I_1 = V_1/X_M = 239.6/12 =$ **19.97 A**, which is $19.97/100.3 =$ **0.1991** of rated current.

(g) Sketch: τ/τ_R starts at 0.94 at 0 rpm, rises to a peak of 1.72 at 2241 rpm, falls steeply through 1.0 at 2790 rpm to 0 at 3000 rpm.

Course answers: (a) 342.4 A, 126.6 kW, 179.2 Nm (b) 100.3 A, 0.9083 lag, 65.47 kW (c) 191.6 Nm, 55.99 kW, 85.52 % (d) 0.9353, 3.414 (e) 2241 rpm, 329.3 Nm, 1.719 (f) 19.97 A, 0.1991.

New questions

New Q1 (easy) [5 marks].

$R_1 = 2.4/2 = \mathbf{1.2\ \Omega}$ (a line-to-line reading in star is two phases in series).

Blocked rotor: $V_1 = 43.30$ V, $P_1 = 200$ W. $R = 200/81 = 2.469\ \Omega$, so $R_2' = \mathbf{1.269\ \Omega}$.
$S_1 = 389.7$ VA, $Q_1 = 334.5$ VAr, $X = 334.5/81 = 4.129\ \Omega$, so $X_1 = X_2' = \mathbf{2.065\ \Omega}$.

No load: $V_1 = 230.9$ V, $P_1 = 83.33$ W, $S_1 = 923.8$ VA, $Q_1 = 920.0$ VAr.
$R_C = 230.9^2/83.33 = \mathbf{640\ \Omega}$ and $X_M = 230.9^2/920.0 = \mathbf{57.97\ \Omega}$.

New Q2 (easy–medium) [6 marks].

$V_1 = 239.6$ V, $n_S = 1000$ rpm, $\omega_S = 104.72$ rad/s, $X = 3.6\ \Omega$, $|0.5 + j3.6| = 3.635\ \Omega$.

a. $s_{\tau max} = 0.6/3.635 = 0.1651$, so $n = 1000(1 - 0.1651) = \mathbf{834.9}$ rpm.
$\tau_{max} = 3(239.6)^2/[2(104.72)(0.5 + 3.635)] = \mathbf{198.9}$ Nm.

b. $\tau_{start} = 3(239.6)^2(0.6)/\{104.72[(1.1)^2 + 3.6^2]\} = \mathbf{69.6}$ Nm.
$I_{START} = 239.6/|1.1 + j3.6| = \mathbf{63.7}$ A.

c. $R_{2EXT}' = 3.635 - 0.6 = \mathbf{3.035\ \Omega}$. The new $\tau_{start} = \tau_{max} = \mathbf{198.9}$ Nm. The new $I_{START} = 239.6/|0.5 + 3.635 + j3.6| = \mathbf{43.7}$ A. So you get almost 3 times the starting torque with about 30 % less current.

New Q3 (medium) [5 marks].

a. $V_L = 415 \times 25/50 = \mathbf{207.5}$ V (phase 119.8 V).

New Q3 circuit at 25 Hz: 119.8 V, 0.5 ohm, j0.9 ohm, j0.9 ohm and R2′/s

b. At 25 Hz: $X = 1.8\ \Omega$, $n_S = 500$ rpm, $\omega_S = 52.36$ rad/s. $|0.5 + j1.8| = 1.868\ \Omega$, so $s_{\tau max} = 0.6/1.868 = 0.3212$ and $n = 500(1 - 0.3212) = \mathbf{339.4}$ rpm.

$\tau_{max} = 3(119.8)^2/[2(52.36)(0.5 + 1.868)] = \mathbf{173.6}$ Nm, compared with 198.9 Nm at 50 Hz.

The lecture says that τ_{max} stays constant. That is exact only if R_1 is neglected: with $R_1 = 0$ both cases give $3V_1^2/(2\omega_S X) = 228.4$ Nm. At 25 Hz, X has halved but R_1 has not, so R_1 takes a bigger share of the impedance and τ_{max} drops by 13 %.

c. The slip speed is $1000 - 976.8 = 23.2$ rpm at 50 Hz and $500 - 475.9 = 24.1$ rpm at 25 Hz. That is almost the same. The curve has slid left with nearly the same shape near

n_S. The slip as a fraction is larger at 25 Hz (0.048 vs 0.023), so the ideal efficiency is slightly lower.

New Q4 (medium) [6 marks].

a. $n = 1040 > n_S = 1000$ rpm, so it is a **generator**.
$s = (1000 - 1040)/1000 = \mathbf{-0.04}$ and $R_2'/s = 0.6/(-0.04) = \mathbf{-15\ \Omega}$.

New Q4 generator circuit: 239.6 V, 0.5 ohm, j3.6 ohm and R2′/s = -15 ohm

b. $Z = 0.5 - 15 + j3.6 = -14.5 + j3.6 = 14.94\angle 166.1^\circ\ \Omega$.
$I_1 = 239.6/14.94\angle 166.1^\circ = \mathbf{16.04\angle -166.1^\circ}$ A.

$P_{IN} = 3(239.6)(16.04)\cos(-166.1^\circ) = \mathbf{-11.19}$ kW, so 11.19 kW is delivered to the grid. $Q_{IN} = \mathbf{+2.78}$ kVAr, so the machine still absorbs reactive power from the grid.

c. $P_{AG} = 3(16.04)^2(-15) = \mathbf{-11.57}$ kW and $\tau = -11574/104.72 = \mathbf{-110.5}$ Nm. The turbine supplies $|P_{AG}|(1 - s) = 11.57 \times 1.04 = \mathbf{12.04}$ kW (check: $110.5 \times 2\pi \times 1040/60 = 12.04$ kW).

Losses: rotor Cu 463 W and stator Cu 386 W. $12.04 - 0.463 - 0.386 = 11.19$ kW, which matches (b).

New Q5 (hard, chained) [10 marks].

$V_1 = 230.9$ V, $n_S = 1500$ rpm, $\omega_S = 157.08$ rad/s. $R_1 = 1.2$, $R_2' = 1.269$, $X = 4.129\ \Omega$, $R_C = 640\ \Omega$, $X_M = 57.97\ \Omega$.

a. $s = 60/1500 = 0.04$ and $R_2'/s = 31.73\ \Omega$.

New Q5 circuit at rated speed: 230.9 V, 640 ohm parallel j57.97 ohm, then 1.2 ohm, j2.065 ohm, j2.065 ohm, 31.73 ohm

$I_2' = 230.9/(1.2 + 31.73 + j4.129) = 6.959\angle -7.15^\circ$ A.

$I_1 = I_2' + 230.9/640 + 230.9/(j57.97) = \mathbf{8.736\angle -33.7^\circ}$ A, so pf $= \mathbf{0.832}$ lag.

$P_{IN} = 3(230.9)(8.736)(0.832) = \mathbf{5.034}$ kW.

$\tau_R = 3(6.959)^2(31.73)/157.08 = \mathbf{29.34}$ Nm. $P_{OUT} = 29.34 \times 150.80 = \mathbf{4.425}$ kW, so $\eta = \mathbf{87.9\ \%}$.

b. $|1.2 + j4.129| = 4.300\ \Omega$, so $s_{\tau max} = 1.269/4.300 = 0.2951$ and $n = \mathbf{1057}$ rpm.
$\tau_{max} = 3(230.9)^2/[2(157.08)(1.2 + 4.300)] = \mathbf{92.6}$ Nm, and $\tau_{max}/\tau_R = \mathbf{3.16}$.

$\tau_{start} = 3(230.9)^2(1.269)/\{157.08[(2.469)^2 + 4.129^2]\} = 55.8$ Nm, so $\tau_{start}/\tau_R = \mathbf{1.90}$.

c. $s = -0.02$ and $R_2'/s = -63.46\ \Omega$. The torque formula gives $\tau = \mathbf{-16.6}$ Nm. The machine is generating: the shaft must be driven against 16.6 Nm.

d. $I_{NL} = |230.9/640 + 230.9/(j57.97)| = 4.00$ A. This matches the test, as it should. $4.00/8.736 = \mathbf{0.46}$ of rated current. That is typical: the lecture gives 25–60 %, which is mostly magnetising current needed because of the airgap.

New Q6 (short answer, words only) [6 marks].

a. At no load the rotor runs at almost synchronous speed, so very little rotor current flows. The stator must still produce the rotating magnetic field. Because the flux has to cross an airgap, this needs a large magnetising current, typically a quarter to over half of rated current. That current is modelled by the magnetising reactance. A smaller in-phase current supplies the hysteresis and eddy-current losses in the stator iron, which are modelled by the core-loss resistance. A further small in-phase current supplies the friction and windage losses, because the rotor is still turning. The current is mostly magnetising current, which lags the voltage by nearly 90 degrees, so the power factor is very low.

b. There would be no rotor copper loss, so efficiency would rise. The slip at maximum torque is proportional to rotor resistance, so it would be zero. The torque–speed curve near synchronous speed would become vertical, like a synchronous machine, and the speed would not drop with load. The starting torque is proportional to rotor resistance, so it would be zero, and the motor could not start by itself.

6. Quick recall check

Question

1. The magnetising branch ($R_C \parallel jX_M$). At low voltage it carries little current compared with the low-impedance series branch. You need R_1 from the DC test to get $R_2' = R - R_1$.
2. $s_{\tau max} = R_2'/\sqrt{R_1^2 + (X_1 + X_2')^2}$. τ_{max} is unchanged. Only $s_{\tau max}$ doubles.
3. $R_{2EXT}' = \sqrt{R_1^2 + X^2} - R_2'$.
4. $n_S = 1500$ rpm and $s = -0.04$. $s < 0$, $R_2'/s < 0$, $\tau < 0$, $P_{IN} < 0$ (generating), and $Q_{IN} > 0$ (still absorbs VArs).
5. V_1, X and ω_S all halve. R_1 and R_2' stay the same.

Exam-style induction problems

Answers and worked solutions · Day 12 · Wed 4 Nov · pack M9

5. Practice

From your course files

PQ 7.11 (practice questions).

Ratings: 2.2 kW, 50 Hz, 1420 rpm, 415 V, 4.9 A, $\eta = 0.80$, pf 0.78.

a. $p = 3000/1420 = 2.11$, so $p = 2$. $n_S = 1500$ rpm.

b. $\omega = 2\pi \times 1420/60 = 148.70$ rad/s. $\tau = 2200/148.70 =$ **14.80 Nm**.

(c)(i) $P_{IN} = \sqrt{3} V_L I_L \cos\phi = \sqrt{3} \times 415 \times 4.9 \times 0.78 =$ **2.747 kW**. (ii) $P_{IN} = P_{OUT}/\eta = 2200/0.80 =$ **2.750 kW**. They agree, which shows the rating plate is self-consistent.

d. $s = 80/1500 = 0.05333$. Ideal efficiency $= 1 - s =$ **94.67%**. The actual 80% is lower because the ideal value only counts rotor copper loss. Stator copper loss, iron loss and friction and windage are extra.

Course answers: 1500 rpm, 14.80 Nm, 2.747 kW, 2.750 kW, 94.67%. All match.

PQ 7.16 (practice questions).

Constant V/f keeps the flux constant: $V = 460 \times f/60$.

- 30 Hz: $V = 460 \times 30/60 =$ **230 V**
- 50 Hz: $V = 460 \times 50/60 =$ **383.3 V**

Course answers: 230 V, 383.3 V.

PQ 7.12 (practice questions; do (c)–(d) here).

415 V, 1455 rpm, 50 Hz, star. From part (a) (method in M8): $R_1 = 3\ \Omega$. Blocked rotor per phase: 37.53 V, 4.8 A, 116.7 W, so $R = 5.064\ \Omega$, $R_2' = 2.064\ \Omega$, $X = 5.957\ \Omega$. No-load per phase: 239.6 V, 3.0 A, 68.33 W, so $R_C = 840.1\ \Omega$, $X_M = 80.23\ \Omega$.

(c) At rated speed. $p = 2$, $n_S = 1500$ rpm, $s = 45/1500 = 0.03$, $R_2'/s = 68.79\ \Omega$.

$$Z_1 = (3 + 68.79) + j5.957 = 71.79 + j5.957 = 72.04\angle 4.743^\circ\ \Omega$$
$$I_2' = 239.6/72.04\angle 4.743^\circ = 3.326\angle -4.743^\circ = 3.315 - j0.2751\ \text{A}$$
$$I_M = 239.6/840.1 - j\,239.6/80.23 = 0.2852 - j2.986\ \text{A}$$
$$I_1 = 3.600 - j3.261 = \mathbf{4.858\angle -42.18^\circ\ A}$$

- pf $= \cos 42.18^\circ =$ **0.7411 lag**
- $P_{IN} = 3 \times 239.6 \times 4.858 \times 0.7411 =$ **2.588 kW**
- $P_{AG} = 3 \times 3.326^2 \times 68.79 = 2283$ W, $\tau = 2283/157.08 =$ **14.53 Nm**
- $\omega = 2\pi \times 1455/60 = 152.37$ rad/s, $P_{OUT} = 14.53 \times 152.37 =$ **2.215 kW**
- $\eta = 2.215/2.588 =$ **85.58%**

(d) Losses. Copper $= 3 \times 3.326^2 \times (3 + 2.064) =$ **168.1 W**. Iron $= 3 \times 239.6^2/840.1 = 205.0$ W (equals the no-load input power, as expected). Total **373.1 W**. $\eta = 2215/(2215 + 373.1) =$ **85.58%**, the same as (c).

Course answers: $I_2' = 3.323\angle -4.738^\circ$ A, 4.855 A, 0.7408 lag, 2.585 kW, 14.52 Nm, 2.212 kW, 85.58%. Copper 167.8 W, iron 205.0 W, total 372.8 W. Mine agree to within 0.1%. The small difference comes from rounding of the part (a) parameters. The efficiency matches exactly.

PQ 7.14 (practice questions; do (a)–(d) and (f) here).

415 V star, 50 Hz, 2790 rpm. $R_1 = 0.2$, $R_2' = 0.16$, $X = 0.3 + 0.3 = 0.6\ \Omega$, $X_M = 12\ \Omega$, no R_C. $p = 1$, $n_S = 3000$ rpm, $\omega_S = 314.16$ rad/s, $V_1 = 239.6$ V.

(a) Locked rotor, $s = 1$, ignore X_M. $Z = 0.36 + j0.6 = 0.6997\angle 59.04^\circ\ \Omega$.

- $I = 239.6/0.6997 =$ **342.4 A**
- $P_{IN} = 3 \times 342.4^2 \times 0.36 =$ **126.6 kW** (all of it is copper loss)
- $\tau = 3 \times 342.4^2 \times 0.16/314.16 =$ **179.2 Nm** (P_{AG} uses only R_2', since $s = 1$)

(b) Rated speed. $s = 210/3000 = 0.07$, $R_2'/s = 2.286\ \Omega$.

$$Z_1 = 2.486 + j0.6 = 2.557\angle 13.57^\circ\ \Omega, \quad I_2' = 93.70\angle -13.57^\circ = 91.08 - j21.99\ \text{A}$$
$$I_M = 239.6/j12 = -j19.97\ \text{A}$$
$$I_1 = 91.08 - j41.95 = \mathbf{100.3\angle -24.73^\circ\ A}$$

pf $= \cos 24.73^\circ =$ **0.9083 lag**. $P_{IN} = 3 \times 239.6 \times 100.3 \times 0.9083 =$ **65.47 kW**.

(c) $P_{AG} = 3 \times 93.70^2 \times 2.286 = 60.20$ kW. $\tau = 60.20\text{k}/314.16 =$ **191.6 Nm**. $\omega = 292.17$ rad/s, $P_{OUT} = 191.6 \times 292.17 =$ **55.99 kW**. $\eta = 55.99/65.47 =$ **85.52%**.

(d) $\tau_{start}/\tau_R = 179.2/191.6 = 0.935$. $I_{start}/I_R = 342.4/100.3 = 3.414$.

(f) No-load ($s \to 0$, $R_2'/s \to \infty$, $I_2' = 0$). $I_1 = I_M = 239.6/12 =$ **19.97 A**, which is $19.97/100.3 = 0.1991$ of rated current.

(e) is the maximum-torque method (M8). Check answers: 2241 rpm, 329.3 Nm, 1.719 × rated. **(g)** is the sketch method in M8, using these three torques.

Course answers: 342.4 A, 126.6 kW, 179.2 Nm; 100.3 A, 0.9083 lag, 65.47 kW; 191.6 Nm, 55.99 kW, 85.52%; 0.9353, 3.414; 19.97 A, 0.1991. All match, except τ_{start}/τ_R: unrounded values give 0.9349. The course's 0.9353 comes from dividing the rounded torques 179.2/191.6, so either is fine.

PQ 7.15 (practice questions; do (b)(i), (ii), (iv) here).

460 V, 60 Hz, 1760 rpm, star. $R_1 = 0.25$, $R_2' = 0.2$, $X = 1.2 + 1.1 = 2.3\ \Omega$. $V_1 = 460/\sqrt{3} = 265.6$ V. $p = 3600/1760 = 2.05$, so $p = 2$, $n_S = 1800$ rpm, $\omega_S = 188.50$ rad/s.

- i. Synchronous: $s = 0$, $R_2'/s \to \infty$, so **τ = 0 Nm**.
- ii. Rated: $s = 40/1800 = 0.02222$, $R_2'/s = 9.0\ \Omega$.
$$\tau = \frac{3 \times 265.6^2 \times 9}{188.50\left[9.25^2 + 2.3^2\right]} = \textbf{111.2 Nm}.$$
- iv. Locked: $s = 1$. $\tau = \dfrac{3 \times 265.6^2 \times 0.2}{188.50\left[0.45^2 + 2.3^2\right]} = \textbf{40.88 Nm}$.

iii. and (d) use $s_{max} = R_2'/|R_1 + jX| = 0.08645$ (M8): 1644 rpm, 218.9 Nm, and $R_2' = 2.314\ \Omega$ for maximum torque at standstill. All match the course answers.

New questions

New Q1 (15 marks, easy–medium).

(a) 6 poles, $p = 3$: $n_S = 3000/3 = \textbf{1000 rpm}$, $\omega_S = 104.72$ rad/s. $s = 30/1000 = \textbf{0.03}$. $f_2 = sf = \textbf{1.5 Hz}$. $R_2'/s = 0.6/0.03 = \textbf{20 Ω}$. [3]

(b) Delta: $V_1 = V_L = 400$ V. [1]

$$Z_1 = (0.8 + 20) + j3 = 20.8 + j3 = 21.02\angle 8.207^\circ\ \Omega$$
$$I_2' = 400/21.02\angle 8.207^\circ = 19.03\angle -8.207^\circ = 18.84 - j2.717\ \text{A}$$
$$I_M = 400/300 - j\,400/60 = 1.333 - j6.667\ \text{A}$$
$$I_1 = 20.17 - j9.384 = \mathbf{22.25\angle -24.95^\circ\ A\ (phase)}$$

Line current $= \sqrt{3} \times 22.25 = \textbf{38.53 A}$. pf $= \cos 24.95^\circ = \textbf{0.9067 lag}$. [4]

(c)

- $P_{IN} = 3 \times 400 \times 22.25 \times 0.9067 = \textbf{24.21 kW}$ [1]
- $P_{AG} = 3 \times 19.03^2 \times 20 = \textbf{21.74 kW}$ [1]
- $\tau = 21.74\text{k}/104.72 = \textbf{207.6 Nm}$ [1]
- $\omega = 2\pi \times 970/60 = 101.58$ rad/s, $P_{OUT} = 207.6 \times 101.58 = \textbf{21.09 kW}$ [1]

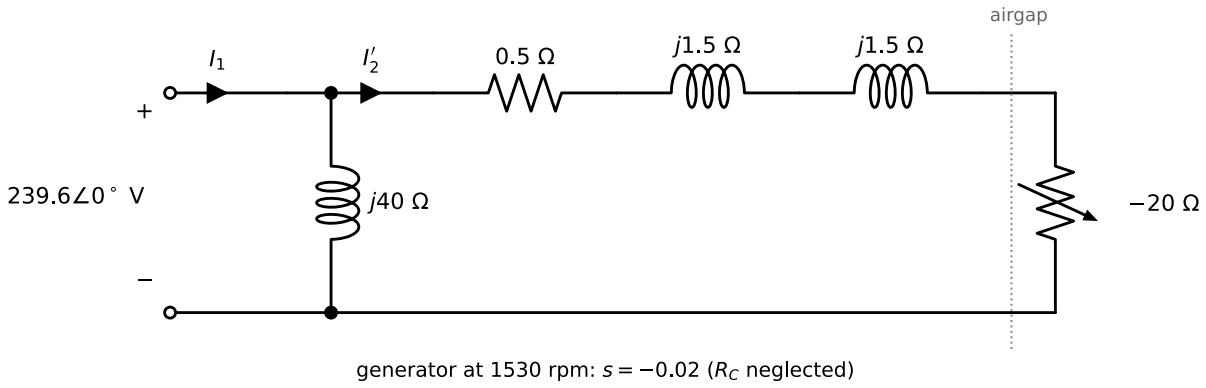
- $\eta = 21.09/24.21 = \mathbf{87.10\%}$ [1]

(d) Stator copper $3 \times 19.03^2 \times 0.8 = 869.5$ W. Rotor copper $3 \times 19.03^2 \times 0.6 = 652.1$ W ($= sP_{AG} = 0.03 \times 21737$ (checks)). Iron $3 \times 400^2/300 = 1600$ W. Total 3122 W. $24.21 - 3.12 = 21.09$ kW $= P_{OUT}$. [2]

New Q2 (17 marks, medium).

(a) $n_S = 1500$ rpm. $n > n_S$, so it is **generating**. $s = (1500 - 1530)/1500 = \mathbf{-0.02}$. $R_2'/s = 0.4/(-0.02) = \mathbf{-20\ \Omega}$. [3]

(b) $V_1 = 239.6$ V.



generator at 1530 rpm: $s = -0.02$ (R_C neglected)

New Q2 generator circuit: 239.6 V across j40 ohm, then 0.5 ohm, j1.5 ohm, j1.5 ohm and R2′/s = -20 ohm

$$Z_1 = (0.5 - 20) + j3 = -19.5 + j3 = 19.73\angle 171.3^\circ\ \Omega$$
$$I_2' = 239.6/19.73\angle 171.3^\circ = \mathbf{12.14\angle -171.3^\circ\ A} = -12.00 - j1.847\ \text{A}$$
$$I_M = 239.6/j40 = -j5.990\ \text{A}$$
$$I_1 = -12.00 - j7.837 = \mathbf{14.33\angle -146.9^\circ\ A}$$

[4]

Using $S = 3V_1 I_1^*$: $P = 3 \times 239.6 \times 14.33 \cos(-146.9^\circ) = \mathbf{-8.628\ kW}$, $Q = 3 \times 239.6 \times 14.33 \sin(146.9^\circ) = \mathbf{+5.633\ kVAr}$. $P < 0$: 8.628 kW is delivered **to** the grid. $Q > 0$: it still **absorbs** 5.633 kVAr from the grid. [2]

(c)

- $P_{AG} = 3 \times 12.14^2 \times (-20) = \mathbf{-8.849\ kW}$ [1]
- $\tau = -8849/157.08 = \mathbf{-56.33\ Nm}$ (the turbine must apply 56.33 Nm) [1]
- $\omega = 2\pi \times 1530/60 = 160.22$ rad/s, $P_{OUT} = \tau\omega = -9.026$ kW, so the mechanical input is **9.026 kW** [1]
- $\eta_{gen} = 8.628/9.026 = \mathbf{95.59\%}$ [1]
- Check: copper loss $3 \times 12.14^2 \times 0.9 = 398.2$ W $= 9026 - 8628$ (checks) [1]

(d) The rotor has no separate excitation. The rotating flux is set up only by the stator magnetising current through X_M, which comes from the supply. So the machine always draws lagging reactive power (V_1^2/X_M per phase, plus the leakage VArs), whether motoring or generating. The grid (or local capacitors) must provide these VArs. A stand-alone induction generator needs a capacitor bank. [3]

New Q3 (18 marks, medium–hard).

$V_1 = 239.6$ V, $n_S = 1500$ rpm, $\omega_S = 157.08$ rad/s, $X = 2.4\ \Omega$, $3V_1^2 = 172{,}225\ \text{V}^2$.

(a) $s = 30/1500 = 0.02$, $R_2'/s = 20\ \Omega$. $\tau = \dfrac{172225 \times 20}{157.08\left[20.5^2 + 2.4^2\right]} =$ **51.47 Nm**.
$|I_1| = 239.6/|20.5 + j2.4| =$ **11.61 A**. [3]

(b) Let $x = R_2'/s$. Rearranging the torque formula with $\tau_L = 40$ Nm:

$$\tau_L\omega_S x^2 + (2R_1\tau_L\omega_S - 3V_1^2)x + \tau_L\omega_S(R_1^2 + X^2) = 0$$
$$6283.2\,x^2 - 165942\,x + 37762 = 0$$

[2]

Roots: $x = 26.18\ \Omega$ or $x = 0.2296\ \Omega$. [1]

These give $s = 0.4/26.18 = 0.01528$ or $s = 0.4/0.2296 = 1.742$. The second is outside $0 < s \leq 1$ (it is braking, with the rotor turning backwards). Choose the larger x, which is the small slip on the stable, steep part of the curve. [1]

$s =$ **0.01528**, $n = 1500(1 - 0.01528) =$ **1477.1 rpm**. [2]

(c) Linear: $s \approx 0.02 \times 40/51.47 = 0.01554$, $n \approx$ **1476.7 rpm**. [2] The error is only 0.4 rpm, because both points are at low slip where $\tau \propto s$ (Lecture 7.8.1). This approximation gets worse near maximum torque. [1]

(d) With $V_1 = 0.9 \times 239.6$, $3V_1^2$ falls by 0.81 to 139,502. The new quadratic is $6283.2x^2 - 133219x + 37762 = 0$, giving $x = 20.92\ \Omega$, $s = 0.01912$, $n =$ **1471.3 rpm**. [3] Estimate: at low slip $\tau \propto V^2 s$, so $s \approx 0.01528/0.81 = 0.01886$, $n \approx 1471.7$ rpm. Close. [1]

(e) $s = 1$: $\tau_{st} = \dfrac{172225 \times 0.4}{157.08\left[0.9^2 + 2.4^2\right]} =$ **66.75 Nm** > 40 Nm, so it **can** start the load.
$I_{st} = 239.6/|0.9 + j2.4| = 93.48$ A. Running current at 40 Nm:
$239.6/|0.5 + 26.18 + j2.4| = 8.944$ A. Ratio $= 10.45$. [2]

New Q4 (20 marks, hard: a full Q3-style chain).

(a) Blocked rotor, per phase: $V = 75/\sqrt{3} = 43.30$ V, $I = 8$ A, $P = 173.3$ W.
$R = 173.3/8^2 = 2.708\ \Omega$, so $R_2' = 2.708 - 1.2 =$ **1.508 Ω**. $S = 346.4$ VA,

$Q = \sqrt{346.4^2 - 173.3^2} = 299.9$ VAr, $X = 299.9/64 =$ **4.686 Ω**. [2]

No load, per phase: $V = 239.6$ V, $I = 4.0$ A, $P = 86.67$ W. $R_C = 239.6^2/86.67 =$ **662.4 Ω**. $S = 958.4$ VA, $Q = 954.5$ VAr, $X_M = 239.6^2/954.5 =$ **60.15 Ω**. [2]

(b) $p = 3000/1440 = 2.08$, so $p = 2$: **4 poles**, $n_S =$ **1500 rpm**, $s = 60/1500 =$ **0.04**. Ideal efficiency $1 - s =$ **96%**. [3]

(c) $R_2'/s = 1.508/0.04 = 37.71\ \Omega$.

> New Q4 circuit at rated speed: 239.6 V, 662.4 ohm parallel j60.15 ohm, then 1.2 ohm, j4.686 ohm and 37.71 ohm

$$Z_1 = (1.2 + 37.71) + j4.686 = 38.91 + j4.686 = 39.19\angle 6.868^\circ\ \Omega$$
$$I_2' = 239.6/39.19\angle 6.868^\circ = 6.114\angle -6.868^\circ = 6.070 - j0.7311\ \text{A}$$
$$I_M = 239.6/662.4 - j\,239.6/60.15 = 0.3617 - j3.984\ \text{A}$$
$$I_1 = 6.432 - j4.715 = \mathbf{7.975\angle -36.24^\circ\ A}$$

[3]

- pf $= \cos 36.24^\circ =$ **0.8065 lag** [1]
- $P_{IN} = 3 \times 239.6 \times 7.975 \times 0.8065 =$ **4.623 kW** [1]
- $P_{AG} = 3 \times 6.114^2 \times 37.71 = 4229$ W, $\tau = 4229/157.08 =$ **26.92 Nm** [1]
- $P_{OUT} = 26.92 \times 150.80 =$ **4.059 kW**, $\eta = 4.059/4.623 =$ **87.81%** [1]

(Check: $P_{CU} = 3 \times 6.114^2 \times 2.708 = 303.7$ W, $P_{FE} = 260$ W,
$4623 - 304 - 260 = 4059$ W (checks).)

(d) $s = (1500 - 1560)/1500 = -0.04$, $R_2'/s = -37.71\ \Omega$. $Z_1 = -36.51 + j4.686$,
$I_2' = 6.509\angle -172.7^\circ$ A $= -6.457 - j0.8288$ A.
$I_1 = I_2' + I_M = -6.095 - j4.812 = 7.766\angle -141.7^\circ$ A. [1]

- $P_{AG} = 3 \times 6.509^2 \times (-37.71) = -4793$ W, $\tau = -4793/157.08 =$ **−30.52 Nm**. So **30.52 Nm** must be applied. [1]
- $P_{IN} = 3 \times 239.6 \times 7.766 \cos(-141.7^\circ) = -4.381$ kW. So **4.381 kW** is delivered to the supply. [1]
- Mechanical input $= 30.52 \times (2\pi \times 1560/60) = 30.52 \times 163.36 = 4.985$ kW. $\eta_{gen} = 4.381/4.985 =$ **87.88%**. [1]

(e) The ideal efficiency $1 - s$ only accounts for rotor copper loss (sP_{AG}). The real machine also has stator copper loss (here $3I_2'^2R_1 = 134.6$ W) and iron loss (260 W), plus friction and windage, which the model does not include. [2]

6. Quick recall check

Question

1. Star: $415/\sqrt{3} = 239.6$ V. Delta: 415 V.
2. Divide by ω_S (synchronous). Multiply by ω (actual rotor speed).
3. s, R_2'/s, P_{IN} and τ are negative. Q_{IN} stays positive, because the machine still needs magnetising VArs.
4. Rotor copper loss $= sP_{AG}$. Mechanical power $= (1-s)P_{AG}$.
5. The larger R_2'/s (smallest slip). It is on the stable, steep part of the curve near n_S.

Per-unit and synchronous reactance

Answers and worked solutions · Day 13 · Thu 5 Nov · pack M10

5. Practice

From your course files

PQ 8.1 (practice questions).

Single-phase, so there is no $\sqrt{3}$.

a. $I_B = S_B/V_B = 2000/240 = \mathbf{8.333}$ A

b. $Z_B = V_B/I_B = 240/8.333 = \mathbf{28.80\ \Omega}$ (also $V_B^2/S_B = 57600/2000 = 28.80\ \Omega$)

c. $R_B = Z_B = \mathbf{28.80\ \Omega}$

d. $X_B = Z_B = \mathbf{28.80\ \Omega}$

Resistance, reactance and impedance all share one base. This matches the course answers.

PQ 8.2 (practice questions).

Convert to pu ($Z_B = 28.80\ \Omega$): $R = 25/28.80 = 0.8681$ pu, $X = 125/28.80 = 4.340$ pu.

(a) $\mathbf{Z} = \dfrac{1}{1/0.8681 + 1/(j4.340)} = 0.8512\angle 11.31^\circ$ pu. (In ohms this is $24.51\angle 11.31^\circ\ \Omega$, and $24.51/28.80$ gives the same value.)

(b) $\mathbf{V} = 240/240 = 1\angle 0^\circ$ pu

(c) $\mathbf{I} = \mathbf{V}/\mathbf{Z} = 1/0.8512\angle 11.31^\circ = \mathbf{1.175\angle -11.31^\circ}$ **pu**. In amps that is $1.175 \times 8.333 = 9.79$ A.

(d) pf $= \cos(11.31^\circ) = \mathbf{0.9806}$ **lag** (the current lags the voltage).

(e) $\mathbf{S} = \mathbf{V}\mathbf{I}^* = 1 \times 1.175\angle +11.31^\circ = \mathbf{1.175\angle 11.31^\circ = 1.152 + j0.2304}$ **pu**

(f) $P = 1.152 \times 2000 = \mathbf{2304}$ W. Check with method (iii): $P = V^2/R = 1^2/0.8681 = 1.152$ pu. You cannot use I^2R because I is the total current, not the resistor current.

(g) $Q = 0.2304 \times 2000 = +\mathbf{460.8}$ VAr (the course gives 460.9 from rounding).

(h) $S = 1.175 \times 2000 = \mathbf{2350}$ VA

The course gives $\mathbf{Z} = 0.8510\angle 11.31^\circ$ pu, which is $24.51/28.80$ with rounded inputs. The unrounded value is 0.8512. Both round to the same current.

PQ 8.3 (practice questions).

Bases: $I_B = 1000/200 = 5$ A. $Z_B = 200/5 = 40\ \Omega$.

Convert: $\mathbf{Z}_1 = 100/40 = 2.5\angle 0^\circ$ pu. $\mathbf{Z}_2 = (10 + j20)/40 = 0.25 + j0.5$ pu.
$\mathbf{V} = 200/200 = 1\angle 0^\circ$ pu.

SOLVE:

$$\mathbf{I}_1 = \frac{1}{2.5} = 0.4\angle 0^\circ \text{ pu}, \qquad \mathbf{I}_2 = \frac{1}{0.25 + j0.5} = 1.789\angle -63.43^\circ \text{ pu}$$

Convert back: $I_1 = 0.4 \times 5 = \mathbf{2.0}$ A, and $I_2 = 1.789 \times 5 = \mathbf{8.944}$ A.

Extra check against Q2.20(a): $\mathbf{S} = \mathbf{V}(\mathbf{I}_1 + \mathbf{I}_2)^* = 1.2 + j1.6$ pu, which is 1200 W and +1600 VAr. pf $= \cos(\tan^{-1}(1.6/1.2)) = 0.600$ lag. This matches the course answers.

PQ 8.4 (practice questions).

a. $V_{B,L} = \mathbf{11}$ kV (the chosen primary base)

b. $V_{B,ph} = 11/\sqrt{3} = \mathbf{6.351}$ kV

c. $I_B = \dfrac{10 \times 10^6}{\sqrt{3} \times 11 \times 10^3} = \mathbf{524.9}$ A

d. $Z_B = \dfrac{(11 \times 10^3)^2}{10 \times 10^6} = \mathbf{12.10\ \Omega}$. Check: $6351/524.9 = 12.10\ \Omega$.

(e), (f) $X_B = R_B = \mathbf{12.10\ \Omega}$

g. $P_B = \mathbf{10}$ MW, (h) $Q_B = \mathbf{10}$ MVAr, (i) $S_B = \mathbf{10}$ MVA

This matches the course answers.

PQ 8.5 (practice questions).

Each part has its own voltage base (the system's nominal line voltage), with the same S_B.

a. $Z_B = 3300^2/750 \times 10^3 = 14.52\ \Omega$. $X = j5/14.52 = \mathbf{j0.3444}$ pu.

b. $Z_B = 415^2/750 \times 10^3 = 0.2296\ \Omega$. $R = 0.05/0.2296 = \mathbf{0.2177}$ pu.

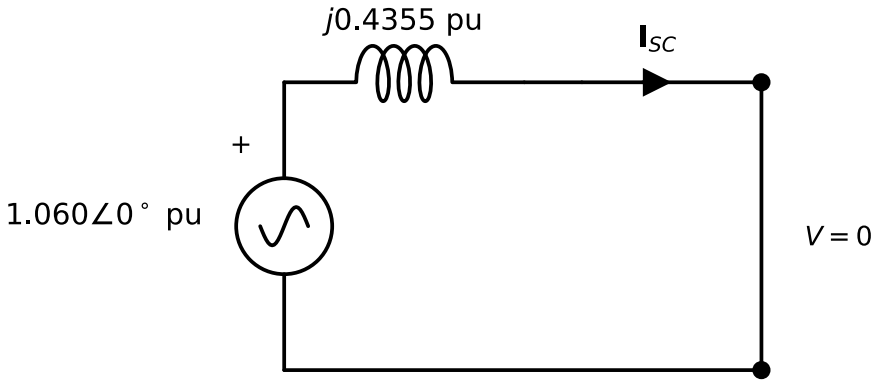
A tiny 0.05 Ω is a large 0.22 pu on a low-voltage system. That is the “perspective” the course talks about. This matches the course answers.

PQ 8.6 (practice questions).

(a) $S_B = 10$ kVA, $V_B = 415$ V (the rated voltage).

$$I_B = \frac{10 \times 10^3}{\sqrt{3} \times 415} = \mathbf{13.91}\ \text{A}, \qquad Z_B = \frac{415^2}{10 \times 10^3} = \mathbf{17.22}\ \boldsymbol{\Omega}$$

(b) $\mathbf{E} = 440/415 = \mathbf{1.060\angle 0^\circ}$ **pu**. $jX = j7.5/17.22 = \mathbf{j0.4355}$ **pu**. Draw a source of 1.060 pu in series with $j0.4355$ pu, with the output terminals shorted for part (c).



Practice Question 8.6 per-unit circuit: 1.060 pu behind j0.4355 pu with the terminals shorted

(c) $I_{SC} = 1.060/0.4355 = \mathbf{2.435}$ pu, so $I_{SC} = 2.435 \times 13.91 = \mathbf{33.87}$ A.

(d) Per phase: $E_1 = 440/\sqrt{3} = 254.0$ V. $I_{SC} = 254.0/7.5 = \mathbf{33.87}$ A, the same.

The course gives $j0.4353$ pu (from rounding Z_B up to 17.23). The unrounded value is 0.4355. The current answers agree exactly.

PQ 8.7 (practice questions).

Use $\mathbf{V} = V\angle 0^\circ$ and $\mathbf{I} = (S/V)\angle \pm \cos^{-1}(\text{pf})$. Use $-$ for lagging and $+$ for leading.

(a) $\mathbf{V} = 1\angle 0^\circ$ pu. $\cos^{-1} 0.9 = 25.84^\circ$. $\mathbf{I} = (1/1)\angle -25.84^\circ = \mathbf{1\angle -25.84^\circ}$ **pu**.

(b) $\mathbf{V} = 1.05\angle 0^\circ$ pu. $\mathbf{I} = (0.5/1.05)\angle +25.84^\circ = \mathbf{0.4762\angle +25.84^\circ}$ **pu**.

(c) $P = 100/150 = 0.6667$ pu. $S = 0.6667/0.85 = 0.7843$ pu. $\mathbf{V} = 1\angle 0^\circ$ pu. $\cos^{-1} 0.85 = 31.79^\circ$. $\mathbf{I} = \mathbf{0.7843\angle -31.79^\circ}$ **pu**. (The course gives 0.7844, a rounding difference.)

(d) $\mathbf{S} = (150 + j50)/200 = 0.75 + j0.25$ pu, so $|\mathbf{S}| = 0.7906$ pu. $\mathbf{V} = 0.95\angle 0^\circ$ pu.

$$\mathbf{I} = \left(\frac{\mathbf{S}}{\mathbf{V}}\right)^* = \frac{0.7906}{0.95}\angle -\tan^{-1}(0.25/0.75) = \mathbf{0.8322\angle -18.43^\circ}\ \text{pu}$$

The course gives −18.44°. The exact value of $\tan^{-1}(1/3)$ is 18.435°, so both roundings are fine.

Jun23 Q4a (past exam, 5 marks).

(i) Same I_F = 60 A for both readings.

$$E_{0,ph} = \frac{4600}{\sqrt{3}} = 2656\ \text{V}, \qquad |Z_S| = \frac{2656}{720} = 3.689\ \Omega$$

$$X_S = \sqrt{3.689^2 - 0.3^2} = \mathbf{3.676\ \Omega}$$

Stop here. The question asks for ohms. (For reference, $Z_B = 14^2/150 = 1.307\ \Omega$, so $X_S = 2.814$ pu.)

(ii) Rated voltage, so $\mathbf{V} = \mathbf{1.0\angle 0^\circ}$ **pu**.

$$\mathbf{S} = \frac{100 + j50}{150} = 0.6667 + j0.3333\ \text{pu}, \quad |\mathbf{S}| = 0.7454\ \text{pu}$$

The generator delivers Q > 0, so its pf is lagging and the current lags:

$$\mathbf{I} = \left(\frac{\mathbf{S}}{\mathbf{V}}\right)^* = 0.7454\angle -\tan^{-1}(0.5) = \mathbf{0.7454\angle -26.57^\circ}\ \text{pu}$$

(In amps: $I_B = 150 \times 10^6/(\sqrt{3} \times 14 \times 10^3) = 6186$ A, so $I = 4611$ A.)

Course answer: $X_S = 3.676\ \Omega$; $\mathbf{V} = 1.0\angle 0^\circ$ pu, $\mathbf{I} = 0.7453\angle -26.57^\circ$ pu. The text copy of the answer file loses the minus sign, but the PDF shows −26.57°. $\sqrt{0.6667^2 + 0.3333^2} = 0.74536$, so 0.7454 is the correctly rounded value; the course truncates to 0.7453.

Jun24 Q4a (past exam, 5 marks).

(i) The OC voltage is a line value (the course convention):

$$|Z_S| = \frac{12000/\sqrt{3}}{6500} = 1.066\ \Omega, \qquad X_S = \sqrt{1.066^2 - 0.2^2} = 1.047\ \Omega$$

$$Z_B = \frac{(14 \times 10^3)^2}{400 \times 10^6} = 0.4900\ \Omega, \qquad X_S = \frac{1.047}{0.4900} = \mathbf{2.137}\ \text{pu}$$

(Also $R = 0.2/0.49 = 0.408$ pu.) The marks are for: $\sqrt{3}$ and $|Z_S|$ (1), removing R (1), and Z_B with the pu conversion (1).

(ii) $\mathbf{V} = \mathbf{0.97}\angle\mathbf{0}^\circ$ pu.

$P = 200/400 = 0.5$ pu. $S = 0.5/0.8 = 0.625$ pu. $I = 0.625/0.97 = 0.6443$ pu. The pf is lagging, so the angle is $-\cos^{-1} 0.8 = -36.87^\circ$.

$$\mathbf{I} = \mathbf{0.6443}\angle\mathbf{-36.87}^\circ \text{ pu}$$

Course answer: $X_S = 2.137$ pu ($X_S = 1.047\ \Omega$); $\mathbf{V} = 0.97\angle 0^\circ$ pu, $\mathbf{I} = 0.6443\angle -36.87^\circ$ pu. This matches exactly.

PQ 8.11 (practice questions; parts (a)–(b) here).

a. $Z_B = \frac{15^2}{250} = 0.9\ \Omega$. $|Z_S| = \frac{5750/\sqrt{3}}{2050} = 1.619\ \Omega$.
$X_S = \sqrt{1.619^2 - 0.15^2} = 1.612\ \Omega$, so $X_S = 1.791$ **pu**.

b. Full load means S = 1 pu at V = 1 pu, so $|I| = 1$ pu. 0.85 lag: $\mathbf{I} = 1\angle -31.79^\circ$ **pu**.

Both match the course answers (1.791 pu, $1\angle -31.79^\circ$ pu). Parts (c)–(f) are worked in M11.

PQ 8.12 (practice questions; parts (a)–(b) here).

a. $Z_B = \frac{16^2}{120} = 2.133\ \Omega$. $|Z_S| = \frac{4400/\sqrt{3}}{600} = 4.234\ \Omega$.
$X_S = \sqrt{4.234^2 - 0.1^2} = 4.233\ \Omega$, so $X_S = 1.984$ **pu**.

b. $P = 102/120 = 0.85$ pu. $S = 0.85/0.95 = 0.8947$ pu. At V = 1 pu:
$\mathbf{I} = 0.8947\angle -18.19^\circ$ **pu**.

Course answers: 1.984 pu, $0.8947\angle -18.2^\circ$ pu. Part (c) is worked in M11.

New questions

New Question 1 (easy) [4 marks].

a. $I_B = \dfrac{50 \times 10^6}{\sqrt{3} \times 11 \times 10^3} = \mathbf{2624}$ A. $Z_B = \dfrac{11^2}{50} = \mathbf{2.420\ \Omega}$.

b. $X = 0.8/2.420 = \mathbf{0.3306}$ pu.

c. Line: $0.95 \times 11 = \mathbf{10.45}$ kV. Phase: $10.45/\sqrt{3} = \mathbf{6.033}$ kV. The pu value is the same for both; only the base differs.

New Question 2 (easy–medium) [5 marks].

a. Take $I_F = 1$ A: $V_{OC} = 400$ V line, so $E_{0,ph} = 230.9$ V, and $I_{SC} = 80$ A.

$$|Z_S| = 230.9/80 = 2.887\ \Omega, \qquad X_S = \sqrt{2.887^2 - 0.05^2} = \mathbf{2.886\ \Omega}$$

$Z_B = 18^2/300 = 1.080\ \Omega$, so $X_S = 2.886/1.080 = \mathbf{2.673}$ pu.

b. $I_F = 18000/400 = \mathbf{45}$ A (this is the field current for $E_0 = 1$ pu).

c. $I_F = 1.8 \times 45 = \mathbf{81}$ A.

New Question 3 (medium) [4 marks].

$\mathbf{V} = 1.03\angle 0^\circ$ pu.

A generator that absorbs Q has $Q < 0$: $\mathbf{S} = (180 - j60)/250 = 0.72 - j0.24$ pu.
$|\mathbf{S}| = 0.7589$ pu.

$$\mathbf{I} = \left(\frac{\mathbf{S}}{\mathbf{V}}\right)^* = \frac{0.7589}{1.03}\angle +\tan^{-1}(0.24/0.72) = \mathbf{0.7368\angle +18.43^\circ}\ \text{pu}$$

pf $= \cos 18.43^\circ = \mathbf{0.9487}$ **leading** (the current leads the voltage, as for a generator absorbing VArs).

New Question 4 (exam-style, chained) [9 marks].

(a) Bases: $I_B = 180 \times 10^6/(\sqrt{3} \times 15 \times 10^3) = 6928$ A. $Z_B = 15^2/180 = 1.250\ \Omega$.

$$|Z_S| = \frac{6000/\sqrt{3}}{2400} = 1.443\ \Omega, \qquad X_S = \sqrt{1.443^2 - 0.12^2} = 1.438\ \Omega$$

$$X_S = 1.438/1.250 = \mathbf{1.151}\ \text{pu}, \qquad R = 0.12/1.250 = \mathbf{0.096}\ \text{pu}$$

(b) $\mathbf{V} = \mathbf{1.02\angle 0^\circ}$ **pu**. $P = 135/180 = 0.75$ pu, so $S = 0.75/0.9 = 0.8333$ pu.

$$\mathbf{I} = \frac{0.8333}{1.02}\angle -25.84^\circ = \mathbf{0.8170\angle -25.84^\circ}\ \text{pu}, \qquad I = 0.8170 \times 6928 = \mathbf{5660}\ \text{A}$$

(c) Generator convention:

New Question 4 circuit in pu: E0 behind 0.096 pu and j1.151 pu, V = 1.02 pu, I = 0.8170 at -25.84 degrees out

$$\mathbf{E}_0 = 1.02 + 0.8170\angle -25.84^\circ \times (0.096 + j1.151) = \mathbf{1.706\angle 28.42^\circ}\ \text{pu}$$

Line value: $1.706 \times 15 = \mathbf{25.59}$ kV.

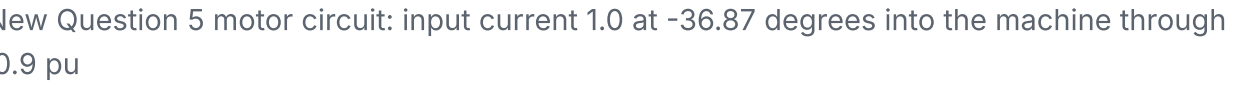
Excitation scale: $I_F(E_0 = 1 \text{ pu}) = 80 \times 15/6.0 = 200$ A. So $I_F = 1.706 \times 200 = \mathbf{341}$ A.

(If you neglect R, you get $\mathbf{E}_0 = 1.661\angle 30.62^\circ$ pu. Only neglect R when the question says so.)

New Question 5 (motor convention) [5 marks].

(a) $I_B = 500 \times 10^3/(\sqrt{3} \times 415) = 695.6$ A. $P = 400/500 = 0.8$ pu, so $S = 0.8/0.8 = 1.0$ pu. The input current lags:

$$\mathbf{I}_m = \frac{1.0}{1.0}\angle -36.87^\circ = \mathbf{1.0\angle -36.87^\circ} \text{ pu} = 0.8 - j0.6 \text{ pu}, \qquad I_m = \mathbf{695.6} \text{ A}$$



New Question 5 motor circuit: input current 1.0 at -36.87 degrees into the machine through j0.9 pu

(b) $\mathbf{E}_0 = \mathbf{V} - jX_S\mathbf{I}_m = 1 - j0.9(0.8 - j0.6) = 1 - 0.54 - j0.72 = 0.46 - j0.72$

$$\mathbf{E}_0 = \mathbf{0.8544\angle -57.43^\circ} \text{ pu}, \qquad E_0 = 0.8544 \times 415 = \mathbf{354.6} \text{ V (line)}$$

(c) $\delta = -57.43^\circ < 0$, so $\mathbf{E}_0$ lags $\mathbf{V}$, which confirms motoring. Also $E_0 < V$, so the machine is under-excited, which goes with a lagging input pf (see M11).

6. Quick recall check

Question

1. $Z_B = 16^2/120 = 2.133\ \Omega$.
2. $|Z_S| = (4400/\sqrt{3})/600 = 4.234\ \Omega$ (this is Q8.12's value).
3. $S = 0.6/0.8 = 0.75$ pu, so $\mathbf{I} = 0.75\angle +36.87^\circ$ pu.
4. Generator: $\mathbf{E}_0 = \mathbf{V} + \mathbf{I}(R + jX_S)$. Motor: $\mathbf{E}_0 = \mathbf{V} - \mathbf{I}_m(R + jX_S)$, with $\delta < 0$.
5. In the SC test the stator field is nearly equal and opposite to the rotor field, so the net flux is small and the iron does not saturate. In the OC test there is no stator current, so the full rotor field is in the iron.

Synchronous phasors and the power angle

Answers and worked solutions · Day 14 · Fri 6 Nov · pack M11

5. Practice

From your course files

PQ 8.8 (practice questions).

a. $V = 1\angle 0°$. Rated VA at unity pf gives $I = 1\angle 0°$ pu.

$$E_0 = 1 + j1.8(1\angle 0°) = 1 + j1.8 = 2.059\angle 60.95° \text{ pu}$$

Excitation = 2.059 pu.

b. **Rotor angle $\delta = 60.95°$.**

c. $P_{max} = \frac{VE_0}{X_S} = \frac{1\times 2.059}{1.8}$ = **1.144 pu** at $\delta = 90°$.

All three match the course answers.

Jun24 Q4b (past exam, 6 marks).

i. [3] $V = 1\angle 0°$ pu. Rated VA means S = 1 pu, so $|I| = 1$ pu. Leading: $\phi = +\cos^{-1} 0.9 = +25.84°$. So $I = 1\angle +25.84°$ pu.

$$E_0 = V + jX_S I = 1\angle 0° + (1.8\angle 90°)(1\angle 25.84°) = 1 + 1.8\angle 115.84° = 0.2155 + j1.620$$

$$E_0 = \mathbf{1.634\angle 82.42°} \text{ pu}$$

ii. [1] **Rotor angle $\delta = 82.42°$.** This is very close to the 90° limit, because the machine is under-excited.

iii. [2] With E_0 and V fixed, $P_{max} = \frac{VE_0}{X_S} = \frac{1\times 1.634}{1.8}$ = **0.9077 pu**, at a **rotor angle of 90°**.

(See Figure 1.) Unrounded working gives 82.43° and 0.9079 pu. The course's 82.42° and 0.9077 pu come from rounding E_0 to 1.634 first. Either is fine.

PQ 8.9 (practice questions).

Given $E_0 = 1.5$ pu, $V = 1.02$ pu, $X_S = 1.8$ pu, $\delta = 70°$:

$$P = \frac{VE_0}{X_S}\sin\delta = \frac{1.02 \times 1.5}{1.8}\sin 70° = 0.7987 \text{ pu}$$

$P = 0.7987 \times 200 =$ **159.7 MW**. This matches the course answer.

PQ 8.11 (practice questions).

a. $Z_B = \frac{15^2}{250} = 0.9\ \Omega$. $|Z_S| = \frac{5750/\sqrt{3}}{2050} = 1.619\ \Omega$.
$X_S = \sqrt{1.619^2 - 0.15^2} = 1.612\ \Omega$, so $X_S = 1.791$ **pu**. $R = 0.15/0.9 = 0.1667$ pu.

b. Full load means S = 1 pu at V = 1 pu, so $|I| = 1$ pu. 0.85 lag: $I = 1\angle -31.79^\circ$ **pu**.

c. $E_0 = 1 + (1\angle -31.79^\circ)(0.1667 + j1.791) = 1 + 1.799\angle 52.89^\circ = 2.531\angle 34.53^\circ$ **pu**.

d. $E_{0,line} = 2.531 \times 15 =$ **37.97 kV**.

e. On the air-gap line, 50 A gives 5.75 kV, so $I_F = 37.97\ \text{kV} \times \frac{50}{5.75\ \text{kV}} =$ **330.2 A**.

f. $\delta = 34.53^\circ$. All six match the course answers.

PQ 8.12 (practice questions).

a. $Z_B = \frac{16^2}{120} = 2.133\ \Omega$. $|Z_S| = \frac{4400/\sqrt{3}}{600} = 4.234\ \Omega$.
$X_S = \sqrt{4.234^2 - 0.1^2} = 4.233\ \Omega$, so $X_S = 1.984$ **pu**. $R = 0.1/2.133 = 0.0469$ pu.

b. $P = 102/120 = 0.85$ pu. $S = 0.85/0.95 = 0.8947$ pu. At V = 1 pu:
$I = 0.8947\angle -18.19^\circ$ **pu**.

c. $E_0 = 1 + (0.8947\angle -18.19^\circ)(0.0469 + j1.984) = 2.311\angle 46.39^\circ$ pu.

The synchronising field current (OC voltage = 16 kV) is $I_F = 50 \times \frac{16}{4.4} = 181.8$ A.

So $I_F = 2.311 \times 181.8 =$ **420.2 A**, and the **ratio is 2.31 : 1** ($= |E_0|$ in pu).

The course gives 420.4 A and $2.312\angle 46.40^\circ$. The 0.2 A difference is only rounding: with unrounded values I get $E_0 = 2.3112$ pu, and $2.3112 \times 181.8 = 420.2$ A. Either is acceptable.

PQ 8.13 (practice questions).

$P_E = 270/0.92 = 293.5$ kW.

$|Q|_{max} = \sqrt{330^2 - 293.5^2} =$ **150.9 kVAr**. It can be either sign: −150.9 kVAr over-excited (supplying Q for PFC) or +150.9 kVAr under-excited. This matches the course answer.

PQ 8.14 (practice questions).

This is worked in full as Example 3 (with the plant figure). The key lines:

1. Induction motor: $P = 200/0.9 = 222.2$ kW, $S = 222.2/0.82 = 271.0$ kVA, $Q = +155.1$ kVAr.
2. Synchronous motor: $P = 80$ kW, $Q = -\sqrt{100^2 - 80^2} = -60$ kVAr (over-excited).
3. Totals: $P = 302.2$ kW, $Q = +95.1$ kVAr, so $\text{pf} = \cos(\tan^{-1}(95.1/302.2)) =$ **0.9539 lag**.

This matches the course answer.

Jun23 Q4b (past exam, 7 marks).

i. [2] Remaining induction motor: $P_{IM} = 200/0.85 =$ **235.3 kW**. $S = 235.3/0.83 = 283.5$ kVA. $Q_{IM} = 283.5\sin(\cos^{-1}0.83) =$ **+158.1 kVAr** (lagging).

ii. [2] Synchronous motor: $P_{SM} = 200/0.90 =$ **222.2 kW**. The most leading Q within the 250 kVA rating is $Q_{SM} = -\sqrt{250^2 - 222.2^2} = -114.5$ kVAr. So the **capability is 114.6 kVAr** (supplied, leading).

iii. [3] $P_{tot} = 235.3 + 222.2 = 457.5$ kW. $Q_{tot} = 158.1 - 114.5 = 43.6$ kVAr.

$$\text{pf} = \cos\left(\tan^{-1}\frac{43.6}{457.5}\right) = \mathbf{0.9955\ lag}$$

All the parts match the answer sheet. The sheet's 114.6 kVAr is a rounding of 114.53.

PQ 8.15 (practice questions).

- Existing load: $P = 2760 \times 0.71 = 1960$ kW, $Q = 2760\sin(\cos^{-1}0.71) = +1944$ kVAr.
- Induction motors (800 kW mechanical): $P = 800/0.89 = 898.9$ kW, $Q = 898.9\tan(\cos^{-1}0.75) = +792.7$ kVAr.
- Synchronous motors (the remaining 700 kW mechanical): $P = 700/0.93 = 752.7$ kW, $Q = -\sqrt{1350^2 - 752.7^2} = -1121$ kVAr.
- Totals: $P = 3611$ kW (course: 3612 kW), $Q = 1616$ kVAr.

$\text{pf} = \cos(\tan^{-1}(1616/3611)) =$ **0.9128 lag**. The course says 0.9129. That is the last-digit rounding of its rounded totals.

PQ 8.10 (practice questions).

a. $I_B = \frac{150\text{ M}}{\sqrt{3}\times 15\text{ k}} =$ **5774 A**. $Z_B = 15^2/150 =$ **1.50 Ω**.

b. At $I_F = 1$ A: $E_{ph} = 500/\sqrt{3} = 288.7$ V and $I_{SC} = 100$ A, so $|Z_S| = 2.887\ \Omega$. $X_S = \sqrt{2.887^2 - 0.3^2} = 2.871\ \Omega$ = **1.914 pu**.

c. Synchronising needs $E_0 = 15$ kV line, so $I_F = 15000/500 =$ **30 A**.

d. **1 pu** (by definition).

e. $I_F = 1.65 \times 30 =$ **49.5 A**. All match the course answers.

Jun23 Q4c (past exam, 8 marks).

These are built from the answer sheets' points. The sheet says you do not need every point for full marks. Aim for about one solid point per mark.

(i) Power-factor correction [3]

- Many loads, such as induction motors, draw reactive power as well as real power. Their current lags the voltage.
- Reactive power is energy oscillating back and forth between the source and the load. It does no useful work, but it adds to the current.
- The extra current loads up generators, transformers and cables, causes extra resistive losses and voltage drop, and raises the VA demand that the supply company charges for.
- Correction means supplying the reactive power locally. Capacitors, or an over-excited synchronous motor, are connected in parallel with the load and generate reactive power. The supply then provides mostly real power, and the pf moves towards unity.
- The useful result: lower current and losses for the same useful power, freed-up capacity, and lower tariffs.

(ii) Transformers vs induction machines [3]

- Similarities: both can run from three-phase AC, have primary (stator) and secondary (rotor) windings on a magnetic core, and transfer power by mutual induction. Their equivalent circuits have the same form: winding resistances, leakage reactances, a magnetising reactance and a core-loss resistance.
- Differences: the induction machine's secondary is a rotating, short-circuited winding. Its rotor frequency is the slip frequency, not the stator frequency. It has an air gap, so its magnetising current is much larger.
- In the induction-machine equivalent circuit, the referred rotor resistance is divided by the slip. That term represents the mechanical output plus the rotor loss.

(iii) Perfectly conducting rotor cage [2]

- The referred rotor resistance would be zero.

- The torque–speed curve would be vertical at synchronous speed, like a synchronous machine. The machine could only produce torque at synchronous speed.
- There would be no rotor copper losses, so the efficiency would be higher.
- The starting torque would be zero, so the motor could not start by itself.

Jun24 Q4c (past exam, 9 marks).

These are built from the answer sheets' points. The sheet says you do not need every point for full marks. Aim for about one solid point per mark.

(i) Real, reactive and apparent power [3]

- Real power: produced by the part of the current that is in phase with the voltage. It is the average of the instantaneous power, it is energy that flows one way from source to load, and it does useful work. Resistors absorb it.
- Reactive power: produced by the part of the current that is 90° out of phase with the voltage. Its average is zero, because energy oscillates between source and load each cycle, so it does no useful work. Inductors absorb it and capacitors generate it. It still adds to the current and the losses.
- Both real and reactive power are scalar quantities.
- Apparent power: the magnitude of the complex power, whose real part is the real power and whose imaginary part is the reactive power. It is the product of rms voltage and total current, so it sets the current and the ratings (VA) of generators, transformers and cables.

(ii) Transformer losses [3]

- Copper (winding) loss: resistive heating in the primary and secondary windings. It is modelled by the series winding resistances. It is proportional to current squared, so it rises with the square of the output power.
- Iron (core) loss: hysteresis and eddy-current loss in the core, caused by the alternating flux. It is modelled by the parallel core-loss resistance. It depends on the voltage and frequency, not the load, so it is roughly constant with output power.

(iii) Why induction-machine no-load current is not zero [3]

- Magnetising current is needed to set up the rotating air-gap flux. Because there is an air gap, this current is large. It is modelled by the magnetising reactance.
- Core-loss current supplies the hysteresis and eddy-current losses in the stator and rotor iron. It is modelled by the core-loss resistance.
- Even at no load, the rotor needs some torque to overcome mechanical losses: bearing friction, and windage (air drag and the cooling fan). So it runs slightly below synchronous speed with a small rotor current. That draws a little real power.
- So the no-load current is mostly reactive (a low pf), plus a small real component.

Practice Questions, Ch8 short answers (j) and (k) (bonus).

- j. Reactive power capability is limited mainly by the stator VA rating, because of stator winding heating. At a real power P, the most Q available is the square root of (rated S squared minus P squared). At full rated real power, no reactive power is left. When absorbing Q (under-excited), the machine is also limited by the rotor angle approaching the stability limit.
- k. Industrial plant pf is usually lagging, because most loads are induction motors that need magnetising current. Synchronous motors are used because, when over-excited, they draw leading current and supply reactive power. That corrects the plant pf and lowers the VA charges, while also driving a constant-speed load.

New questions

N1.

With P = 0, the current is at ±90° to V and $\delta = 0$, so everything is in line.

a. Supplying Q from the machine means a lagging output current: $I = 0.8\angle -90^\circ = -j0.8$ pu.

$E_0 = 1 + j1.5(-j0.8) = 1 + 1.2 =$ **2.2 pu (over-excited)**.

Check with the derived formula: $Q = (1 \times 2.2 - 1)/1.5 = 0.8$ pu.

b. Absorbing Q means a leading output current: $I = +j0.5$, so $E_0 = 1 + j1.5(j0.5) = 1 - 0.75 =$ **0.25 pu (under-excited)**.

N2.

a. Per phase: $P_{ph} = \frac{(11000/\sqrt{3})(14000/\sqrt{3})}{2.4}\sin 25^\circ = 9.039$ MW. So **P = 3 × 9.039 = 27.12 MW**.

With line quantities: $P = \frac{11000 \times 14000}{2.4}\sin 25^\circ = 27.12$ MW, the same.

b. $P_{max} = \frac{11000 \times 14000}{2.4} =$ **64.17 MW** at $\delta = 90^\circ$.

In pu: $Z_B = 11^2/50 = 2.42\ \Omega$, so $X_S = 0.9917$ pu and $E_0 = 14/11 = 1.273$ pu. Then $P_{max} = \frac{1 \times 1.273}{0.9917} =$ **1.283 pu** (= 64.17/50). The pu formula already gives three-phase power.

N3.

a. $I = 0.8\angle -31.79^\circ$ pu. $E_0 = 1 + j1.6(0.8\angle -31.79^\circ) = 1 + 1.28\angle 58.21^\circ = 1.997\angle 33.02^\circ$ **pu**.

b. Same P = 0.8 × 0.85 = 0.68 pu, so $|I|$ stays 0.8 pu, but now $I = 0.8\angle +31.79^\circ$.

$E_0 = 1 + 1.28\angle 121.79^\circ = 1.136\angle 73.33^\circ$ **pu**.

c. Lagging: $P_{max} = 1.997/1.6 =$ **1.248 pu**. Leading: $P_{max} = 1.136/1.6 =$ **0.710 pu**.

Check: $\frac{1.136}{1.6}\sin 73.33^\circ = 0.68$ pu, as required.

Comment: reducing the excitation (to absorb Q) cuts P_{max} from 1.25 to 0.71 pu, and pushes δ from 33° to 73°. That is right at the typical 75° operating limit. The machine has only about 4% power margin before pull-out, compared with about 84% when lagging.

N4.

a. $E_{0,min} = \frac{PX_S}{V\sin 75^\circ} = \frac{0.6\times 1.8}{\sin 75^\circ} =$ **1.118 pu**.

b. $I = \frac{E_0 - V}{jX_S} = \frac{1.118\angle 75^\circ - 1}{j1.8} = 0.718\angle +33.34^\circ$ **pu** (leading).

$S = 0.718$ pu. $Q = S\sin(-\phi) = 0.718\sin(-33.34^\circ) =$ **−0.395 pu**, so the machine absorbs 0.395 pu.

Check with the derived formula: $Q = \frac{1.118\cos 75^\circ - 1}{1.8} = -0.395$ pu.

pf = 0.6/0.718 = **0.835 leading**.

N5.

a. $P_E = 300/0.95 =$ **315.8 kW**. $Q_{SM} = -\sqrt{400^2 - 315.8^2} =$ **−245.5 kVAr** (supplied).

b. Plant: $Q_p = 600\tan(\cos^{-1}0.75) = +529.2$ kVAr.

$P_{tot} = 915.8$ kW, $Q_{tot} = 529.2 - 245.5 = 283.6$ kVAr.

pf = $\cos(\tan^{-1}(283.6/915.8)) =$ **0.9552 lag** (it was 0.75 before).

N5(c) motor circuit: input current 1.000 at +37.86 degrees through j1.0 pu

c. $S_m = (315.8 - j245.5)/400 = 0.7895 - j0.6138$ pu, with $|S_m| = 1$ pu. At V = 1 pu, $I_m = (S_m/V)^* = 1.000\angle +37.86^\circ$ **pu** (leading). That is the rated current, $\frac{400\text{ k}}{\sqrt{3}\times 415} =$ **556.5 A**.

$E_0 = V - jX_S I_m = 1 - (1\angle 90^\circ)(1\angle 37.86^\circ) = 1.797\angle -26.07^\circ$ **pu**, which is $1.797 \times 415 =$ **745.6 V (line)**. δ is negative, so it is motoring.

d. Unity pf: $I_m = 0.7895\angle 0^\circ$, so $E_0 = 1 - j0.7895 = 1.274\angle -38.29^\circ$ **pu**.

The motor now contributes Q = 0, so plant pf = $\cos(\tan^{-1}(529.2/915.8)) =$ **0.866 lag**. Lower excitation means less correction and also a larger $|\delta|$.

6. Quick recall check

Question

1. At $\delta = 90^\circ$ (the theoretical stability limit). Past it, the machine loses synchronism (pull-out), with large currents and oscillating torques. In practice it is run at about 75° or less.
2. Under-excited. It absorbs Q.
3. Three-phase. Multiply by S_{base} to get watts.
4. $\sqrt{100^2 - 60^2} = 80$ kVAr, which is −80 kVAr in the course sign convention.
5. At the bottom of the V. The pf there is unity (Q = 0), so $|I| = P/V$.

ELECTRICAL MACHINES · PAPER

Jun 2023 exam: full worked solutions

Answers and worked solutions · Day 18 · Tue 10 Nov · pack MX1

Question 1. AC and DC Power Circuit Analysis [20 marks]

1a) Battery Modelling [5 marks]

(i) With no current there is no drop across R_S, so

$$E = V_{oc} = \mathbf{13.8\ V}$$

(ii) KVL around the loop: $V = E - IR_S$, so

$$R_S = \frac{E - V}{I} = \frac{13.8 - 9.5}{300} = \mathbf{0.01433\ \Omega}\ (14.33\ \text{m}\Omega)$$

$$P_{R_S} = I^2 R_S = 300^2 \times 0.01433 = \mathbf{1290\ W}$$

$$P_{motor} = VI = 9.5 \times 300 = \mathbf{2850\ W}$$

Check: $EI = 13.8 \times 300 = 4140$ W $= 1290 + 2850$ W.

(iii)

$$V = E - IR_S = 13.8 - 20 \times 0.01433 = \mathbf{13.51\ V}$$

1b) Star-Delta Equivalent Circuits [6 marks]

(i) Star. Each resistor sees the phase voltage.

$$V_1 = \frac{V_L}{\sqrt{3}} = \frac{460}{\sqrt{3}} = 265.6\ \text{V}$$

$$I_1 = \frac{V_1}{R} = \frac{265.6}{70} = \mathbf{3.794\ A}, \qquad I_L = I_1 = \mathbf{3.794\ A}$$

$$P = 3V_1 I_1 = 3 \times 265.6 \times 3.794 = \mathbf{3.023\ kW}$$

(ii) Delta. Each resistor sees the full line voltage.

$$I_1 = \frac{V_L}{R} = \frac{460}{70} = \mathbf{6.571\ A}, \qquad I_L = \sqrt{3}\, I_1 = \mathbf{11.38\ A}$$

$$P = 3V_L I_1 = 3 \times 460 \times 6.571 = \mathbf{9.069\ kW}$$

Check: the delta power is exactly 3 times the star power ($9.069/3.023 = 3$), as it must be for the same resistors.

1c) Phasor Calculations [9 marks]

Multiply magnitudes and add angles. $j = 1\angle 90^\circ$ and $-j = 1\angle -90^\circ$.

(i)

$$\mathbf{V}_R = R\,\mathbf{I}_1 = 10 \times 13\angle 20^\circ = \mathbf{130\angle 20^\circ\ V}$$

$$\mathbf{V}_L = jX_L\,\mathbf{I}_1 = 30\angle 90^\circ \times 13\angle 20^\circ = \mathbf{390\angle 110^\circ\ V}$$

$$\mathbf{V}_C = -jX_C\,\mathbf{I}_1 = 50\angle -90^\circ \times 13\angle 20^\circ = \mathbf{650\angle -70^\circ\ V}$$

(ii)

$$Z = R + jX_L - jX_C = 10 + j30 - j50 = 10 - j20 = \mathbf{22.36\angle -63.43^\circ\ \Omega}$$

$$\mathbf{V}_1 = Z\,\mathbf{I}_1 = 22.36\angle -63.43^\circ \times 13\angle 20^\circ = \mathbf{290.7\angle -43.43^\circ\ V}$$

Check by KVL: $\mathbf{V}_R + \mathbf{V}_L + \mathbf{V}_C = 211.1 - j199.9 = 290.7\angle -43.43^\circ$ V.

(iii)

$$\mathbf{I}_1^* = \mathbf{13\angle -20^\circ\ A}$$

$$S = \mathbf{V}_1\mathbf{I}_1^* = 290.7\angle -43.43^\circ \times 13\angle -20^\circ = 3779\angle -63.43^\circ = \mathbf{1690 - j3380\ VA}$$

$$P = \mathbf{1690\ W}, \qquad Q = \mathbf{-3380\ VAr}\ \text{(the load supplies reactive power: net capacitive)}$$

Check: $P = I^2R = 13^2 \times 10 = 1690$ W and
$Q = I^2(X_L - X_C) = 169 \times (-20) = -3380$ VAr. The power factor is
$\cos 63.43^\circ = 0.4472$ leading.

Question 2. Magnetic Circuits and Transformers [20 marks]

2a) Electromagnetics [9 marks]

(i) Use $\mathcal{R} = l/(\mu_0\mu_r A)$ with $A = 20 \times 10^{-4}$ m² and $\mu_0 = 4\pi \times 10^{-7}$ H/m.

$$\mathcal{R}_G = \frac{g}{\mu_0 A} = \frac{1.5 \times 10^{-3}}{4\pi \times 10^{-7} \times 20 \times 10^{-4}} = \mathbf{596.8\ kA/Wb}$$

$$\mathcal{R}_C = \frac{l_c}{\mu_0\mu_r A} = \frac{0.5}{4\pi \times 10^{-7} \times 3000 \times 20 \times 10^{-4}} = \mathbf{66.31\ kA/Wb}$$

The two are in series:

$$L = \frac{N^2}{\mathcal{R}_G + \mathcal{R}_C} = \frac{32^2}{663.1 \times 10^3} = \mathbf{1.544\ mH}$$

The gap is 1.5 mm out of 500 mm of path, but it has 9 times the reluctance of the core. (The course uses the full 50 cm for the core. Subtracting the gap changes $\mathcal{R}_C$ by 0.3% and does not change the answer at 4 figures.)

(ii) The energy density in the gap is $B^2/(2\mu_0)$ J/m³, and the gap volume is Ag.

$$W_g = \frac{\hat{B}^2}{2\mu_0} A\,g = \frac{1.8^2}{2 \times 4\pi \times 10^{-7}} \times 20 \times 10^{-4} \times 1.5 \times 10^{-3} = \mathbf{3.867\ J}$$

(The total energy $\frac{1}{2}LI^2$ at 1.8 T is 4.297 J. The question asks for the **gap only**, so 3.867 J.)

(iii) Split first: $P_h = 0.3 \times 12 = 3.6$ W and $P_e = 0.7 \times 12 = 8.4$ W. Then scale each with its own law, $P_h \propto \hat{B}^2 f$ and $P_e \propto \hat{B}^2 f^2$.

$$P_h' = 3.6 \times \left(\frac{1.2}{1.5}\right)^2 \times \frac{100}{60} = \mathbf{3.84\ W}$$

$$P_e' = 8.4 \times \left(\frac{1.2}{1.5}\right)^2 \times \left(\frac{100}{60}\right)^2 = \mathbf{14.93\ W}$$

$$P_{FE}' = 3.84 + 14.93 = \mathbf{18.77\ W}$$

2b) Transformer Parameters [5 marks]

All values are referred to the **primary**, because both tests were measured on the primary side.

SC test (series branch; the shunt branch is negligible at 27 V):

$$|Z| = \frac{V_{SC}}{I_{SC}} = \frac{27}{9.7} = 2.784\ \Omega$$

$$R = R_1 + R_2' = \frac{P_{SC}}{I_{SC}^2} = \frac{55}{9.7^2} = \mathbf{0.5845\ \Omega}$$

$$X = X_1 + X_2' = \sqrt{|Z|^2 - R^2} = \sqrt{2.784^2 - 0.5845^2} = \mathbf{2.721\ \Omega}$$

OC test (shunt branch at rated voltage):

$$R_C = \frac{V_{OC}^2}{P_{OC}} = \frac{240^2}{45} = \mathbf{1280\ \Omega}$$

$$Q_{OC} = \sqrt{(V_{OC}I_{OC})^2 - P_{OC}^2} = \sqrt{312^2 - 45^2} = 308.7\ \text{VAr}$$

$$X_M = \frac{V_{OC}^2}{Q_{OC}} = \frac{240^2}{308.7} = \mathbf{186.6\ \Omega}$$

Sanity check: rated primary current is $2400/240 = 10$ A, so the SC test was run near rated current, and the SC voltage is 11% of rated.

2c) Transformer Analysis [6 marks]

(i) KVL from the primary terminals, through the series branch, to the load. The shunt branch sits directly across $\mathbf{V}_1$, so it does not appear.

$$\boxed{\mathbf{V}_2' = \mathbf{V}_1 - \mathbf{I}_2' \left[(R_1 + R_2') + j(X_1 + X_2')\right]}$$

Use $\mathbf{I}_2'$, **not** $\mathbf{I}_1$. $\mathbf{I}_1$ also includes the current in R_C and X_M.

(ii) With R_L' connected, $\mathbf{V}_1$ is across the series branch plus the load, so

$$\boxed{\mathbf{V}_2' = \mathbf{V}_1 \frac{R_L'}{R_L' + R_1 + R_2' + j(X_1 + X_2')}}$$

(iii) Lagging.

- The series path has reactance $X_1 + X_2'$, so the load current $\mathbf{I}_2' = \mathbf{V}_1/(R_L' + R + jX)$ **lags** $\mathbf{V}_1$.
- The load is purely resistive, so $\mathbf{V}_2' = R_L'\mathbf{I}_2'$ is **in phase with** $\mathbf{I}_2'$.
- Therefore $\mathbf{V}_2'$ lags $\mathbf{V}_1$ by the angle $\tan^{-1}\left[X/(R_L' + R)\right]$.

(The official sheet writes $\mathbf{V}_1'$ for the primary voltage. On the primary side $\mathbf{V}_1' = \mathbf{V}_1$.)

Question 3. AC and Induction Machines [20 marks]

3a) AC Machines [6 marks]

(i) The rated speed must be just below a synchronous speed $n_S = 60f/p$ (with p = pole-pairs).

$$p = \frac{60f}{n} = \frac{3000}{560} = 5.36 \Rightarrow p = 5, \qquad n_S = \frac{60 \times 50}{5} = 600 \text{ rpm}$$

So $p = 5$ pole-pairs, **10 poles**.

(ii) Induction machine, because the rated speed (560 rpm) is not a synchronous speed. It is below $n_S = 600$ rpm, so it runs with slip. A synchronous machine runs at exactly n_S.

(iii) The rotor copper loss is sP_{AG}, so even with no other losses $P_{OUT} \leq (1 - s)P_{AG}$.

$$s = \frac{n_S - n}{n_S} = \frac{600 - 560}{600} = 0.06667, \qquad \eta_{max} = 1 - s = \frac{n}{n_S} = \frac{560}{600} = \mathbf{93.33\%}$$

(iv) On the linear part of the curve, $\tau \propto (n_S - n)$, the slip speed. Rated slip speed is $600 - 560 = 40$ rpm, so at one-quarter torque it is 10 rpm.

$$n = 600 - \frac{40}{4} = \mathbf{590\ rpm}$$

$$\frac{P}{P_{rated}} = \frac{\tau\,\omega}{\tau_{rated}\,\omega_{rated}} = 0.25 \times \frac{590}{560} = \mathbf{26.34\%}$$

3b) Induction Machine Torque vs Speed [5 marks]

(i) From the simplified circuit, maximum torque occurs where R_2'/s equals the magnitude of the rest of the series impedance:

$$s_{max} = \frac{R_2'}{\sqrt{R_1^2 + (X_1 + X_2')^2}}$$

For maximum torque at standstill, $s_{max} = 1$, so the total rotor resistance must equal that magnitude:

$$R_2' + R_{EXT}' = \sqrt{0.4^2 + (1.3 + 1.5)^2} = 2.828\ \Omega$$

$$R_{EXT}' = 2.828 - 0.5 = \mathbf{2.328\ \Omega}\ (\text{referred to the stator})$$

X_M and R_C are not needed: in the simplified circuit they sit across the supply and do not affect I_2'.

(ii) Key facts for the sketch:

- All three curves go to **zero torque at** n_S.
- They have the **same peak torque**. τ_{max} does not depend on R_2'.
- The peak moves to lower speed as resistance increases: $s_{max} = 0.177$ (peak at $0.82\,n_S$) for $R_{EXT} = 0$; $s_{max} = 0.588$ ($0.41\,n_S$) for 0.5 pu (1.164 Ω); $s_{max} = 1$ (peak **at** $n = 0$) for 1 pu.
- Starting torque rises with R_{EXT}. Relative to the 1 pu curve it is 37% (0 pu), 89% (0.5 pu) and 100% (1 pu).

Torque against speed for the Jun23 Q3b machine with three values of external rotor resistance (simplified circuit, torque in relative units).

3c) Induction Machine Torque Calculation [9 marks]

(i) $p = 60f/n = 3000/1450 = 2.07 \Rightarrow p = 2$ pole-pairs, **4 poles**.

$$n_S = \frac{60f}{p} = \frac{3000}{2} = \mathbf{1500\ rpm}$$

(ii) Generator, because $n = 1520 > n_S = 1500$ rpm, so $s = (1500 - 1520)/1500 = -0.01333 < 0$. The rotor is driven faster than the field, and R_2'/s is negative.

(iii) Per-phase quantities (star): $V_1 = 415/\sqrt{3} = 239.6$ V.

$$s = \frac{1500 - 1450}{1500} = 0.03333, \qquad \frac{R_2'}{s} = \frac{0.2}{0.03333} = 6.0\ \Omega, \qquad \omega_S = \frac{2\pi \times 1500}{60} = 157.08\ \text{rad/s}$$

Torque formula:

$$\tau = \frac{3V_1^2\,(R_2'/s)}{\omega_S\left[(R_1 + R_2'/s)^2 + (X_1 + X_2')^2\right]} = \frac{3 \times 239.6^2 \times 6.0}{157.08 \times \left[6.3^2 + 5.0^2\right]} = \mathbf{101.7\ Nm}$$

(iv) Ignoring the shunt branch, $\mathbf{I}_1 = \mathbf{I}_2'$:

$$\mathbf{I}_1 = \frac{\mathbf{V}_1}{(R_1 + R_2'/s) + j(X_1 + X_2')} = \frac{239.6\angle 0^\circ}{6.3 + j5.0} = \frac{239.6\angle 0^\circ}{8.043\angle 38.44^\circ} = \mathbf{29.79\angle -38.44^\circ\ A}$$

Check: $P_{AG} = 3I_2'^2 R_2'/s = 3 \times 29.79^2 \times 6.0 = 15.97$ kW and $\tau = P_{AG}/\omega_S = 101.7$ Nm, matching (iii).

Question 4. Synchronous Machines [20 marks]

4a) Synchronous Machines [5 marks]

(i) Use the OC and SC readings at the **same field current**, with the OC voltage converted to a phase value.

$$|Z_S| = \frac{V_{OC}/\sqrt{3}}{I_{SC}} = \frac{4600/\sqrt{3}}{720} = 3.689\ \Omega$$

$$X_S = \sqrt{|Z_S|^2 - R^2} = \sqrt{3.689^2 - 0.3^2} = \mathbf{3.676\ \Omega}$$

(For reference, $Z_B = 14^2/150 = 1.307\ \Omega$, so $X_S = 2.81$ pu. The question asks for ohms.)

(ii) Bases: $S_B = 150$ MVA, $V_B = 14$ kV. At rated voltage, take the terminal voltage as reference:

$$\mathbf{V} = \mathbf{1.0\angle 0^\circ\ pu}$$

$$S = \frac{100 + j50}{150} = 0.6667 + j0.3333\ \text{pu}$$

In per-unit, $S = \mathbf{V}\mathbf{I}^*$, so

$$\mathbf{I} = \left(\frac{S}{\mathbf{V}}\right)^* = 0.6667 - j0.3333 = \mathbf{0.7454\angle -26.57^\circ\ pu}$$

The current lags because the generator is delivering reactive power (pf 0.894 lag). In amps this is $0.7454 \times 6186 = 4611$ A.

4b) Power Factor Correction [7 marks]

(i) 200 kW is the **output**. Input:

$$P_{im} = \frac{P_{out}}{\eta} = \frac{200}{0.85} = \mathbf{235.3\ kW}$$

$$Q_{im} = P_{im}\tan(\cos^{-1}0.83) = 235.3 \times 0.6720 = \mathbf{158.1\ kVAr}\ \text{(absorbed)}$$

(ii)

$$P_{syn} = \frac{200}{0.90} = \mathbf{222.2\ kW}$$

The 250 kVA rating limits $|S|$. The rest of the rating is available as reactive power, which an over-excited synchronous motor can **supply** (leading pf):

$$Q_{syn,max} = \sqrt{S_{rated}^2 - P_{syn}^2} = \sqrt{250^2 - 222.2^2} = \mathbf{114.5\ kVAr}\ \text{(supplied)}$$

(iii) Add the plant's powers with signs (absorbed reactive power positive):

$$P = 235.3 + 222.2 = 457.5\ \text{kW}, \qquad Q = 158.1 - 114.5 = 43.6\ \text{kVAr}$$

$$\text{pf} = \frac{P}{\sqrt{P^2+Q^2}} = \frac{457.5}{\sqrt{457.5^2 + 43.6^2}} = \mathbf{0.9955\ lag}$$

It is still lagging, because the synchronous motor cannot cancel all 158.1 kVAr. For comparison, the original plant (two induction motors) ran at 0.83 lag.

4c) Short-Answer Questions [8 marks]

(i) Power-factor correction. Many loads, such as induction motors and transformers, draw reactive power as well as real power. Reactive power is energy that oscillates back and forth between the supply and the load each cycle. It does no useful work, but it increases the current. This extra current loads up generators, transformers and lines, uses up their rating, and causes extra copper losses and voltage drop. Power-factor correction means adding a source of reactive power close to the load, usually a capacitor bank connected in parallel with it (or an over-excited synchronous machine). The capacitor generates the reactive power the load needs, so less reactive power is drawn from the supply. The supply current

falls for the same real power, the power factor moves towards one, and losses and charges for poor power factor are reduced.

(ii) Transformers and induction machines. *Similarities:* both can operate from a three-phase AC supply. Both have a primary (stator) winding and a secondary (rotor) winding coupled by a magnetic core. Both have nearly the same equivalent circuit, with winding resistances, leakage reactances, a magnetising reactance and a core-loss resistance, and both can be analysed per phase with a simplified circuit. *Differences:* in an induction machine the secondary winding is on the rotor, which turns, and it is short-circuited rather than feeding a load. The rotor frequency is the slip times the stator frequency, so it differs from the supply frequency. An induction machine also has an airgap, so its magnetising current is much larger. In the induction machine equivalent circuit, the referred rotor resistance is divided by the slip. This term represents both the rotor copper loss and the mechanical output power.

(iii) Perfectly conducting rotor. The rotor resistance would be zero. There would be no rotor copper loss, so efficiency would be higher. The rotor would carry whatever current is needed at almost exactly synchronous speed. The torque–speed curve would become vertical at synchronous speed, like a synchronous machine, so the speed would hardly change with load. But the machine would produce no torque away from synchronous speed, so it would have zero starting torque and could not start on its own. (This is the limit of the Q3b sketch as rotor resistance goes to zero: the peak moves right up to synchronous speed.)

ELECTRICAL MACHINES · PAPER

Jun 2024 exam: full worked solutions

Answers and worked solutions · Day 21 · Fri 13 Nov · pack MX2

Question 1. AC and DC Power Circuit Analysis [20 marks]

1a) Fill-In-The Blank Questions [4 marks]

(i) The turns ratio is $n = N_1/N_2 = V_1/V_2$.

- $240/24$: $n = \mathbf{10}$ (10:1), **step down**.
- $240/480$: $n = \mathbf{0.5}$ (1:2), **step up**.

(ii)

$$P_{OUT} = P_{IN} - P_{LOSS} = 1000 - 300 = \mathbf{700\ W}, \qquad \eta = \frac{700}{1000} = \mathbf{70\,\%}$$

- More loss at the same input: η **decreases**.
- More input at the same loss: η **increases**, because the fixed loss becomes a smaller fraction of the input. For example, at 2000 W input, $\eta = 1700/2000 = 85\,\%$.

All answers match the course.

1b) Electrical and Mechanical Power [3 marks]

(i) The 415 V and 6.8 A are line values.

$$S = \sqrt{3}\,V_L I_L = \sqrt{3} \times 415 \times 6.8 = 4888\ \text{VA}$$

$$\text{pf} = \frac{P}{S} = \frac{4000}{4888} = \mathbf{0.8184\ lagging}$$

(ii)

$$\omega = \frac{2\pi n}{60} = \frac{2\pi \times 1420}{60} = 148.7\ \text{rad/s}$$

$$P_{OUT} = \tau\omega = 23 \times 148.7 = \mathbf{3.420\ kW}, \qquad \eta = \frac{3420}{4000} = \mathbf{85.50\,\%}$$

"4 pole, 50 Hz" is not needed here ($n_S = 1500$ rpm). All answers match the course.

1c) AC Circuit Analysis [5 marks]

Figure values. The paper's figure gives $R = 100\ \Omega$ and $jX_L = j40\ \Omega$. As a check, the official answers ($P_R = 400$ W and $Q_L = 160$ VAr at 2 A) give $R = 400/2^2 = 100\ \Omega$ and

$X_L = 160/2^2 = 40\ \Omega$.

(i) In a series circuit the same current flows through both elements.

Element	P	Q
Resistor	$I^2R = 2^2 \times 100 = \mathbf{400\ W}$	$\mathbf{0\ VAr}$
Inductor	$\mathbf{0\ W}$	$I^2X_L = 2^2 \times 40 = \mathbf{160\ VAr}$

(ii) V_R is in phase with I_1, and V_L leads I_1 by 90°. So the two phasors are perpendicular:

$$|V_1| = \sqrt{V_R^2 + V_L^2} = \sqrt{50^2 + 20^2} = \mathbf{53.85\ V}$$

(This is consistent with the figure values: $I_1 = 50/100 = 0.5$ A, so $V_L = 0.5 \times 40 = 20$ V.)

All answers match the course.

1d) Power Factor Correction [8 marks]

(i) [2] Power triangle: horizontal side $P = 80$ W, vertical side $Q = 60$ VAr (upwards, because the load is inductive), hypotenuse $S = 100$ VA. The angle between P and S is $-\phi = 36.87^\circ$.

$$S = \sqrt{80^2 + 60^2} = 100\ \text{VA}, \qquad \text{pf} = \frac{P}{S} = \frac{80}{100} = \mathbf{0.8\ lagging}$$

(ii) [4] An ideal capacitor absorbs no real power, so the supply real power is unchanged:

$$P_S = \mathbf{80\ W}$$

At 0.9 lagging, $-\phi = \cos^{-1} 0.9 = 25.84^\circ$:

$$Q_S = P_S \tan(25.84^\circ) = 80 \times 0.4843 = \mathbf{38.75\ VAr}$$

Conservation of reactive power: $Q_S = Q_{LOAD} + Q_C$, so

$$Q_C = 38.75 - 60 = \mathbf{-21.25\ VAr}$$

The negative sign means the capacitor generates reactive power.

(iii) [2] Use the new value, $Q_C = -30$ VAr. The capacitor sits directly across 115 V.

$$|Q_C| = \frac{V^2}{X_C} = \omega C V^2 \ \Rightarrow\ C = \frac{|Q_C|}{\omega V^2} = \frac{30}{2\pi \times 60 \times 115^2} = \mathbf{6.017\ \mu F}$$

All answers match the course.

Question 2. Magnetic Circuits and Transformers [20 marks]

2a) Electromagnetics [7 marks]

$$A = \pi r^2 = \pi(0.0125)^2 = 4.909 \times 10^{-4}\ \text{m}^2$$

(i) [3] The core has infinite μ, so $\mathcal{R}_{core} = 0$ and only the gap counts.

$$\mathcal{R}_g = \frac{l_g}{\mu_0 A} = \frac{0.5 \times 10^{-3}}{4\pi \times 10^{-7} \times 4.909 \times 10^{-4}} = \mathbf{810.6\ kA/Wb}$$

$$L = \frac{N^2}{\mathcal{R}_g} = \frac{120^2}{810.6 \times 10^3} = \mathbf{17.77\ mH}$$

(ii) [2] Back-emf equation (no reluctance needed):

$$V_{max} = 4.44\, f N \hat{B} A = 4.44 \times 50 \times 120 \times 1.5 \times 4.909 \times 10^{-4} = \mathbf{19.62\ V\ rms}$$

(With the exact factor $2\pi/\sqrt{2} = 4.443$ you get 19.63 V. Both are fine.)

(iii) [2] One gap, so one area:

$$F = \frac{B^2 A}{2\mu_0} = \frac{1.5^2 \times 4.909 \times 10^{-4}}{2 \times 4\pi \times 10^{-7}} = \mathbf{439.5\ N}$$

All answers match the course.

2b) Ideal Transformers [5 marks]

$n = 6$, so $n^2 = 36$.

(i) [2]

$$Z_1 = \mathbf{0.3\ \Omega}, \qquad Z_2 = n^2 Z_1 = 36 \times 0.3 = \mathbf{10.8\ \Omega}, \qquad Z_3 = 15 + 10.8 = \mathbf{25.8\ \Omega}$$

(ii) [3]

$$I_1 = \frac{V_S}{Z_3} = \frac{12}{25.8} = \mathbf{0.4651\ A}, \qquad I_2 = nI_1 = 6 \times 0.4651 = \mathbf{2.791\ A}$$

$$V_1 = I_1 Z_2 = 0.4651 \times 10.8 = \mathbf{5.023\ V}, \qquad V_2 = \frac{V_1}{n} = \mathbf{0.8372\ V}$$

CHECK: $V_1 I_1 = V_2 I_2 = 2.336$ W, as it must be for an ideal transformer. All answers match the course.

2c) Three-Phase Transformers [8 marks]

(i) [2] Simplified circuit, per phase, referred to the primary. Draw the shunt branch $R_C = 250\ \Omega$ in parallel with $jX_M = j14\ \Omega$ at the input terminals. Then draw one series branch:

$$R = R_1 + R_2' = 0.14\ \Omega, \qquad jX = j(X_1 + X_2') = j0.28\ \Omega$$

Label it $0.14\ \Omega$ and $j0.28\ \Omega$, leading to the V_2' terminals.

(ii) [4] Assume star. Per phase:

$$V_1 = \frac{16\,000}{\sqrt{3}} = 9238\ \text{V}$$

On open circuit $I_2' = 0$, so only the shunt branch carries current:

$$Z = 250 \parallel j14 = \frac{1}{\frac{1}{250} + \frac{1}{j14}} = 13.98\angle 86.79^\circ\ \Omega$$

$$|I_1| = \frac{9238}{13.98} = \mathbf{660.9\ A}$$

$$P = 3\frac{V_1^2}{R_C} = 3 \times \frac{9238^2}{250} = \mathbf{1.024\ MW}$$

> **CHECK:** $I_R = 36.95$ A and $I_M = 659.8$ A, and $\sqrt{36.95^2 + 659.8^2} = 660.9$ A. The rated current is 5413 A, so the OC current is 12.2 % of rated.

(iii) [2] The circuit is linear and Z is fixed.

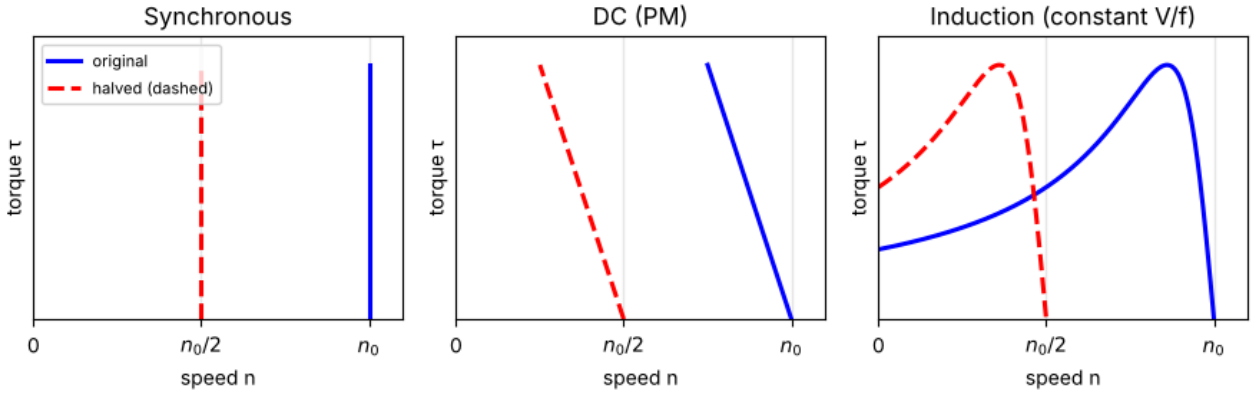
Quantity	Law	Factor
$\lvert I\rvert$	$V/\lvert Z\rvert$	**×0.5**
P	$3V^2/R_C$	**×0.25**
Q	$3V^2/X_M$	**×0.25**
pf	$\cos\angle Z$, independent of V	**×1**

All answers match the course.

Question 3. Electric Machines [20 marks]

3a) Torque vs. Speed Characteristics [6 marks]

The answer sheet names the curves, in order: **synchronous, DC, induction** (Lecture Sections 4.1 and 7.9).



τ–n curves (solid) and the curves after halving the no-load or synchronous speed (dashed). The induction curve is drawn for constant V/f with R_1 neglected.

(i) [3] Name and sketch.

1. **Synchronous machine:** a vertical line at n_S. Halved, it becomes a vertical line at $n_S/2$.
2. **DC machine:** a straight line falling from n_{NL} at $\tau = 0$. Halved, it is a **parallel** line starting at $n_{NL}/2$. The slope (R/k^2) is unchanged.
3. **Induction machine:** zero torque at n_S, a steep, nearly linear region just below n_S, a peak (τ_{max}) and then the starting torque at $n = 0$. Halved at constant V/f, the curve keeps its shape and shifts left to start at $n_S/2$. The slip speed at a given torque and the peak torque are unchanged.

(ii) [3] How to halve the speed safely.

- **Synchronous:** halve f, which halves $n_S = 60f/p$. **Also halve** V, so that V/f (and hence the peak flux) stays constant. Otherwise the core saturates and the current becomes very large.
- **DC: halve the armature voltage** V. The no-load speed is V/k.
- **Induction: halve both** f **and** V (constant V/f), for the same reason as the synchronous machine.

AC machines need a variable-voltage, variable-frequency inverter for this. A DC machine needs a variable DC supply.

3b) Induction Machine Parameters [4 marks]

$R_1 = \mathbf{2\ \Omega}$. It is given per phase, so no halving is needed.

The blocked-rotor test gives the series branch. Per phase:

$$V_1 = \frac{80}{\sqrt{3}} = 46.19\ \text{V}, \qquad I_1 = 5\ \text{A}, \qquad P_1 = \frac{390}{3} = 130\ \text{W}$$

$$R_1 + R_2' = \frac{P_1}{I_1^2} = \frac{130}{25} = 5.2\ \Omega \ \Rightarrow\ R_2' = 5.2 - 2 = \mathbf{3.2\ \Omega}$$

$$S_1 = 46.19 \times 5 = 230.9\ \text{VA}, \quad Q_1 = \sqrt{230.9^2 - 130^2} = 190.9\ \text{VAr}, \quad X = \frac{Q_1}{I_1^2} = \mathbf{7.635\ \Omega}$$

If the split is wanted: $X_1 = X_2' = 3.818\ \Omega$.

The no-load test is not needed for these three. (It gives $R_C = 637.9\ \Omega$ and $X_M = 68.85\ \Omega$.)

All answers match the course.

3c) Induction Machine Calculation [10 marks]

(i) [3] The rated speed of 960 rpm sits just below a synchronous speed. $p = 60f/n = 3000/960 = 3.125$, so $p = 3$ pole-pairs (6 poles).

$$n_S = \frac{60f}{p} = \frac{60 \times 50}{3} = \mathbf{1000\ rpm}, \qquad \omega_S = 104.72\ \text{rad/s}$$

$$s = \frac{n_S - n}{n_S} = \frac{1000 - 1025}{1000} = \mathbf{-0.025}, \qquad \frac{R_2'}{s} = \frac{3}{-0.025} = \mathbf{-120\ \Omega}$$

(ii) [5] Circuit: V_1, then $R_1 = 2.5\ \Omega$, then $jX = j12\ \Omega$, then $R_2'/s = -120\ \Omega$, all in series. There is no shunt branch.

$$V_1 = \frac{415}{\sqrt{3}}\angle 0^\circ = 239.6\angle 0^\circ\ \text{V}$$

$$Z = 2.5 + j12 - 120 = -117.5 + j12 = 118.1\angle 174.2^\circ\ \Omega$$

$$I_1 = \frac{239.6\angle 0^\circ}{118.1\angle 174.2^\circ} = \mathbf{2.029\angle -174.2^\circ\ A}$$

$$P_{IN} = 3V_1 I_1 \cos\phi = 3 \times 239.6 \times 2.029 \times \cos(-174.2^\circ) = \mathbf{-1451\ W}$$

$P_{IN} < 0$, so 1.451 kW flows into the supply: the machine is generating. ($Q_{IN} = +148$ VAr, so it still absorbs reactive power.)

(iii) [2]

$$P_{AG} = 3I_2'^2 \frac{R_2'}{s} = 3 \times 2.029^2 \times (-120) = \mathbf{-1.482\ kW}$$

$$\tau = \frac{P_{AG}}{\omega_S} = \frac{-1482}{104.72} = \mathbf{-14.15\ Nm}$$

The negative torque opposes the prime mover.

Power-flow check: shaft input $= |P_{AG}|(1-s) = 1519$ W. Rotor copper loss $= 3I^2R_2' = 37$ W. Airgap $= 1482$ W. Stator copper loss $= 3I^2R_1 = 31$ W. Electrical output $= 1451$ W, which equals $|P_{IN}|$.

Against the course: all agree. Unrounded, $P_{AG} = -1481.5$ W, which rounds to −1.481 kW. The course's −1.482 kW uses the rounded $I = 2.029$ A ($3 \times 2.029^2 \times 120 = 1482.1$ W). Either gets the mark. The torque is −14.15 Nm either way. (Pack M9 reproduced the same values.)

Question 4. Synchronous Machines [20 marks]

4a) Synchronous Machines [5 marks]

(i) [3] The OC voltage is a line value. At the same field current:

$$|Z_S| = \frac{E_{0,ph}}{I_{SC}} = \frac{12\,000/\sqrt{3}}{6500} = 1.066\ \Omega$$

$$X_S = \sqrt{|Z_S|^2 - R^2} = \sqrt{1.066^2 - 0.2^2} = 1.047\ \Omega$$

$$Z_B = \frac{V_B^2}{S_B} = \frac{(14\times10^3)^2}{400\times10^6} = 0.4900\ \Omega, \qquad X_S = \frac{1.047}{0.49} = \mathbf{2.137\ pu}$$

(ii) [2]

$$\mathbf{V} = \mathbf{0.97\angle 0^\circ\ pu}$$

$$P = \frac{200}{400} = 0.5\ \text{pu}, \quad |S| = \frac{0.5}{0.8} = 0.625\ \text{pu}, \quad |I| = \frac{0.625}{0.97} = 0.6443\ \text{pu}$$

Lagging, so $\phi = -\cos^{-1}0.8 = -36.87^\circ$:

$$\mathbf{I} = \mathbf{0.6443\angle -36.87^\circ\ pu}$$

All answers match the course.

4b) Synchronous Machines [6 marks]

(i) [3] $\mathbf{V} = 1\angle 0^\circ$ pu. Rated VA means $|S| = 1$ pu, so $|I| = 1$ pu. Leading, so $\phi = +\cos^{-1}0.9 = +25.84^\circ$:

$$\mathbf{I} = 1\angle +25.84^\circ\ \text{pu}$$

Generator convention:

$$\mathbf{E}_0 = \mathbf{V} + jX_S\mathbf{I} = 1 + 1.8\angle 115.84^\circ = 0.2154 + j1.620 = \mathbf{1.634\angle 82.43^\circ\ pu}$$

The excitation is $\mathbf{1.634}$ **pu**.

Q4b phasor diagram: I leads V by 25.84 degrees, jXsI leans left, E0 = 1.634 at 82.43 degrees

(ii) [1] The rotor angle is $\delta = \angle\mathbf{E}_0 = \mathbf{82.42^\circ}$ (82.43° unrounded). It is close to 90° because a leading-pf generator is under-excited.

(iii) [2]

$$P_{max} = \frac{VE_0}{X_S} = \frac{1 \times 1.634}{1.8} = \mathbf{0.9078\ pu} \quad \text{at} \quad \delta = \mathbf{90^\circ}$$

Against the course: the course gives 82.42° and 0.9077 pu. Unrounded values are 82.43° and 0.9079 pu, and $1.634/1.8 = 0.90778$. The differences come only from intermediate rounding and truncation. Any of these values earns the marks.

CHECK: $P = (1 \times 1.634/1.8)\sin 82.43^\circ = 0.900$ pu, which matches 0.9×1 pu.

4c) Short-Answer Questions [9 marks]

- **Real power** comes from the part of the current that is in phase with the voltage. It is the average value of the instantaneous power. It represents energy flowing one way, from the source to the load, and it is the power that does useful work, such as heat or mechanical output. Resistors absorb it.
- **Reactive power** comes from the part of the current that is a quarter-cycle (90 degrees) out of phase with the voltage. Its average is zero: energy surges back and forth between the source and the inductors or capacitors each cycle. It does no useful work, but it still adds to the current, and so to the losses and voltage drops. Inductors absorb it and capacitors generate it.
- Both real and reactive power are scalar quantities.
- **Apparent power** is the magnitude of the complex power, whose real part is the real power and whose imaginary part is the reactive power. It is the rms voltage times the total rms current. It sets the current that generators, transformers and cables must carry, so equipment is rated in volt-amperes.

- **Copper (winding) loss:** resistive heating of the primary and secondary windings by the load current. In the simplified equivalent circuit it is modelled by the series winding resistances (the primary resistance plus the referred secondary resistance). It is proportional to the square of the current, so at constant voltage it rises with the **square of the output power**. It is small at light load and dominant at full load.

- **Iron (core) loss:** hysteresis loss (energy to repeatedly reverse the magnetisation of the steel) and eddy-current loss (currents induced in the core laminations by the alternating flux). It is modelled by the core-loss resistance in parallel across the supply. It depends on the supply voltage and frequency, through the peak flux density, and **not on the load**, so it is essentially constant with output power.
- As a result, efficiency is low at light load. It peaks at the load where the copper loss equals the iron loss.

- **Magnetising current.** Even with no load, the stator must set up the rotating magnetic field in the air gap. This needs a magnetising current, which is modelled by the magnetising reactance. An induction machine has an air gap, so its magnetising current is large, typically a quarter to over half of rated current. It is much larger than in a transformer.
- **Core-loss current.** The alternating flux causes hysteresis and eddy-current losses in the stator (and rotor) iron. These need an in-phase current component, which is modelled by the core-loss resistance.
- **Mechanical losses.** Even unloaded, the rotor must overcome bearing friction and windage (air drag, including the cooling fan). So it runs slightly below synchronous speed. The small slip induces a small rotor current that supplies this power, and the stator copper loss adds a little more.
- The no-load current is therefore mostly reactive, so the no-load power factor is very low.

ELECTRICAL MACHINES · MOCK

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Marking guide

All numbers below were computed in Python and are given to 4 significant figures. Accept answers within about 0.5 % that come from sensible intermediate rounding.

Question 1 marking guide

(i) KVL around the loop:

$$V_t = E - IR_{cable} = 48 - 40(0.12) = \mathbf{43.2\ V}$$

$$P_{cable} = I^2 R_{cable} = 40^2(0.12) = \mathbf{192\ W}$$

(ii)

$$P_{in,motor} = V_t I = 43.2 \times 40 = \mathbf{1728\ W}$$

$$\eta = \frac{P_{out}}{P_{in}} = \frac{1500}{1728} = \mathbf{86.81\ \%}$$

$$\omega = \frac{2\pi \times 1200}{60} = 125.7\ \text{rad/s}, \qquad \tau = \frac{P_{out}}{\omega} = \frac{1500}{125.7} = \mathbf{11.94\ Nm}$$

(Check: the source supplies $48 \times 40 = 1920$ W $= 1728 + 192$.)

MARKS: (i) 1 for V_t, 1 for the cable loss. (ii) 1 each for input power, efficiency and torque.

> COMMON MISTAKES:

- Using $EI = 1920$ W as the motor input. That is the source output. The motor only receives $V_t I$. (The overall efficiency from the source, 78.13 %, is not what was asked.)
- Using rpm instead of rad/s in $\tau = P/\omega$.

Method from Lecture 2.8.3 (single-phase analysis, delta): convert to phase quantities, analyse one phase, convert back.

(i) In delta the phase voltage equals the line voltage: $\mathbf{V}_1 = 415\angle 0^\circ$ V.

$$Z = 24 + j18 = 30\angle 36.87^\circ\ \Omega$$

$$\mathbf{I}_1 = \frac{\mathbf{V}_1}{Z} = \frac{415\angle 0^\circ}{30\angle 36.87^\circ} = \mathbf{13.83\angle -36.87^\circ\ A}$$

$$I_L = \sqrt{3} I_1 = \sqrt{3} \times 13.83 = \mathbf{23.96\ A}$$

$$\text{pf} = \cos 36.87^\circ = \mathbf{0.8\ lagging}$$

(ii)

$$P = 3I_1^2 R = 3(13.83)^2(24) = \mathbf{13.78\ kW}, \qquad Q = 3I_1^2 X = 3(13.83)^2(18) = \mathbf{10.33\ kVAr}$$

(Check: $P = \sqrt{3} V_L I_L \cos\phi = \sqrt{3}(415)(23.96)(0.8) = 13.78$ kW.)

(iii) In star each impedance sees $415/\sqrt{3} = 239.6$ V, so $I_1 = 239.6/30 = 7.987$ A and

$$P_{star} = 3(7.987)^2(24) = \mathbf{4.593\ kW}$$

This is one-third of the delta value, because the phase voltage drops by $\sqrt{3}$ and power goes as V^2.

MARKS: (i) 1 each for $\mathbf{I}_1$, I_L and pf. (ii) 1 each for P and Q. (iii) 1.

> **COMMON MISTAKES:**

- Dividing 415 V by $\sqrt{3}$ for a delta load. In delta, $V_1 = V_L$ and it is the current that has the $\sqrt{3}$.
- Reporting $I_L = I_1$ in delta.

(i) Motor input. The 18 kW is the shaft output, so

$$P_m = \frac{18}{0.9} = 20\ \text{kW}, \qquad Q_m = P_m \tan(\cos^{-1} 0.75) = 20 \times 0.8819 = 17.64\ \text{kVAr}$$

Add the delta load from 1b (powers add, currents do not):

$$P_T = 13.78 + 20 = \mathbf{33.78\ kW}, \qquad Q_T = 10.33 + 17.64 = \mathbf{27.97\ kVAr}$$

$$S_T = \sqrt{P_T^2 + Q_T^2} = 43.86\ \text{kVA}, \qquad \text{pf} = \frac{33.78}{43.86} = \mathbf{0.7702\ lagging}$$

$$I_L = \frac{S_T}{\sqrt{3} V_L} = \frac{43\,856}{\sqrt{3} \times 415} = \mathbf{61.01\ A}$$

(ii) The capacitors do not change P. The target supply reactive power is

$$Q_S = P_T \tan(\cos^{-1} 0.95) = 33.78 \times 0.3287 = 11.10\ \text{kVAr}$$

$$Q_C = Q_S - Q_T = 11.10 - 27.97 = \mathbf{-16.87\ kVAr}$$

(negative: the capacitors generate reactive power). The new apparent power is $S = 33.78/0.95 = 35.56$ kVA, so

$$I_L = \frac{35\,556}{\sqrt{3} \times 415} = \mathbf{49.47\ A}$$

The line current falls by about 19 %.

(iii) Each delta capacitor sees the line voltage, 415 V, and absorbs one-third of Q_C:

$$|Q_{C,ph}| = \frac{16.87}{3} = 5.623\ \text{kVAr} = \frac{V^2}{X_C} = V^2 \omega C$$

$$C = \frac{5623}{415^2 \times 2\pi \times 50} = \mathbf{103.9\ \mu F\ \text{per capacitor}}$$

(iv) For 0.95 leading, $Q_S = -11.10$ kVAr, so $Q_C = -11.10 - 27.97 = \mathbf{-39.07\ kVAr}$ (2.3 times as much). It is not done because the larger capacitors cost more for no reduction in line current (the line current is the same 49.47 A at 0.95 lead as at 0.95 lag), and a leading load can raise the supply voltage.

MARKS: (i) 1 for the motor input P and Q (dividing by efficiency), 1 for the totals, 1 for pf and I_L. (ii) 1 for Q_S, 1 for Q_C with sign, 1 for the new I_L. (iii) 1 for using 415 V and $Q_C/3$, 1 for C. (iv) 1 for the value with a valid reason.

COMMON MISTAKES:

- Using 18 kW as the motor input power. Always divide the shaft power by efficiency first (as in Jun23 Q4b).
- Adding power factors, or adding current magnitudes, instead of adding P and Q separately.
- Using 239.6 V for delta capacitors. That gives 311.8 µF, three times too big. 239.6 V is only correct if the capacitors are in star.
- Using Q_S (11.10 kVAr) as the capacitor size. The capacitor supplies the *difference*, $Q_T - Q_S$.

Question 2 marking guide

$A = 0.04 \times 0.03 = 12 \times 10^{-4}$ m², $\mu_0 = 4\pi \times 10^{-7}$ H/m.

(i)

$$\mathcal{R}_c = \frac{l_c}{\mu_0 \mu_r A} = \frac{0.36}{4\pi \times 10^{-7}(2500)(12 \times 10^{-4})} = \mathbf{95.49\ kA/Wb}$$

$$\mathcal{R}_g = \frac{g}{\mu_0 A} = \frac{0.4 \times 10^{-3}}{4\pi \times 10^{-7}(12 \times 10^{-4})} = \mathbf{265.3\ kA/Wb}\ \text{(each gap)}$$

The iron and the two gaps are in series:

$$\mathcal{R}_T = \mathcal{R}_c + 2\mathcal{R}_g = 95.49 + 2(265.3) = \mathbf{626.0\ kA/Wb}$$

$$L = \frac{N^2}{\mathcal{R}_T} = \frac{200^2}{626\,009} = \mathbf{63.90\ mH}$$

(ii) $\Phi = BA = 1.2 \times 12 \times 10^{-4} = 1.44$ mWb. From $NI = \Phi \mathcal{R}_T$:

$$I = \frac{\Phi \mathcal{R}_T}{N} = \frac{1.44 \times 10^{-3} \times 626\,009}{200} = \mathbf{4.507\ A}$$

Energy in the gaps (two gaps, each of volume Ag):

$$E_g = 2 \times \frac{B^2}{2\mu_0} \times Ag = 2 \times \frac{1.2^2}{2(4\pi \times 10^{-7})} \times (12 \times 10^{-4})(0.4 \times 10^{-3}) = \mathbf{0.5500\ J}$$

(Check: the total stored energy is $\frac{1}{2}LI^2 = 0.6490$ J. The gaps hold 85 % of it, which matches $2\mathcal{R}_g/\mathcal{R}_T$.)

(iii) Each gap has two facing surfaces of area A, and each gap pulls with

$$F_1 = \frac{B^2}{2\mu_0} A = \frac{1.2^2}{2(4\pi \times 10^{-7})}(12 \times 10^{-4}) = 687.5\ \text{N}$$

There are two gaps between the halves, so the total force is $F = 2F_1 = \mathbf{1375\ N}$.

MARKS: (i) 1 for $\mathcal{R}_c$, 1 for $\mathcal{R}_g$ and the total with **two** gaps, 1 for L. (ii) 1 for I, 1 for the gap energy. (iii) 1 for the formula per gap, 1 for doubling.

COMMON MISTAKES:

- Counting only one airgap. The question says the flux crosses two gaps in series.
- Using the formula-sheet force with $2A$ but then doubling again (4 times too big), or forgetting to double (half the answer). Think about it as “one force per gap, two

gaps".
- Using $\frac{1}{2}LI^2$ for "energy in the airgaps". That is the total energy, including the iron.

Turns ratio $a = 3300/240 = 13.75$, so $a^2 = 189.1$. Rated HV current $= 20\,000/3300 = 6.061$ A, so the SC test is at rated current.

OC test (LV side). The shunt branch is found on the side where the test was done, then referred to the HV side.

$$R_{C,LV} = \frac{V^2}{P} = \frac{240^2}{135} = 426.7\ \Omega$$

$S = VI = 240 \times 2.6 = 624$ VA, $\quad Q = \sqrt{S^2 - P^2} = \sqrt{624^2 - 135^2} = 609.2$ VAr, $\quad X_{M,LV} = \frac{V^2}{Q} = \frac{240^2}{609.2} = 94.55\ \Omega$

Refer to the HV side by multiplying by a^2:

$$R_C = 426.7 \times 189.1 = \mathbf{80.67\ k\Omega}, \qquad X_M = 94.55 \times 189.1 = \mathbf{17.88\ k\Omega}$$

SC test (HV side). These are already on the HV side.

$$R = \frac{P}{I^2} = \frac{290}{6.06^2} = \mathbf{7.897\ \Omega}, \qquad |Z| = \frac{V}{I} = \frac{145}{6.06} = 23.93\ \Omega$$

$$X = \sqrt{|Z|^2 - R^2} = \sqrt{23.93^2 - 7.897^2} = \mathbf{22.59\ \Omega}$$

MARKS: 1 each for R_C and X_M on the LV side, 1 for referring both to HV, 1 each for R and X.

COMMON MISTAKES:

- Forgetting that the OC test was done on the LV side, and quoting 426.7 Ω and 94.55 Ω as the HV values. They are 189 times too small.
- Referring the SC results as well. They were measured on the HV side already.
- $X_M = V/I$. That is the magnitude of the parallel pair, $|R_C \parallel jX_M|$ (92.3 Ω on the LV side), not X_M. Split P and Q first.

All on the HV side. $\mathbf{V}_2' = aV_2 = 13.75 \times 240 = 3300\angle 0^\circ$ V.

(i) Rated kVA at rated voltage means $I_2' = 20\,000/3300 = 6.061$ A. A leading load puts the current **ahead** of the voltage:

$$\mathbf{I}_2' = \mathbf{6.061\angle +36.87^\circ\ A} = 4.848 + j3.636\ \text{A}$$

KVL through the series branch:

$$\mathbf{V}_1 = \mathbf{V}_2' + \mathbf{I}_2'(R + jX) = 3300 + (4.848 + j3.636)(7.897 + j22.59)$$

$$= 3300 + (38.29 - 82.13) + j(109.5 + 28.71) = 3256.2 + j138.2 = \mathbf{3259\angle 2.43^\circ\ V}$$

(ii)

$$VR = 100\,\frac{|\mathbf{V}_1| - |\mathbf{V}_2'|}{|\mathbf{V}_2'|} = 100\,\frac{3259.1 - 3300}{3300} = \mathbf{-1.24\ \%}$$

It is negative because the leading current turns the $jX\mathbf{I}_2'$ drop so that it partly opposes $\mathbf{V}_2'$. The output voltage therefore rises above its no-load value as load is applied. (Check with the approximate formula $I_2'(R\cos\phi - X\sin\phi)/V_2' = -1.33$ %: same sign, similar size.)

(iii) $P_{out} = 20\,000 \times 0.8 = 16$ kW.

$$P_{CU} = I_2'^2 R = 6.061^2(7.897) = 290.1\ \text{W}, \qquad P_{FE} = \frac{|\mathbf{V}_1|^2}{R_C} = \frac{3259.1^2}{80\,667} = 131.7\ \text{W}$$

$$\eta = \frac{16\,000}{16\,000 + 290.1 + 131.7} = \mathbf{97.43\ \%}$$

(Using the 135 W OC-test value for the iron loss gives 97.41 %. Either gets the mark if the method is stated.)

(iv) Maximum efficiency is where copper loss equals iron loss. The full-load copper loss is 290.1 W and the iron loss at rated voltage is 135 W:

$$x = \sqrt{\frac{P_{FE}}{P_{CU,FL}}} = \sqrt{\frac{135}{290.1}} = 0.6822 \quad \Rightarrow \quad S = 0.6822 \times 20 = \mathbf{13.64\ kVA}$$

$$\eta_{max} = \frac{13\,644 \times 0.8}{13\,644 \times 0.8 + 2(135)} = \mathbf{97.59\ \%}$$

MARKS: (i) 1 for $\mathbf{I}_2'$ with the **+** angle, 1 for the phasor equation, 1 for $\mathbf{V}_1$ in polar form. (ii) 1 for the value with its sign, 1 for the reason. (iii) 1. (iv) 1 for the kVA, 1 for η_{max}.

COMMON MISTAKES:

- Writing $\mathbf{I}_2' = 6.061\angle -36.87^\circ$ for a leading load. That is the lagging answer: it gives $\mathbf{V}_1 = 3421\angle 1.35^\circ$ V and +3.68 % regulation.
- Adding magnitudes: $3300 + 6.061 \times 23.93$. This is a phasor sum. Go to rectangular form.
- Using I_1 (which includes the magnetising current) in the copper loss.
- Putting maximum efficiency at full load. It is where $P_{CU} = P_{FE}$.

Question 3 marking guide

(i) $n_S = 60f/p$ with p = pole-pairs. $60 \times 60/1164 = 3.09$, so p = 3 and $n_S = 3600/3 = 1200$ rpm. The machine has **6 poles**.

It is an **induction machine**, because 1164 rpm is not a synchronous speed for 60 Hz. It runs below $n_S = 1200$ rpm, with slip. A synchronous machine would run at exactly 1200 rpm.

(ii)

$$s = \frac{n_S - n}{n_S} = \frac{1200 - 1164}{1200} = \mathbf{0.03}, \qquad f_2 = sf = 0.03 \times 60 = \mathbf{1.8\ Hz}$$

(iii) Rotor copper loss is sP_{AG}, so the output can be at most $(1 - s)P_{AG}$:

$$\eta_{ideal} = 1 - s = \frac{n}{n_S} = \mathbf{97\ \%}$$

(iv) On the linear part of the curve, torque is proportional to the slip speed. Rated slip speed is $1200 - 1164 = 36$ rpm, so at half torque it is 18 rpm:

$$n = 1200 - 18 = \mathbf{1182\ rpm}$$

$$\frac{P}{P_{rated}} = \frac{\tau}{\tau_{rated}} \times \frac{n}{n_{rated}} = 0.5 \times \frac{1182}{1164} = \mathbf{50.77\ \%}$$

MARKS: (i) 1 for 6 poles, 1 for induction with the reason. (ii) 1 for both s and f_2. (iii) 1. (iv) 1 for the speed, 1 for the power.

COMMON MISTAKES:

- Using 50 Hz out of habit. This nameplate says 60 Hz.
- Answering "3 poles". p in $60f/p$ is pole-**pairs**.
- Saying the output power is exactly 50 %. The speed also rose a little.
- Halving the speed instead of halving the slip speed.

Work per phase of the **delta**: $V_1 = V_L$, $I_1 = I_L/\sqrt{3}$, $P_1 = P/3$.

Blocked-rotor test (series branch only):

$$V_1 = 92\ \text{V}, \quad I_1 = \frac{10}{\sqrt{3}} = 5.774\ \text{A}, \quad P_1 = \frac{620}{3} = 206.7\ \text{W}$$

$$R = R_1 + R_2' = \frac{P_1}{I_1^2} = \frac{206.7}{5.774^2} = 6.200\ \Omega \quad \Rightarrow \quad R_2' = 6.200 - 2.8 = \mathbf{3.400\ \Omega}$$

$$|Z| = \frac{V_1}{I_1} = \frac{92}{5.774} = 15.93\ \Omega, \qquad X = \sqrt{15.93^2 - 6.200^2} = \mathbf{14.68\ \Omega}$$

(If a split is needed, $X_1 = X_2' = 7.34\ \Omega$.)

No-load test (shunt branch only):

$$V_1 = 400\ \text{V}, \quad I_1 = \frac{4.2}{\sqrt{3}} = 2.425\ \text{A}, \quad P_1 = \frac{320}{3} = 106.7\ \text{W}$$

$$R_C = \frac{V_1^2}{P_1} = \frac{400^2}{106.7} = \mathbf{1500\ \Omega}$$

$S_1 = 400 \times 2.425 = 969.9\ \text{VA}$, $Q_1 = \sqrt{969.9^2 - 106.7^2} = 964.1\ \text{VAr}$, $X_M = \dfrac{400^2}{964.1} = \mathbf{166.0\ \Omega}$

(The no-load pf is only 0.11. The no-load current is almost all magnetising current. R_C here also includes friction and windage.)

MARKS: 1 for converting to delta phase quantities, 1 for R_2', 1 for X, 1 for R_C, 1 for X_M.

COMMON MISTAKES:

- Mixing star and delta: using star-equivalent test values ($V_1 = 92/\sqrt{3}$, $I_1 = 10$ A) with the delta winding's $R_1 = 2.8\ \Omega$. This gives $R = 2.07\ \Omega$, which is less than 2.8 Ω, so R_2' comes out negative. A negative resistance is a sure sign of a star/delta mix-up. (A consistent star-equivalent working, with $R_{1Y} = 2.8/3 = 0.933\ \Omega$, is valid: it gives $R_2' = 1.133\ \Omega$, $X = 4.893\ \Omega$, $R_C = 500\ \Omega$ and $X_M = 55.32\ \Omega$, exactly one-third of the delta values. But the question asks for per-phase-of-delta values.)
- Forgetting to subtract R_1 from the blocked-rotor R.
- $X_M = V_1/I_1$. Separate P and Q first.

(i) Six poles, so p = 3:

$$n_S = \frac{60 \times 50}{3} = \mathbf{1000\ rpm}, \qquad s = \frac{1000 - 970}{1000} = \mathbf{0.03}, \qquad \frac{R_2'}{s} = \frac{0.45}{0.03} = \mathbf{15.00\ \Omega}$$

(ii) Star, so $\mathbf{V}_1 = 415/\sqrt{3} = 239.6\angle 0^\circ$ V. Series branch: $X = X_1 + X_2' = 2.4\ \Omega$.

$$\mathbf{I}_2' = \frac{\mathbf{V}_1}{R_1 + R_2'/s + jX} = \frac{239.6}{15.6 + j2.4} = \mathbf{15.18\angle -8.75^\circ\ A}$$

Shunt branch at the terminals:

$$\mathbf{I}_C = \frac{239.6}{300} = 0.7987\ \text{A}, \qquad \mathbf{I}_M = \frac{239.6}{j40} = -j5.990\ \text{A}$$

$$\mathbf{I}_1 = \mathbf{I}_2' + \mathbf{I}_C + \mathbf{I}_M = (15.00 - j2.308) + 0.799 - j5.990 = 15.80 - j8.298 = \mathbf{17.85\angle -27.71^\circ\ A}$$

$$\text{pf} = \cos 27.71^\circ = \mathbf{0.8854\ lagging}$$

(iii)

$$P_{IN} = 3V_1 I_1 \cos\phi = 3(239.6)(17.85)(0.8854) = \mathbf{11.36\ kW}$$

$$P_{AG} = 3I_2'^2 \frac{R_2'}{s} = 3(15.18)^2(15) = \mathbf{10.37\ kW}$$

$$\omega_S = \frac{2\pi \times 1000}{60} = 104.7\ \text{rad/s}, \qquad \tau = \frac{P_{AG}}{\omega_S} = \frac{10\,370}{104.7} = \mathbf{99.03\ Nm}$$

(The formula-sheet torque formula gives the same 99.03 Nm.)

$$P_{OUT} = (1 - s)P_{AG} = 0.97 \times 10\,370 = 10.06\ \text{kW}, \qquad \eta = \frac{10.06}{11.36} = \mathbf{88.56\ \%}$$

Loss check: stator Cu $= 3(15.18)^2(0.6) = 414.8$ W, iron $= 3(239.6)^2/300 = 574.1$ W, rotor Cu $= sP_{AG} = 311.1$ W. $10\,059 + 414.8 + 574.1 + 311.1 = 11\,359$ W $= P_{IN}$.

(iv)

$$s_{\tau\max} = \frac{R_2'}{|R_1 + jX|} = \frac{0.45}{\sqrt{0.6^2 + 2.4^2}} = \mathbf{0.1819}, \qquad n = (1 - 0.1819)(1000) = \mathbf{818.1\ rpm}$$

(Not asked, but useful: $\tau_{max} = 267.5$ Nm, 2.7 times the torque at 970 rpm.)

MARKS: (i) 1 for n_S, 1 for s and R_2'/s. (ii) 1 for $\mathbf{I}_2'$, 1 for adding the shunt currents to get $\mathbf{I}_1$, 1 for pf. (iii) 1 for P_{IN}, 1 for P_{AG} and τ, 1 for η. (iv) 1 for both.

COMMON MISTAKES:

- Using 415 V as the phase voltage of a star machine. Torque comes out 3 times too big.
- Using the rotor speed ω with P_{AG}. P_{AG} goes with ω_S. P_{OUT} goes with ω.
- Forgetting the shunt branch in $\mathbf{I}_1$. It does not change the torque, but it does change I_1, the pf and P_{IN}.
- In (iv), using $R_1 + X$ or X alone instead of the magnitude $|R_1 + jX|$.

Question 4 marking guide

(i)

$$Z_B = \frac{V_B^2}{S_B} = \frac{(13.8\times10^3)^2}{60\times10^6} = \mathbf{3.174\ \Omega}, \qquad I_B = \frac{S_B}{\sqrt{3}V_B} = \frac{60\times10^6}{\sqrt{3}\times 13\,800} = \mathbf{2510\ A}$$

(ii) At the same field current, the OC phase voltage and the SC current give the synchronous impedance:

$$|Z_S| = \frac{V_{OC}/\sqrt{3}}{I_{SC}} = \frac{9200/\sqrt{3}}{1250} = 4.249\ \Omega$$

$$X_S = \sqrt{|Z_S|^2 - R^2} = \sqrt{4.249^2 - 0.05^2} = \mathbf{4.249\ \Omega}, \qquad X_S = \frac{4.249}{3.174} = \mathbf{1.339\ pu}$$

(R is only 0.0158 pu, so it barely changes X_S.)

(iii) OC voltage is proportional to field current on the linear part:

$$I_F = 180 \times \frac{13.8}{9.2} = \mathbf{270\ A}$$

MARKS: (i) 1 for both bases. (ii) 1 for $|Z_S|$ with the $\sqrt{3}$, 1 for X_S in Ω, 1 for pu. (iii) 1.

COMMON MISTAKES:

- $X_S = 9200/1250 = 7.36\ \Omega$. The OC voltage is a line value. The circuit is per phase.
- Using $Z_B = V_B^2/(3S_B)$ or V_B/I_B with the line voltage. With line voltage and three-phase power, $Z_B = V_B^2/S_B$.

(i) $\mathbf{V} = 1\angle 0^\circ$ pu.

$$S = \frac{45}{0.85} = 52.94\ \text{MVA} = 0.8824\ \text{pu}$$

The generator delivers at a leading pf, so (generator convention, current out of the machine) the current leads the voltage:

$$\mathbf{I} = \frac{S}{V}\angle +\cos^{-1} 0.85 = \mathbf{0.8824\angle +31.79^\circ\ pu} = 0.75 + j0.4648\ \text{pu}$$

$$I = 0.8824 \times 2510 = \mathbf{2.215\ kA}$$

(ii) Generator: $\mathbf{E}_0 = \mathbf{V} + jX_S\mathbf{I}$ (R neglected).

$\mathbf{E}_0 = 1 + j1.339(0.75 + j0.4648) = 1 - 0.6224 + j1.004 = 0.3778 + j1.004 = \mathbf{1.073\angle 69.38^\circ\ pu}$

Rotor angle $\delta = \mathbf{69.38}^\circ$.

Q4b phasor diagram: I = 0.8824 pu leading by 31.79 degrees, E0 = 1.073 at 69.38 degrees

Field current, using the excitation scale from 4a(iii) (270 A per pu of E_0):

$$I_F = 1.073 \times 270 = \mathbf{289.6\ A}$$

Check: $P = E_0 V \sin\delta / X_S = 1.073 \times \sin 69.38^\circ / 1.339 = 0.75$ pu = 45 MW.

(iii) With E_0 and V fixed, the maximum power is at $\delta = 90^\circ$:

$$P_{max} = \frac{E_0 V}{X_S} = \frac{1.073 \times 1}{1.339} = 0.8013\ \text{pu} = \mathbf{48.08\ MW}$$

60 MW is not possible. The generator would lose synchronism (pole slip) before reaching it. To make 60 MW (1 pu) even theoretically possible, $E_0 \geq X_S \times 1/V = 1.339$ pu, so

$$I_F \geq 1.339 \times 270 = \mathbf{361.4\ A}$$

(In practice you would want even more field current, so that δ stays well below 90°.)

MARKS: (i) 1 for $\mathbf{I}$ with the + angle, 1 for the kA value. (ii) 1 for the phasor equation, 1 for E_0 and δ, 1 for I_F. (iii) 1 for P_{max} in MW with the conclusion, 1 for the minimum field current.

COMMON MISTAKES:

- $\mathbf{I} = 0.8824\angle -31.79^\circ$. That is lagging. It gives $E_0 = 1.908\angle 31.75^\circ$ pu, a different machine state.
- Using $\mathbf{E}_0 = \mathbf{V} - jX_S\mathbf{I}$. That is the motor equation with input current. This is a generator, and the course's circuit has the current flowing out of $\mathbf{E}_0$.
- Assuming that an under-excited (leading pf) generator must have $E_0 < 1$. Here $E_0 = 1.073$ pu. "Under-excited" means $E_0 \cos\delta < V$ (here 0.378 < 1), so the machine absorbs reactive power.
- In (iii), using $E_0 = 1$ pu or forgetting to convert the pu answer to MW.

You do not need every point for full marks. Roughly one mark per well-explained point.

(i) Over- and under-excitation [3 marks]

- The field current sets the excitation voltage. When a generator is connected to a large grid, the grid fixes the terminal voltage and the frequency.
- **Over-excited:** the field current is higher than needed for the terminal voltage. The generator delivers reactive power to the grid and runs at a lagging pf, like a capacitor seen from the grid.
- **Under-excited:** the field current is lower. The generator absorbs reactive power from the grid and runs at a leading pf. This weakens the magnetic coupling, so the maximum power before losing synchronism is lower (as in 4b(iii)).
- **Mechanical input power (prime mover) controls real power:** more input torque advances the rotor angle, and more real power is delivered. The speed stays at synchronous speed.
- **Field current controls reactive power:** at a fixed real power, changing the field current mainly changes the reactive power and the pf.

(ii) Rotor resistance vs V/f speed control [3 marks]

- In an induction machine, the airgap power splits into rotor copper loss (the slip fraction) and mechanical power (the rest). So the ideal efficiency is one minus the slip, the ratio of rotor speed to synchronous speed.
- Extra rotor resistance lowers the speed by increasing the slip, while the synchronous speed stays the same. At large slip, a large part of the airgap power is wasted as heat in the rotor circuit and the external resistance. At half speed, at least half the airgap power is lost.
- An inverter that lowers the frequency lowers the synchronous speed itself. The torque–speed curve slides to lower speed with the same shape, so the motor still runs at a small slip and the rotor losses stay small. Efficiency stays high.
- The voltage is reduced in proportion to the frequency to keep the flux constant. This avoids saturation and keeps the maximum torque available.

(iii) Per-unit system [2 marks]

- Quantities are expressed as fractions of chosen base values, so the normal operating values are near 1 and abnormal values are easy to see.
- If the base voltages on each side of a transformer are chosen in the ratio of its turns, the per-unit impedance is the same from both sides, and an ideal transformer disappears from the per-unit circuit. You do not have to refer impedances across each transformer.
- Machine and transformer parameters fall in narrow per-unit ranges regardless of size, so values are easy to check and compare. The single-phase and three-phase per-unit values are the same, which removes the factors of 3 and root 3.

> **COMMON MISTAKES:**

- Using equations or symbols. The question says words only.
- In (i), mixing up which knob controls what. Field current is for reactive power. Prime mover is for real power.
- In (ii), saying the first method is inefficient “because of the resistor” without saying why: the slip is large and rotor loss is the slip fraction of the airgap power.

Score interpretation

Mark each part strictly, then add up each question out of 20. Use the table below to decide what to do in your final 24 hours. Do not try to re-learn everything. Fix the one or two weakest areas, and then do the “everyone” list.

Your result	What it means	What to do
64–80 (80 %+)	Exam-ready.	Light review only. Re-do one part you lost marks on, then read through your reference sheet and the 4c model points. Sleep.
48–63 (60–79 %)	Solid. One or two topics are leaking marks.	Use the per-question rows below for any question under 15/20. Spend about 60 minutes on each weak question, not more.
32–47 (40–59 %)	Method is there but errors cost a lot.	Look at *why* you lost marks. If it was mostly signs, $\sqrt{3}$ and star/delta, re-do the traps sections of M2, M7 and M10. If it was not knowing where to start, work through the playbooks (section 2) of the packs for your two lowest questions.
Below 32	Big gaps.	Pick the two questions where you scored lowest, and do only those packs' method and worked examples. A clean 15/20 on two questions is worth more than 10/20 on four.

Per question, if you scored under 15/20:

Lost marks in	Most likely cause	Go to
Q1a, 1b	Power flow, delta vs star, $\sqrt{3}$ in the wrong place	**M1** (power, phasors, complex power), **M2** section on star/delta
Q1c	PFC: forgetting efficiency, wrong sign of Q_C, delta capacitor voltage	**M2** (PFC playbook), then redo Jun24 Q1d and Jun23 Q4b in **MX1/MX2**
Q2a	Reluctance in series, gap counting, energy and force	**M3**
Q2b	Test data: which side, referring with a^2, splitting P and Q	**M4**
Q2c	Leading vs lagging $\mathbf{I}_2'$, phasor sum, regulation and efficiency	**M5**
Q3a	Poles and n_S, slip, ideal efficiency, linear τ–n curve	**M6**
Q3b	No-load and blocked-rotor tests, per-phase delta conversion	**M8** (section 1.2), then M2's equivalent-circuit tests
Q3c	Equivalent-circuit solve, power flow, torque, $s_{\tau\max}$	**M7** (method), then **M9** (exam-style problems)
Q4a	X_S from OC/SC tests, base values	**M10**
Q4b	Generator convention, leading current, E_0, δ, P_{max}	**M11**
Q4c	Short answers too vague, or with equations	Re-read the short-answer sections in **M5**, **M8** (section 1.8) and **M11**, and the official 4c model points in **MX1**/**MX2**. Practise writing 3 bullet points per question in 4 minutes.

Everyone, in the final 24 hours (about 2 hours in total):

1. Re-do, without looking, every part where you lost marks in this mock (30 min).
2. Check your printed reference sheet has: star/delta rules, the PFC steps, $\mathcal{R} = l/(\mu_0 \mu_r A)$, the OC/SC test formulas, the transformer phasor equation, the IM simplified circuit and power flow, $s_{\tau\,\max}$, the OC/SC X_S method and $\mathbf{E}_0 = \mathbf{V} + jX_S\mathbf{I}$ (15 min).
3. Write a one-line "sign check" on the sheet: lagging current is $\angle -\phi$, leading is $\angle +\phi$; generator slip is negative; generator δ is positive (5 min).
4. Do the **R2** quick recall questions on Ch7 and Ch8 (20 min), since these two chapters were not in the mid-semester test.
5. Read the 4c model points from Jun23, Jun24 and this mock out loud once (10 min). Then stop and rest.

Short-answer bank and final pass

Answers and worked solutions · Day 24 · Mon 16 Nov · pack MXF

1. The 20 calculations that come up every year

Q1 territory (Ch1–2): power, AC circuits, three-phase

- 2023 quiz Q1: $P_{OUT} = 1.1$ kW, $P_{IN} = 1.341$ kW, loss 0.241 kW, $\tau = 7.346$ Nm (149.7 rad/s).
- Jun24 Q1b: pf = 0.8184 lag, $P_{OUT} = 3.420$ kW, $\eta = 85.50\%$.

Jun23 Q1a: $E = 13.8$ V, $R_S = 0.01433\ \Omega$, loss 1290 W, motor 2850 W, $V(20\ \text{A}) = 13.51$ V.

Jun23 Q1b (70 Ω, 460 V): star 3.794 A phase and line, 3.023 kW. Delta 6.571 A phase, 11.38 A line, 9.069 kW.

- Jun23 Q1c ($I_1 = 13\angle 20^\circ$ A, 10 Ω, $j30$ Ω, $-j50$ Ω): $V_R = 130\angle 20^\circ$, $V_L = 390\angle 110^\circ$, $V_C = 650\angle -70^\circ$ V. $Z = 22.36\angle -63.43^\circ\ \Omega$, $V_1 = 290.7\angle -43.43^\circ$ V. $S = 1690 - j3380$ VA (sheet: $-j3379$).
- Jun24 Q1c: resistor 400 W and 0 VAr, inductor 0 W and 160 VAr. $V_1 = \sqrt{50^2 + 20^2} = 53.85$ V.
- 2024 test Q3 (parallel, 240 V): $I_R = 2\angle 0^\circ$, $I_L = 4\angle -90^\circ$, $I_1 = 4.472\angle -63.43^\circ$ A. P = 480 W, Q = 960 VAr.
- 2023 quiz Q2c: $V_L = 1.5\angle -20^\circ \times (20 + j5) = 30.93\angle -5.96^\circ$ V.

2023 quiz Q2: $R = 49.31\ \Omega$, $X = 0.9028\ \Omega$. It is inductive (absorbs Q), so $L = 2.874$ mH.

Jun24 Q1d: load pf 0.8 lag. $P_S = 80$ W, $Q_S = 38.75$ VAr, $Q_C = -21.25$ VAr. For $Q_C = -30$ VAr at 115 V, 60 Hz: $C = 6.017\ \mu$F.

Q2 territory (Ch3): magnetics and transformers

- Jun23 Q2a(i): $\mathcal{R}_g = 596.8$ kA/Wb, $\mathcal{R}_c = 66.31$ kA/Wb, $L = 1.544$ mH.
- Jun24 Q2a(i): $\mathcal{R}_g = 810.6$ kA/Wb, $L = 17.77$ mH.
- 2023 quiz Q3b: $\mathcal{R} = 100\,000$ A/Wb, $L = 0.4$ H, $X = 125.7\ \Omega$. Then $I(100\ \text{V}) = 0.7958$ A. The sheet gives 0.7955 A because it rounded X to 125.6 Ω.

- Jun23 Q2a(ii): $E = 3.867$ J.
- Jun24 Q2a(iii): $F = 439.5$ N.
- 2023 quiz Q4: $I = 46.69$ A, $F = 962.9$ N.
- 2024 test Q4: $B = 0.3519$ T (no gap), 0.1086 T (with paper), $B_{min} = 0.0860$ T.

- 2023 quiz Q3a: $N = 360.4$, so 360 or 361 turns.

- 2024 test Q5: $N_1 = 168.9$, so 169 turns. $R = 3.453\ \Omega$, $N_2 = 33.8$, so 34 turns.
- Jun24 Q2a(ii): $V_{max} = 19.62$ V.

Jun23 Q2a(iii): $P_H = 3.84$ W, $P_E = 14.93$ W, total 18.77 W.

Jun24 Q2b (6:1): $Z_1 = 0.3\ \Omega$, $Z_2 = 10.8\ \Omega$, $Z_3 = 25.8\ \Omega$. $I_1 = 0.4651$ A, $I_2 = 2.791$ A, $V_1 = 5.023$ V, $V_2 = 0.8372$ V. These are consistent with a 0.3 Ω load and a 15 Ω series resistance on the primary (see the figure in the PDF).

- Jun23 Q2b: $R = 0.5845\ \Omega$, $X = 2.721\ \Omega$, $R_C = 1280\ \Omega$, $X_M = 186.6\ \Omega$.
- 2024 test Q6: $X = 1.005\ \Omega$ and $R = 0.3068\ \Omega$. **The official sheet prints 0.3086 Ω, which is a digit swap.** $46\,667/390^2 = 0.3068$.

- 2023 quiz Q5: $P_{CU} = 1.179$ kW, $P_{FE} = 1.369$ kW, $\eta = 97.78\%$ (sheet 97.79%, a rounding difference).
- 2023 quiz Q6: (a) $V_2' = 1.905\angle 0^\circ$ kV, $I_2' = 6.056\angle -38.74^\circ$ A. (b) $V_1 = 2.254\angle 5.89^\circ$ kV, so the line voltage is **3.904 kV**. **The sheet's 3.883 kV does not match its own phase value** ($\sqrt{3} \times 2.254 = 3.904$). Read "21000V" as $2100\angle 0^\circ$ V.
- 2024 test Q7: $V_2' = 8.083\angle 0^\circ$ kV, $I_2' = 6.186\angle +13.49^\circ$ kA.
- Jun24 Q2c(ii): $I_1 = 660.9$ A, $P = 1.024$ MW.

Q3 territory (Ch4–5 basics in part a, Ch7 induction machines)

Jun23 Q3a: 10 poles ($n_S = 600$ rpm). Induction, because 560 < 600. Ideal efficiency 93.33%. At ¼ torque: 590 rpm, 26.34% of rated power.

Jun24 Q3b: $R_1 = 2\ \Omega$, $R_2' = 3.2\ \Omega$, $X = 7.635\ \Omega$.

- Jun23 Q3c: 4 poles, 1500 rpm. At 1520 rpm it is a generator. At 1450 rpm, $\tau = 101.7$ Nm and $I_1 = 29.79\angle -38.44^\circ$ A.
- Jun24 Q3c: $n_S = 1000$ rpm, $s = -0.025$, $R_2'/s = -120\ \Omega$. $I_1 = 2.029\angle -174.2^\circ$ A, $P_{IN} = -1451$ W, $P_{AG} = -1.482$ kW, $\tau = -14.15$ Nm.

Jun23 Q3b(i): $R'_{EXT} = 2.828 - 0.5 = 2.328\ \Omega$. With no external resistance, $s_{\tau,max} = 0.177$.

Q4 territory (Ch8 and per-unit)

- Jun23 Q4a: $X_S = 3.676\ \Omega$. $V = 1.0\angle 0^\circ$ pu, $I = 0.7454\angle -26.57^\circ$ pu. The sheet prints 0.7453∠26.57°: it truncates, and the OCR lost the minus sign. Delivering Q means a lagging current.
- Jun24 Q4a: $X_S = 1.047\ \Omega = 2.137$ pu. $V = 0.97\angle 0^\circ$ pu, $I = 0.6443\angle -36.87^\circ$ pu.

> Jun24 Q4b: $I = 1\angle +25.84^\circ$ pu, $E_0 = 1.634$ pu, $\delta = 82.43^\circ$ (sheet 82.42°). $P_{max} = 0.9079$ pu at 90° (sheet 0.9077, from the rounded E_0).

> Jun23 Q4b: induction motor 235.3 kW, 158.1 kVAr. Synchronous motor 222.2 kW, 114.5 kVAr (sheet 114.6, from the rounded 222.2). Plant pf = 0.9955 lag.

2. Short-answer bank (Q4c style)

Past-paper short answers (official points)

S1.

- Many loads, especially induction motors, draw reactive power as well as real power. Reactive power is energy that oscillates back and forth between the source and the load each cycle, and does no useful work.
- It still needs current. So the supply current and the apparent power are larger than the real power alone needs. This can overload generators, transformers and cables, and causes extra resistive losses.
- Power-factor correction connects a capacitor in parallel with the load. The capacitor generates reactive power, so the load's reactive power is supplied locally and the reactive power drawn from the supply falls.
- The real power and the load's own operation are unchanged. Only the supply current falls, and the supply power factor rises towards one. Supply companies also charge for peak volt-amps, so the bill drops.

`OFFICIAL POINTS:` loads draw reactive power, which is energy oscillating in the system that can overload generators and transformers and produce extra losses. A parallel capacitor generates reactive power and reduces the reactive power drawn from the supply.

S2.

- **Similar:** both can run from three-phase AC. Both have a primary winding (the stator), a secondary winding (the rotor) and an iron core. Both transfer power by electromagnetic induction, so their equivalent circuits have the same elements.
- **Different:** the induction machine's secondary rotates and is short-circuited. A transformer's secondary is stationary and feeds an external load.
- The rotor currents are at slip frequency, which changes with speed. In a transformer both windings are at supply frequency.
- In the induction machine equivalent circuit the rotor resistance is divided by the slip. This represents both the rotor copper loss and the mechanical output.
- The induction machine has an airgap, so its magnetising current is a much larger fraction of rated current.

`OFFICIAL POINTS:` both 3ph AC, with primary and secondary windings, a magnetic core and similar equivalent circuits. But the IM secondary is rotating and short-circuited,

and the rotor frequency differs from the stator frequency. In the IM equivalent circuit the rotor resistance is divided by slip.

S3.

- The referred rotor resistance would be zero.
- There would be no rotor copper loss, so the efficiency would rise.
- The slip at maximum torque is proportional to rotor resistance, so it would be zero. The torque–speed curve would be vertical at synchronous speed, like a synchronous machine: no speed drop with load.
- The starting torque would be zero. At standstill the effective rotor resistance absorbs no power, so the motor could not start by itself.

`OFFICIAL POINTS:` zero rotor resistance; the torque–speed slope at synchronous speed becomes vertical; no rotor losses, so higher efficiency; zero starting torque. (Think of the Q3b(ii) curves.)

S4.

- **Real power** is the average of the instantaneous power. It comes from the current component in phase with the voltage. It is the net energy delivered to the load and does useful work (heat, light, shaft power). Resistances absorb it.
- **Reactive power** comes from the current component 90° out of phase with the voltage. It is energy that oscillates between source and load and averages to zero. Inductors absorb it and capacitors generate it. It does no useful work but still needs current.
- **Apparent power** is the magnitude of the complex power, whose real part is the real power and whose imaginary part is the reactive power. It is the rms voltage times the total rms current.
- **Significance:** apparent power sets the current, and so the rating of generators, transformers and cables. Real power is the energy that is paid for. Reactive power is why power-factor correction is used. Real and reactive power are both scalars.

`OFFICIAL POINTS:` for real and reactive power, any of: phase of current relative to voltage, link to instantaneous power, energy flow between source and load, ability to do useful work, which elements are associated, both scalar. For apparent power: magnitude of complex power, related to total current.

S5.

- **Copper loss:** resistive heating of the windings by the load current. It is modelled by the series winding resistance (primary plus referred secondary).
- **Iron loss:** hysteresis loss (energy to cycle the core's magnetisation) and eddy-current loss (currents induced in the core). It is modelled by the core-loss resistance in the shunt branch across the supply.

- **Variation:** copper loss goes with the square of the current, so at constant voltage it goes with the square of the output power. Iron loss depends on the flux, which is set by the supply voltage and frequency, so it stays roughly constant with load.
- So iron loss dominates at light load and copper loss at heavy load. Maximum efficiency occurs where they are equal.

OFFICIAL POINTS: explain copper and iron losses, their physical mechanism and their equivalent-circuit parameters. Copper loss is proportional to the square of output power, and iron loss does not vary with output power.

S6.

- **Magnetising current:** the stator must set up the rotating field even with no load. This current flows in the magnetising reactance. The airgap makes the reluctance high, so this current is large (about 25–60% of rated current, against about 10% in a transformer) and almost purely reactive. That is why the no-load power factor is low.
- **Core loss:** the rotating flux causes hysteresis and eddy-current loss in the iron. A small in-phase current, modelled by the core-loss resistance, supplies this loss.
- **Mechanical loss:** bearing friction and windage (air drag, including the cooling fan) need some shaft power even with no load. So the rotor runs slightly below synchronous speed, and a small rotor current flows.
- These currents also cause a small stator copper loss.

OFFICIAL POINTS: magnetising current and core-loss resistance (their equivalent-circuit parameters and physical reasons), and mechanical losses needing power even at no load, with the types of mechanical loss.

S7.

- **Process:** for a given voltage and frequency, the back-emf equation fixes the product of turns and peak flux. So the number of turns sets the peak flux density. Choose the turns so that the peak flux density is just below the saturation value of the core.
- **Too few turns:** the peak flux density needed is too high, so the core saturates. The magnetising current rises sharply and becomes distorted, the input current is high, and the winding may overheat.
- **Too many turns:** the flux density is lower than needed, so the core is under-used. The longer winding (and thinner wire, to fit the window) has more resistance. Copper loss rises, efficiency falls, and the winding may overheat.

OFFICIAL POINTS: the turns change the peak core flux (back-emf equation), and are chosen so the flux density is below saturation. Too few: too high a flux density, saturation, high input current, possible overheating. Too many: low flux density, higher winding resistance, higher copper loss and lower efficiency, possible overheating.

S8.

- **Ideal:** the output voltage does not change. It is set only by the input voltage and the turns ratio, and the load does not affect it.
- **Ideal:** the output current halves, by Ohm's law, because the voltage is fixed and the resistance has doubled.
- **Real:** the output voltage rises slightly. A real transformer has series winding resistance and leakage reactance. The lower load current gives a smaller voltage drop across them.
- **Real:** the output current still falls, but to slightly more than half, because the output voltage is now slightly higher.

`OFFICIAL POINTS:` the same four points.

S9.

- It **lags**.
- The series impedance of the transformer is mainly leakage reactance, so the load current lags the primary voltage that drives it.
- The load is resistive, so the load voltage is in phase with the load current. It therefore lags the primary voltage.

`OFFICIAL POINTS:` lagging, because the series reactance makes the current lag the primary voltage, and the load voltage is in phase with the current.

S10.

- It is an **induction** machine. 560 rpm is not a synchronous speed at 50 Hz: the nearest one above it is 600 rpm (10 poles).
- A synchronous machine runs exactly at synchronous speed. An induction machine must run slightly below it, so that there is slip and therefore rotor current and torque.

`OFFICIAL POINTS:` induction, as the rated speed is less than the synchronous speed.

S11.

- **DC machine:** halve the armature voltage. The no-load speed is proportional to voltage, and the slope of the torque–speed line stays the same.
- **Synchronous machine:** halve the frequency and halve the voltage. The frequency sets the synchronous speed, and halving the voltage as well keeps the volts-per-hertz ratio, and so the flux, at its rated value.
- **Induction machine:** halve the frequency and halve the voltage, for the same reason. The curve keeps its shape and slides to half speed.

- Never reduce only the frequency on an AC machine: the flux would double, the core would saturate and the current would become very large.

`OFFICIAL POINTS:` synchronous: halve both V and f. DC: halve V. Induction: halve both V and f.

S12.

- On open circuit, only the shunt branch (core-loss resistance in parallel with magnetising reactance) carries current, and it is a fixed impedance in the model.
- Current ×0.5, because current is proportional to voltage.
- Real power ×0.25 (voltage squared over the core-loss resistance). Reactive power ×0.25 (voltage squared over the magnetising reactance).
- Power factor ×1, because both powers scale equally.

`OFFICIAL POINTS:` ×0.5, ×0.25, ×0.25, ×1.

Likely new short answers

S13.

- **Speed:** a synchronous machine runs exactly at synchronous speed at any load. An induction machine runs slightly below it as a motor (above it as a generator), and the slip grows with load.
- **Rotor field:** in a synchronous machine the rotor field comes from DC field current (through slip rings) or permanent magnets. In an induction machine, the rotor currents are induced by the slip between the rotor and the rotating stator field, so no rotor supply is needed.
- **Reactive power:** an induction machine always absorbs reactive power to magnetise its airgap, so it runs at a lagging power factor. A synchronous machine's reactive power is controlled by the field current. It can generate or absorb reactive power.
- **Starting and cost:** an induction motor is self-starting, cheap and robust. A synchronous motor is not self-starting on a fixed-frequency supply and costs several times more.

S14.

- Speed is changed by changing the supply frequency from an inverter, because frequency sets the synchronous speed.
- By the back-emf equation, the peak flux is proportional to voltage divided by frequency (neglecting stator resistance).
- If the frequency is reduced at constant voltage, the flux rises above its rated value and the core saturates. The magnetising current then becomes very large and the machine overheats.

- If the voltage is reduced too much, the flux falls and so does the available torque.
- Keeping the volts per hertz at the rated value keeps the flux constant. So the maximum torque stays about the same, and the torque–speed curve slides along the speed axis. At low speed, the drop across the stator resistance means a small voltage boost is needed.

S15.

- The secondary of both is short-circuited. But in the induction machine the rotor resistance is divided by the slip, so the effective rotor resistance changes with speed.
- This is because the rotor moves. The voltage induced in the rotor, and the rotor frequency, are proportional to the slip. Referring the rotor to the stator frequency turns this into the rotor resistance divided by the slip.
- This resistance represents two things: the real rotor copper loss and the mechanical output power.
- The magnetising reactance is much smaller (the magnetising current much larger), because the flux must cross the airgap. So the shunt branch cannot be ignored the way it often can for a transformer.

S16.

- An airgap has a relative permeability of one, so even a short gap has a reluctance far larger than the iron. The gap usually dominates the total reluctance.
- So the inductance falls, and a larger current is needed for the same flux. Because the gap reluctance is constant, the inductance also becomes more linear and less dependent on the iron's permeability and saturation.
- Most of the magnetic energy is stored in the gap, and the force on the poles comes from the field in the gap.
- In an induction machine the gap is unavoidable, because the rotor must turn. It is made as small as mechanically possible, because a larger gap means more magnetising current, a lower power factor and more stator copper loss.

S17.

- Each quantity is expressed as a fraction of a base value, usually the machine's rating. So you can see at once whether a value is large or small: one per-unit voltage is rated voltage, and one per-unit current is rated current.
- Per-unit values of similar machines fall in a narrow range regardless of size, for example a synchronous reactance of about one to two per-unit. This makes it easy to compare machines and to spot errors.
- In three-phase per-unit, the line and phase values are equal and the total and per-phase powers are equal. The square-root-of-three and factor-of-three terms, and the star–delta conversions, disappear from the working.

- With bases chosen consistently on each side, a transformer's impedances are the same in per-unit whichever side they are referred to. This simplifies power-system calculations.

S18.

- Industrial plants are dominated by induction motors. These must absorb reactive power to magnetise their airgaps, so the plant draws a lagging current.
- Supply companies charge for peak volt-amps as well as energy, so a low power factor costs money. It also loads cables and transformers.
- A synchronous motor's reactive power is set by its field current. If it is over-excited, it generates reactive power (a leading input current) while still driving its mechanical load at constant speed.
- That generated reactive power offsets the induction motors' lagging reactive power, so the total plant power factor rises towards one.
- The amount available is limited by the motor's volt-amp rating less its real power. Synchronous motors cost several times more than induction motors, but the lower tariffs can justify the cost.

S19.

- The machine is connected to the supply and driven by a prime mover above synchronous speed. The slip is then negative.
- The effective rotor resistance becomes negative, so the airgap power and the torque reverse sign. Mechanical power from the shaft is converted to electrical power and fed back to the supply.
- It still needs magnetising current, so it still absorbs reactive power from the supply (or from capacitors) even while it delivers real power. It cannot control its own reactive power or voltage.
- Its output depends on how far above synchronous speed it runs, so it suits wind turbines and regenerative braking. Its losses still have to be supplied, so the electrical output is less than the shaft input.

S20.

- Balanced three-phase stator currents produce a magnetic field of constant magnitude that rotates at synchronous speed.
- When the rotor turns slower than the field, its conductors see a changing flux. This induces voltages in them at slip frequency, and because the rotor is short-circuited, rotor currents flow.
- These currents, in the stator field, produce forces on the rotor conductors, and so a torque in the direction of the field's rotation.

- At synchronous speed the rotor moves with the field and sees no change of flux. There is no induced voltage, no rotor current and no torque. Some slip is always needed.

S21.

- **DC test:** measure the stator winding resistance with DC. This gives the stator resistance per phase.
- **No-load test:** run unloaded at rated voltage. The slip is nearly zero, so the rotor branch draws almost no current, and the shunt branch dominates. Measuring voltage, current and power gives the core-loss resistance and magnetising reactance. The core-loss figure also includes friction and windage.
- **Blocked-rotor test:** hold the rotor still (slip of one) and apply a reduced voltage that gives about rated current. The voltage is low, so the shunt branch can be ignored. The measurements give the total series resistance and leakage reactance.
- Subtract the stator resistance from the total series resistance to get the referred rotor resistance. The leakage reactance is usually split equally between stator and rotor.

S22.

- **Squirrel cage:** bars (usually aluminium) shorted by end rings. Its resistance is fixed. It is cheap, robust and the most common type.
- **Wound rotor:** a three-phase winding brought out through slip rings, so external resistance can be added.
- More rotor resistance increases the slip at which maximum torque occurs, but the maximum torque itself does not change. This raises the starting torque and lowers the starting current.
- The penalty is higher rotor copper loss and lower efficiency while it is in circuit. So the resistance is cut out once the motor is up to speed.

S23.

- At standstill the motor produces a **starting torque**, usually above rated torque, while drawing a large current (several times rated).
- As speed rises, the torque rises to a peak, the **breakdown (pull-out) torque**, typically two to three times rated. It occurs at the slip set by the rotor resistance divided by the stator impedance.
- Between the breakdown point and synchronous speed the curve is almost a straight line, and torque is roughly proportional to slip. This is the stable operating region, and **rated torque** sits in it at a few percent slip.
- The torque falls to zero at synchronous speed and reverses above it (the generating region). If the load exceeds the breakdown torque, the motor slows down and stalls.

S24.

- Three-phase stator currents produce a field that rotates at synchronous speed. The rotor carries its own field, from DC field current or magnets, which is fixed relative to the rotor.
- Steady torque is only produced when the two fields rotate together. So the rotor turns at exactly synchronous speed, locked to the supply frequency.
- The torque depends on the angle between the rotor and stator fields (the rotor angle). As a generator the rotor field leads and drags the stator field. As a motor the stator field leads and drags the rotor.
- Torque rises with the rotor angle up to a maximum at 90°. Beyond that the machine loses synchronism (pull-out), which causes large currents and torque pulsations. Rated load is at about 60° to leave a margin.

S25.

- A low-speed generator, such as a hydro machine, needs many poles to make 50 Hz. Salient (projecting) poles make it easy to fit many poles on a large-diameter rotor, and at low speed the centrifugal stresses are modest.
- A high-speed generator, such as a steam or gas turbine at 3000 rpm, has only two or four poles and very high centrifugal forces. A round, solid rotor with the winding set in slots is mechanically strong and has low windage. Its long, thin shape keeps the diameter, and so the stress, down.

S26.

- The main limit is the stator's volt-amp rating, which is set by stator current heating. The apparent power is the hypotenuse of real and reactive power, so the reactive power available is the square root of rated volt-amps squared minus real power squared.
- So with no real power load (a synchronous compensator), the full rating is available as reactive power. At full real power, no reactive power is available.
- Generating a lot of reactive power needs high excitation, so the rotor field current (rotor heating) also sets a limit.
- Absorbing reactive power needs low excitation. That raises the rotor angle towards 90° and reduces the stability margin, which limits the leading side.

S27.

- The terminal voltage and frequency are fixed by the power system. The real power output is set by the mechanical input from the turbine, not by the field current.
- The field current sets the internal emf. At one particular excitation the output power factor is unity, and the stator current is a minimum for that power.

- **Over-excited:** the generator delivers reactive power and runs at a lagging output power factor.
- **Under-excited:** the generator absorbs reactive power, runs at a leading power factor and has a larger rotor angle, so it is closer to losing synchronism.
- Either way from unity, the apparent power and so the stator current rise. A plot of stator current against field current at constant power is the “V-curve”.

S28.

- Both are a voltage source in series with an impedance, connected to the terminals. The source represents the conversion between electrical and mechanical power.
- In both, the internal emf is proportional to the speed and to the field (excitation) flux, and power flows from the source to the terminals when generating and the other way when motoring.
- The DC machine’s quantities are DC and the series element is the armature resistance. The synchronous machine’s quantities are AC phasors, and the series element is mainly the synchronous reactance, with only a small resistance.
- So in the synchronous machine the phase angle of the emf (the rotor angle) controls real power, and its magnitude controls reactive power. The DC machine has no reactive power.

S29.

- The alternating flux induces voltages in the core itself, which drive eddy currents and cause loss.
- Building the core from thin, insulated laminations breaks up the eddy-current paths and greatly reduces the eddy-current loss. Thinner laminations mean less loss.
- Adding silicon to the steel raises its resistivity, which further reduces eddy currents. It also reduces hysteresis loss.

S30.

- For a given power, the line current is inversely proportional to the transmission voltage.
- The resistive loss in the line is proportional to the square of the current, so raising the voltage greatly reduces the loss. Doubling the voltage quarters the loss.
- The lower current also allows thinner conductors and gives a smaller voltage drop. Transformers make it cheap to step the voltage up for transmission and down again for use.

S31.

- A balanced three-phase winding naturally produces a rotating magnetic field of constant magnitude. AC machines work on this principle, so a three-phase machine is

self-starting. A single-phase machine only produces a pulsating field and needs special starting arrangements.
- The total instantaneous power of a balanced three-phase system is constant. A single-phase machine's power pulsates at twice the supply frequency. So three-phase machines run more smoothly, with less vibration and noise.
- Three-phase also uses the machine material and the supply conductors more efficiently. Single-phase is used only at low power in homes, for simplicity and cost.

S32.

- At standstill the stator field sweeps past the rotor at synchronous speed. The rotor field is first pulled one way and then the other, every half cycle.
- The average torque is zero, and the rotor's inertia stops it following, so the rotor stays still and vibrates.
- So it must be brought up to near synchronous speed by other means before it can lock in. Examples are a pony motor, a starting cage on the rotor (induction start), or an inverter that ramps up the frequency.